Space Challenges Program 2026

Technical Design Document

Armonia

A Three-Axis Reaction-Wheel Balancing Cube

Project Team

Athanasia Nikolova	Andrea Liang	Deyan Nikolaev Vlaev
Niccolo Tonetto	Suvanna Viriyanti Wu	Pablo Urioste

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1 Introduction

1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [**gajamohan2012cubli**, **gajamohan2013cubli**], whose original prototype (15 × 15 × 15 cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

The cube described in this document is Armonia, a Cubli of that class built to the requirements of Section 2. Throughout, *Cubli* names the platform class and the body of work behind it, while *Armonia* names the article built here; where a statement holds for any cube of the class it is written of the Cubli, and where it is a property of this build it is written of Armonia.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilise a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

Alongside the technical problem sits an educational one. Attitude determination and control is taught almost everywhere as mathematics on a whiteboard, because the hardware it describes is unreachable: flight units are expensive, single-shot and locked behind qualification procedures, so a student may complete an entire aerospace curriculum without ever commanding a reaction wheel and watching a body respond. What is missing is not theory but a system a student is allowed to build, mis-wire, crash and fix. The Cubli is exactly that system — it runs the same momentum-exchange physics as a real satellite, on a desk, at a cost a school can absorb.

The problem this project addresses is therefore threefold: (i) derive and validate a dynamic model and controller capable of stabilising this underactuated system about its unstable equilibria and driving it through commanded reorientations; (ii) realise this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing; and (iii) do both in a form that high-school and university students can reproduce, so that the mechanical build, the electronics, the firmware and the control law are documented as *reasoned decisions* a learner can follow and re-derive, not as finished numbers to be copied. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, jump-up, and walking (Section 1.3).

No single subsystem is exotic, but each sits close to a physical bound, and the bounds are coupled: relieving one tightens another. Four constraints drive the design, and sizing is therefore treated as the first engineering activity rather than a detail settled once parts arrive.

- **The jump-up has no energy margin.** The required per-wheel inertia scales as $ML^{3/2}$ (Section 5.1.1), so mass added anywhere demands a larger wheel, and the wheel is itself part of M. Sizing is a fixed-point iteration, not a one-pass calculation.
- **Braking torque fights the structure.** Momentum accumulated over seconds of spin-up

is shed in tens of milliseconds, through one shaft, hub and motor mount — the parts the mass budget wants thin. The structural load case is set by the manoeuvre that proves the machine works.

- **Sensing, not the control law, sets the balancing limit.** Gyroscope bias drifts, and the accelerometer — the only absolute tilt reference — is corrupted by vibration the controller itself generates. Wheel balance is therefore an estimation requirement, not a mechanical nicety.
- **The plant is nonlinear and constrained.** Corner balancing does not decouple into three single-axis problems: the axes are coupled through the contact point and the inertia tensor. The actuators saturate in both torque and speed, so stored momentum must be actively managed. Linearising about upright is adequate for balancing and inadequate for anything more [**muehlebach2017cubli**].

1.2 Purpose, Application & Mission Context

The purpose of this project is educational: to make space engineering something a student can hold, build and debug. The intended application of the hardware described in this document is as a teaching platform for high-school and university students, through which a complete satellite attitude control chain — structure, electronics, firmware, estimation and control — is learned by constructing it rather than by reading about it. Every other claim in this section is subordinate to that one: the spacecraft analogy matters because it is what makes the exercise *real*, and the engineering rigour of the sections that follow matters because a teaching platform that does not actually work teaches nothing.

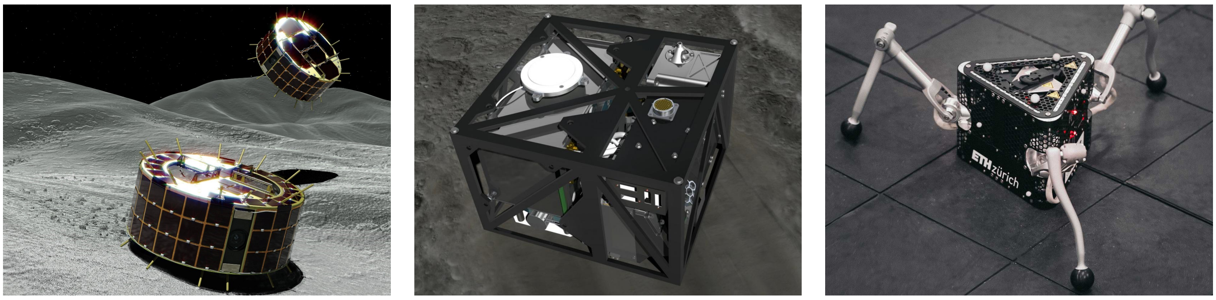
The Cubli is well suited to this role because it is not a simulation of a satellite subsystem; it is one, operated under gravity. The cube is therefore not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilised satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli's three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem.

Reaction wheels are the actuator choice made here, and the alternative is worth stating explicitly because it is the standard smallsat comparison. Since this is a proof-of-concept demonstration of the control loop, a system actuated with orthogonal reaction wheels is well-suited to represent principles applicable to smallsat ADCS: the momentum-exchange actuation principle and closed-loop control architecture are the same as those used on real spacecraft, even though the Cubli's dynamics (balancing against gravity) differ from a satellite's free-fall attitude control problem. Reaction wheels are also structurally simpler and require less specialised hardware than control moment gyros, which offer higher torque density but at the cost of gimbal mechanisms and more complex control software — complexity that is typically only justified on larger spacecraft. For a low-cost, resource-limited project, a reaction-wheel Cubli therefore demonstrates the core actuation and control principles of smallsat ADCS in a practical and relevant way.

Framed as an Attitude Determination and Control System (ADCS) problem, the Cubli's core behaviours map directly onto a spacecraft's day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.

- **Controlled rotation** → a slew manoeuvre: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.
- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies where wheeled traction is impossible (Figure 1). Walking, as a sequence of controlled directional hops, extends single-hop capabilities and represents an active research frontier (Section 1.3).



(a) MINERVA-II1 (b) MASCOT (c) SpaceHopper

Figure 1: Momentum-exchange actuation in flight and research hardware: (a) JAXA's MINERVA-II1 rovers [**yoshimitsu1999minerva**] and (b) the MASCOT lander [**ho2017mascot**], deployed by Hayabusa2 onto asteroid Ryugu (2018), hop across the surface and self-right by spinning up and braking an internal flywheel or eccentric mass; (c) ETH Zurich's SpaceHopper [**spiridonov2024spacehopper**] extends this principle to controlled, directional hopping in low gravity.

The subsystem exercised here is therefore the one a spacecraft uses for pointing: momentum-exchange attitude control under actuator saturation, with desaturation, on an underactuated body whose inertia is imperfectly known. Under gravity the destabilising torque never relents, so an inadequate controller falls over immediately and unambiguously — the demonstration is self-evaluating, which a free-fall attitude testbed is not. That property is what makes the platform teachable: the student needs no instructor and no test verdict to know whether the design is correct, because the cube either stands up or it does not.

Because the cube is a complete system rather than a single experiment, building it forces a student through the whole engineering chain, each discipline entered because the next one depends on it:

- **Design & CAD** → parametric modelling, tolerance and fit, centre-of-mass placement, and design for 3D printing (Section 6, Appendix C).
- **Physics & sizing** → conservation of angular momentum turned into numbers: wheel inertia, torque and speed budgets, and the fixed-point iteration between mass and actuator that makes engineering trade-offs concrete (Section 5).
- **Structures & building** → printing, assembly, fastening and balancing real parts, and discovering that the braking manoeuvre which proves the machine works is also the load case that breaks it.
- **Electronics & power** → motor drivers, battery and power budgeting, harnessing, grounding and the noise a switching motor injects into a sensor sitting centimetres away (Section 7, Appendix D).
- **Coding & embedded systems** → real-time firmware, fixed-rate control loops, sensor drivers, telemetry and debugging on hardware where a software bug becomes a falling cube.
- **ADCS, estimation & control** → state estimation and sensor fusion from a noisy IMU, feedback design, saturation and momentum management — the same problems, and largely the same equations, as a CubeSat's pointing loop (Section 8, Appendix E).

- **Systems engineering** → requirements traced to capabilities, budgets that must close, staged verification, and the experience of one subsystem's decision propagating into three others (Sections 3.2 and 4).

The nominal Cubli problem was solved a decade ago on precisely manufactured hardware [**gajamohan2012cubli**, **gajamohan2013cubli**, **muehlebach2017cubli**]; reproducing it is an engineering exercise, and this document claims no more — but reproducibility is the point when the deliverable is a platform other students are meant to rebuild. A commodity build is adopted for four reasons. It can be crashed repeatedly, so the controller can be tested at its limits and a learner can fail without consequence; its loose tolerances, imperfectly balanced wheels and drifting IMU supply the parameter uncertainty that robust control methods are intended to tolerate; a fixed motor, print volume and budget constrain the sizing, since a marginal design cannot be rescued by specifying a larger component; and every part is a catalogue item or a printed file, so the cost of entry stays within reach of a school or student team rather than a laboratory. The third of these motivates the adjustable ballast scheme of Section 5.3.

1.3 Goal & Objectives

Goal. *To build a low-cost, reproducible reaction-wheel cube that teaches high-school and university students the engineering of a real satellite — design, structures, electronics, embedded coding and ADCS — by having them build, program and fly one, and in doing so make space engineering an accessible, hands-on discipline rather than an abstract one.*

The goal decomposes into two sets of objectives that must both be met: **technical** objectives, which the hardware and controller must demonstrate, and **educational** objectives, which the platform and this document must deliver.

Technical objectives. Table 1 lists each technical objective, the capability it imposes and the requirements that verify it; thresholds and verification methods are given in Section 2.

Edge balance, corner balance and controlled rotation are committed deliverables; jump-up and walking are stretch objectives, contingent on the jump-up feasibility check and on final hardware selection.

Educational objectives. Table 2 lists what the student must be able to do and what the platform must provide for it.

These provisions are constraints on the engineering, not an additional workstream: the commodity build, the parametric CAD (Section 6), the adjustable ballast (Section 5.3), the staged validation (Section 4) and the bill of materials (Appendix B) all follow from them.

Table 1: Objective-to-requirement traceability. Requirement identifiers refer to Table 4.

Objective	Required capability	Requirements
Edge balance	The system must actively maintain dynamic stability and regulate its posture about a single line-contact equilibrium position.	FUN-01, FUN-04, PER-01, PER-03, PER-04
Corner balance	The system must maintain multi-axis closed-loop balance and attitude control around a highly unstable point-contact equilibrium.	FUN-02, FUN-04, PER-02, PER-04
Controlled rotation	The controller must accurately track dynamic reference trajectories to execute smooth slew maneuvers between distinct equilibrium states.	FUN-03
Jump-up (stretch)	The system must store sufficient wheel angular momentum and shed it near-impulsively through a braking event to tip itself from a face rest onto an edge or corner, and then capture the resulting attitude into balance.	FUN-10, PER-06, PER-07, PER-08, PER-10, CON-02
Walking (stretch)	The system must execute a sequential series of dynamic jump-up maneuvers and rapidly re-stabilize itself upon each landing impact.	FUN-11

Table 2: Educational objectives and what the platform must provide to deliver them.

Learning objective	Required provision
Build it	The mechanical design must be printable on hobby-grade equipment and assembled with catalogue fasteners and standard tools, so that construction is a student activity rather than a workshop service.
Wire it	The electrical architecture must be documented to schematic, pinout and harness level, with a bring-up procedure that isolates faults one subsystem at a time.
Program it	The firmware must expose the control loop, the estimator and the tunable gains as editable, readable code, with telemetry sufficient for a student to see why a run failed.
Understand it	Every sizing and control result must be derived from stated assumptions and reproducible from the equations given, so the numbers can be re-derived and challenged rather than trusted.
Break and fix it	The build must tolerate repeated falls and be repairable from spare printed parts and stocked components, so that experimentation carries no penalty beyond time.
Reproduce it	The design must be transferable: parametric CAD, a complete bill of materials and staged verification tests, so another school or team can rebuild the platform from this document.

2 System Requirements and Verification

Based on the objectives established in Section 1.3, a set of technical requirements has been developed to ensure the final design aligns with overall project goals. While a portion of these requirements flows directly from top-level objectives, others were derived during preliminary sizing to accommodate constraints identified during modeling and trade studies.

The complete requirement set is summarized in Table 4. To parse this matrix, Table 3 defines the verification codes used to demonstrate compliance, while full testing procedures, acceptance criteria, and evidence are detailed in Appendix F.

Table 3: Verification methods used in Table 4.

Code	Name	Meaning
A	Analysis	Closure by calculation or simulation against a stated model.
I	Inspection	Confirmation against drawings, a parts list, or a measurement of the built article.
T	Test	Instrumented measurement against a numerical criterion.
D	Demonstration	Observed execution of a behaviour, without instrumented measurement.

Table 4: System requirements, with parent objective and verification method. (D) marks a derived requirement. Methods: A analysis, I inspection, T test, D demonstration.

ID	Requirement	Parent objective	V
Functional			
FUN-01	The cube shall maintain balance about an edge equilibrium using only internal reaction-wheel actuation.	Edge balance	T
FUN-02	The cube shall maintain balance about a corner equilibrium under three-axis feedback.	Corner balance	T
FUN-03	The cube shall execute a commanded slew between two specified attitudes and re-capture balance at the target.	Controlled rotation	T
FUN-04	The cube shall reject an impulse disturbance applied to the body and return to the balancing equilibrium.	Edge, corner balance	T
FUN-05	The system shall estimate body attitude and rate from onboard sensing alone, with no external reference.	All	T
FUN-06	The system shall bound stored wheel momentum by commanding a desaturation torque above a defined wheel-speed threshold.	All	T
FUN-07	The system shall sequence operating modes through a supervisory state machine with hysteresis on every threshold.	All	D
FUN-08	The system shall provide a disarm transition, reachable from every operating mode, that de-energises the drivers.	All	D

Continued on next page

Table 4 (continued)

ID	Requirement	Parent objective	V
FUN-09	On loss of the balancing equilibrium the system shall enter a safe state with the wheels commanded down, rather than continuing to actuate.	All	D
FUN-10	The cube shall transfer stored wheel momentum through a braking event to tip from a face rest onto an edge, and capture the resulting attitude into balance.	Jump-up	T
FUN-11	The cube shall execute consecutive jump-up manoeuvres, re-stabilising after each landing.	Walking	T
FUN-12	The system shall record telemetry without perturbing control-loop timing.	All	D
Performance			
PER-01	The system shall maintain edge balance for 60 s or more unassisted, with no operator contact and no evident wheel-speed drift, on the 1-DoF rig and on the integrated cube.	Edge balance	T
PER-02	The cube shall maintain corner balance for 60 s or more unassisted, with no evident wheel-speed drift.	Corner balance	T
PER-03	The 1-DoF rig shall recover to the balancing equilibrium from an initial tilt of 12° or more, released from rest, re-establishing PER-01.	Edge balance	T
PER-04	The cube shall recover to the balancing equilibrium from an initial tilt of 5° or more in any direction, re-establishing PER-01 or PER-02 as applicable.	Edge, corner balance	T
PER-05	The control and estimation loop shall execute at 1 kHz with a worst-case cycle time not exceeding 1 ms.	All	T
PER-06 (D)	Each reaction wheel shall provide an inertia of at least $6.2352 \times 10^{-4}\,\mathrm{kg\,m^2}$ about its spin axis.	Jump-up	A, I
PER-07	The drive shall sustain the design wheel speed of 4000 rpm at end-of-discharge pack voltage.	Jump-up	A, T
PER-08	The design wheel speed shall not exceed 80 % of the achievable maximum wheel speed at end-of-discharge pack voltage.	Jump-up	A
PER-09	Commanded motor current shall remain within the driver's 9 A continuous rating throughout balancing.	All	T
PER-10	The brake shall arrest a wheel from the design wheel speed within 100 ms.	Jump-up	T
PER-11	Telemetry shall record the state estimate, the control command and the wheel speeds at 50 Hz or faster.	All	T
PER-12 (D)	The drive shall accelerate a reaction wheel from rest to the design wheel speed within 2 s while operating within the driver's continuous current rating.	Jump-up, walking	A, T

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Table 4 (continued)

ID	Requirement	Parent objective	V
PER-13 (D)	The reaction torque applied to the body during wheel spin-up shall not exceed half the gravitational tip-over torque $MgL/2$, so that the cube cannot leave its face rest before the brake event.	Jump-up	A, T
Design constraint			
CON-01	The cube outer edge length shall be 150 mm.	All	I
CON-02	Total cube mass shall not exceed 1.8 kg.	Jump-up	I
CON-03	The system shall be actuated by exactly three mutually orthogonal reaction wheels.	All	I
CON-04	The cube shall operate from a single onboard 6S lithium-polymer pack, with no tether or external power during a demonstration.	All	I, D
CON-05	Driver communication shall use a single linear CAN-FD chain, terminated at exactly the two physical ends.	All	I
CON-06	The 5 V logic rail shall carry its coincident worst-case load within the regulator's derated rating.	All	A, T
CON-07	The electronics shall fit within the frozen board outline, mounting pattern and keep-out heights of the electronics-mount interface.	All	I

Reading the table. The three blocks answer different questions. The functional requirements state what the cube must do, each written so that it can be seen to happen or not: balance on an edge and on a corner, reject a disturbance, sequence its own modes, and stand down safely when the equilibrium is lost. The performance requirements attach numbers to those behaviours — durations, angles, loop times, inertias, currents — and are settled by a measurement rather than by a judgement, which is why they are verified almost entirely by test or by analysis. The design constraints are boundaries fixed before any analysis begins — edge length, mass, wheel count, power source, bus topology — and are checked by inspection of the built article; the sizing of Section 5 works inside them rather than against them.

Status at the time of writing. PER-01 and PER-03 are verified on the single-axis rig (Section 9.1.1); PER-01 therefore stands verified for the rig and open for the integrated cube, since it is one statement covering both articles. PER-03 is demonstrated to 24°, which is the rig's mechanical hard stop rather than a limit of the controller, so the requirement is stated at the largest angle for which a recovery time was recorded. PER-06 and PER-09 are closed by analysis and by rig measurement respectively, pending confirmation on the integrated cube. Every remaining requirement is open. Appendix F carries the per-requirement status and is the record updated as testing proceeds.

Relationship to the milestone gates. The gates of Table 45 are scheduling events; the requirements above are properties of the system. They meet where a gate's pass criterion is a requirement: G3 closes on PER-01 for the rig, G6 on PER-01 for the cube and on PER-04, G7 on FUN-02, FUN-03 and PER-02, and G8 on FUN-10 and PER-10. A gate can slip without a requirement changing, and a requirement can fail at a gate that the schedule nonetheless passes; keeping the two separate is what allows the jump-up to be dropped as sacrificial scope without any other requirement being renegotiated.

Basis of PER-06 and CON-02. PER-06 is derived at the converged sizing point of Table 12 — $M = 1.8000\,\mathrm{kg}$, $L = 149\,\mathrm{mm}$, $\omega_{\max} = 4000\,\mathrm{rpm}$, corner case — and not at any lighter mass the build may achieve. That is deliberate: a derived requirement has to hold across the mass range the cube is permitted to occupy, so it is written against the heaviest case the design admits rather than the lightest it might reach. CON-02 is raised to 1.8 kg in this revision, from 1.7 kg, which is exactly the sizing basis and not a round number above it; at the previous cap the analysis of Section 5 would have documented a design heavier than its own constraint permitted. Constraint and analysis therefore state the same figure, which is the intended relation rather than an oversight: the cap is not spare allowance the build may grow into, it is the mass the wheel was sized to carry, and a cube exceeding it would not merely breach a constraint but fall outside the case PER-06 was derived for. The two requirements are coupled, and neither can be moved without re-deriving the other. Margin is taken in the build rather than in the requirement — the integrated cube weighs 1.626 kg (Appendix C.6), 174 g under the cap — and that realised margin is deliberately left out of the sizing, which continues to run at 1.8 kg.

Open items. Several requirements rest on figures not yet fixed. PER-07 and PER-08 depend on the achievable wheel speed at end-of-discharge voltage, currently derived rather than measured; the spin-up test of WBS task E10 closes this. The reduction of the design speed to 4000 rpm relaxes PER-08 substantially — the design value falls from 79.5 % to 57.8 % of the derived ceiling — so the risk now sits with PER-06 rather than with PER-08. PER-12 and PER-13 are closed by analysis — 1.2 s to design speed at the continuous current limit, and a spin-up reaction torque 17.2 % of the tip-over torque (Section 5.1.1) — and ride on the same E10 run for their measured half. PER-10 has no measurement and is scoped with the jump-up.

CON-01 is stated at 150 mm to match the frame freeze of gate M2, while the sizing analysis of Section 5 is currently run at 149 mm. This discrepancy has become *material* at the present design point and is no longer merely noted. Since $h_w \propto ML^{3/2}$, re-running the closure at 150 mm raises $I_{w,\mathrm{target}}$ by 1.22 %, which against the 100.6 % margin of Table 16 would leave the selected wheel at 99.3 % — short of its own requirement. The fifteen-station wheel is therefore compliant at the analysis value and marginally non-compliant at the frozen one. Reconciliation is no longer a tidying exercise for CAD closure but a decision that changes the ballast count, and it is to be taken before the wheel is committed to manufacture.

3 Project Organisation

3.1 Team Organisation

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of Armonia: mathematical modelling, control, embedded software, mechanical and three-dimensional design, electrical build, and system integration and validation. Table 5 states each member's role and the scope they are accountable for.

Table 5: Team members, contact details and primary roles.

Member	Contact	Role and accountable scope
Pablo Urioste Alarcón	+34 654 491 174	**Project management and hardware-facing software.** Schedule, gate criteria and task allocation; motor-driver configuration, the flight-computer–driver link, the sensor drivers and their calibration, and the fixed-rate loop that carries them. Owns the hardware–software seam. Assembles and edits this document.
Deyan Nikolaev Vlaev	+359 877 082 499	**Dynamics and technical documentation.** Equations of motion, linearisation and jump-up energy sizing; drafting and deepening of this document against the model.
Niccolo Tonetto	+39 327 569 4498	**Control stack and build support.** Single owner of the control and estimation software — estimator, safety and mode logic, and the consolidated control release — written against the sensor interface Pablo maintains. Fabricates the distribution board and cube harness; leads the sacrificial brake and jump-up scope.
Andrea Liang	+31 06 5796 2976	**Plant identification and control design.** Sizing calculations and the inertia and control budgets; sole owner of rig parameter measurement, from the instrumented measurement through to the validated plant model the balancing controllers are designed on.
Suvanna Viriyanti Wu	+39 329 074 7550	**1-DoF model and mechanical build.** Builds and iterates the single-axis prototype and its testbench, fabricates the reaction wheels, and supports the 3D design work.
Athanasia Nikolova	+30 697 232 4491	**3D design and commercialisation.** 3D CAD, frame design and mechanical enclosures; monitors the design against the frozen mass and interface constraints; prototype assembly and the commercial pitch.

3.2 Work Breakdown Structure

This plan runs from project start to the final presentation on 20 August 2026. The work is grouped into five streams — investigation, requirements and documentation; control and software; hardware design and build; three-dimensional design; and electrical and electronics — led by the owners assigned in Table 5. The investigation is kept to the essentials needed to substantiate the design: a focused literature review, the dynamics derivation, and the wheel sizing that follows from it. Table 6 breaks the streams into individual tasks, each with its scope, single accountable lead and dates.

Table 6: Work breakdown structure, broken into individual tasks with scope, lead and dates, grouped by workstream.

ID	Task	Scope	Lead	Dates
Investigation, Requirements & Documentation				
A1	Literature & state of the art	Review the ETH Cubli, the one-wheel Cubli, CubeSat ADCS practice and the hopping-mission precedents	Suvanna	23–25 Jul
A2	Problem statement & framing	Lock the ADCS application narrative that the rest of the document refers back to	Pablo	23–24 Jul
A3	Scope & requirements	Define scope and build the requirements table (functional, performance, design constraint) with verification methods	Pablo	24–25 Jul
A4	Milestones & roles	Prototype staging, binary gate criteria, and a one-page RACI of owners	Pablo	23–25 Jul
A5	Risk register	Trigger, impact, owner and mitigation for the leading risks	Pablo	28–29 Jul
A6	1-DoF equations of motion	Lagrangian derivation of the reaction-wheel pendulum with friction terms	Deyan	24–25 Jul
A7	3D equations of motion	Coupled three-wheel dynamics for the edge and corner pivots	Deyan	27–28 Jul
A8	Linearisation & controllability	Equilibria, linear models, controllability/observability checks	Deyan	29–31 Jul
A9	Jump-up sizing & budget	Energy-method wheel sizing and the parametric mass/inertia/CoM budget	Deyan	30 Jul–2 Aug
A10	TDD draft (all sections)	Every member drafts their own section; placeholder figures acceptable	Deyan	25–28 Jul
A11	TDD deepen + rig evidence	Modelling and control sections in full, with the rig evidence and measured parameters available at the time of writing	Deyan	29 Jul–2 Aug
A12	TDD v1 assemble & submit	Consistency pass, figures, references, submission of the first assessed deliverable	Pablo	3 Aug
A13	Assessor feedback triage	Triage the v1 assessment, assign each comment to an owner, and re-baseline roles and task allocation for v2	Pablo	4–7 Aug
A14	TDD v2 corrections & second draft	Address the assessor comments and fold in the build, bring-up and test results since v1. Ends 12 Aug: no work 13 Aug	Pablo	8–12 Aug
A15	TDD v2 final submission	Final consistency pass and submission of the corrected document	Pablo	14 Aug
Control & Software				
B1	Sim environment + plant	Simulation setup and the 1-DoF plant, validated by energy conservation	Andrea	23 Jul–3 Aug
B2	LQR design (1-DoF, sim)	Tune the regulator and sweep robustness to inertia error, delay and saturation	Andrea	26 Jul–6 Aug
B3	State estimator	Complementary tilt filter, tested against injected noise and bias	Niccolo	27 Jul–7 Aug
B3b	Software interface contract	Fix the sensor and command structs passed between the sensor loop and the controller, with the units and sign convention of every field	Pablo	3–4 Aug

ID	Task	Scope	Lead	Dates
B4	Flight computer & comms	Teensy 4.1 architecture, loop rates, driver configuration and the CAN-FD command/reply protocol	Pablo	24 Jul–7 Aug
B5	Sensor drivers + cal	IMU and encoder drivers, gyro/accel calibration and the lever-arm correction	Pablo	28 Jul–7 Aug
B6	Safety & mode logic	Watchdog, tilt-limit disarm, wheel-speed cap and the mode state machine	Niccolo	30 Jul–7 Aug
B7	1-DoF control design complete	Consolidated 1-DoF control stack — plant, regulator, estimator and safety logic — released as the rig baseline	Niccolo	8–9 Aug
B8	System identification	Measure K_t, wheel inertia, friction and loop latency on the rig. Single owner with E6	Andrea	7–9 Aug
B9	3D model + LQR (sim)	3D plant and the edge/corner regulators with the CAD inertia tensor	Andrea	10–14 Aug
B10	EKF / MEKF estimator	3D estimator with gyro-bias states; characterise the yaw drift	Niccolo	12–15 Aug
B11	Edge balance (S2a)	Retune the edge controller on the assembled cube; log disturbance recovery	Andrea	15–17 Aug
B12	Corner balance + slew (S2b/c)	Coupled three-axis corner balance and a commanded slew; report ADCS metrics	Andrea	17–19 Aug
B13	Jump-up (S3)	Spin-up profile and brake release for a face-to-edge jump. Sacrificial scope: yields to G6 and G7 if the schedule is contested	Niccolo	19–20 Aug
Hardware Design & Build				
C1	Envelope & layout study	Packaging of wheels, drivers, battery and electronics with a central CoM	Nasia	23–24 Jul
C2	Reaction-wheel design	Wheel to the inertia target: radius, rim, hub bolting to the motor bell	Suvanna	24–26 Jul
C3	Mounts & brackets	Motor mount, ring-magnet carrier and adjustable encoder bracket	Nasia	27–29 Jul
C4	1-DoF rig CAD + jigs	Single-axis rig and the balancing/assembly tooling	Suvanna	25–28 Jul
C6	Procurement order (round 1)	Order the long-lead items (rims, frame stock, connectors, spares)	Pablo	23–24 Jul
C7	Bench PSU & bring-up	Current-limited supply, safety kit, and single-axis electrical bring-up	Pablo	25–27 Jul
C8	Fabricate wheel rings	Print the PET-CF rings and hubs to the frozen wheel geometry	Suvanna	27 Jul–2 Aug
C8b	Wheel ballast & completion	Install the M6 bolt/nut ballast stations to the sizing specification and complete each wheel	Suvanna	4–6 Aug
C9	Balance & spin-test	Static-balance and containment spin-test every wheel, including a per-wheel ballast-count verification	Suvanna	5–6 Aug
C10	Assemble 1-DoF rig	Motor, balanced wheel, encoder gap, IMU and hard stops, on reinforced rig supports	Pablo	31 Jul–2 Aug
C11	Print structure + frame	Structural print set and the fabricated, orthogonal frame	Suvanna	8–12 Aug

ID	Task	Scope	Lead	Dates
C12	Perfboard + harness	Power/signal board and the full power and CAN harness	Pablo	8–12 Aug
C13	Cube integration + power-on	Assemble the three axes, then bench and battery smoke test	Pablo	13–15 Aug
C14	Brake mechanism build	Fabricate and fit the three servo brakes; bench-test the deceleration	Suvanna	15–18 Aug
3D Design				
D1	Concept sizing & fit study	Confirm that every component fits the 150 mm envelope and lay out the internal structure	Nasia	27 Jul–2 Aug
D2	Frame design freeze	Freeze the inertia-bearing frame geometry. Later features that do not affect the mass properties may still be added	Nasia	2 Aug
D3	Mass properties & budget closure	Export the inertia tensor and CoM and close the mass budget against the sizing analysis	Nasia	3–4 Aug
D4	Detail CAD	Brackets, electronics bay, harness routing and the electrical keep-outs to the frozen board outline	Suvanna	3–5 Aug
D5	Procurement order (round 2)	Consolidate the bill of materials arising from the 3D design and place the second order	Pablo	5 Aug
D6	Tolerance & clearance analysis	Fit and tolerance study, including the wheel sweep envelope and fastener access	Nasia	5–6 Aug
D7	Manufacturing package release	Issue the drawings, print set and assembly instructions for fabrication	Nasia	7 Aug
Electrical & Electronics				
E1	Wiring & placement design	Electrical architecture, wiring scheme and component placement for the single-axis rig, issued as the build reference	Pablo	28–31 Jul
E2	Rail, driver & encoder bring-up	5 V rail soldered and trimmed to 5.00 V under load; driver phase-wired and FOC-calibrated; MA600 encoder mounted at $\leq$ 1.5 mm air gap with verified read-back. Open-loop operation on the bench supply	Pablo	1–3 Aug
E3	Rig bus & sensor integration	Two-node CAN-FD bus with split termination at both physical ends, Teensy and IMU integrated on the rig harness	Pablo	4–5 Aug
E4	Board outline & floorplan	Driver placement from the encoder cable limit, CAN routing, and the perfboard floorplan issued as a frozen outline to CAD	Pablo	4–5 Aug
E5	1-DoF closed-loop bring-up	Closed loop on the rig with the disarm path and watchdog verified before any balance attempt	Pablo	6–7 Aug
E6	Rig parameter measurement	Instrumented measurement of K_t, wheel inertia, friction and loop latency. Merged with B8: one owner measures and models	Andrea	7–9 Aug

ID	Task	Scope	Lead	Dates
E6b	Mechanical & performance validation (1-DoF)	Reconcile the measured torque, inertia, friction and achievable momentum against the jump-up sizing (A9). Deviation beyond tolerance forces a re-sizing decision that day, while the wheel geometry can still change	Andrea	9 Aug
E7	Perfboard + battery power chain	Populate the distribution perfboard (two ground domains, one star point, split 5 V rail); XT60 chain with anti-spark, fuse and ≥ 50 V bulk capacitance; replicate the remaining actuation sets	Niccolo	8–11 Aug
E8	4-node bus + cube harness	Bus scaled to four nodes with stubs < 10 cm and an error soak at 5 Mbit/s; full power and signal harness fabricated and routed; Wi-Fi telemetry for untethered operation	Niccolo	11–12 Aug
E9	Cube power-on, 48 A + thermal	First battery power-on and the three-axis 48 A transient; driver thermals in the enclosed volume; all axes addressable	Pablo	14–15 Aug
E10	Brake servo rail + measurement	Servo on an isolated 5 V rail with PWM drive and flyback; rail stability under stall, brake closure time and spin-down brake torque measured. Sacrificial with C14/B13	Niccolo	16–19 Aug
Deliverables				
F1	Results, demo & final presentation	Slow-motion capture, results plots, deck, and the final presentation	Andrea	18–20 Aug

3.3 Schedule and Contingency Plan

The full-page schedule in Figure 2 places every task of Section 3.2 on the 23 July–20 August 2026 calendar, and Table 45 states the gate that closes each phase.

Structure of the plan. The first two weeks establish the analytical and design foundation: the dynamics derivation and wheel sizing on the documentation side, the single-axis rig and its wiring design on the hardware and electrical sides, and the simulation environment on the control side. These converge on 2 August, which carries two objectives: the first powered bring-up of the single-axis rig, taking the 5 V rail, the driver and the encoder to verified open-loop operation (M1); and the freeze of the three-dimensional frame geometry, which closes the mass and inertia budget against the sizing analysis (M2). The frame freeze releases the second procurement round on 5 August (M3), which is the last point at which parts can be ordered with useful delivery margin before cube assembly begins on 13 August.

The documentation stream carries two assessed submissions. The first is delivered on 3 August (G4) with the evidence available at that date; the assessor's comments are triaged during the following week and the corrected second draft is submitted on 14 August (M8), by which point the rig results and the integrated cube can both be reported.

Critical path and contingency. The critical path runs from the 2 August bring-up through the closed-loop rig (M4), the measured plant parameters (M6) and the single-axis control design (M7), into the three-dimensional controller and the balancing demonstrations. The single-axis rig is the protected item, since its balance result is the experimental evidence in the document. The three-dimensional design proceeds in parallel on the mechanical side, so that fabrication is not held by the control work.

The plan carries the full scope through to the jump-up demonstration. The final week is nonetheless tightly constrained: edge balance begins two days after cube assembly completes, and the jump-up gate falls on the day of the final presentation. The jump-up (S3) is therefore designated sacrificial scope by standing decision. If the balancing demonstrations require the time, the jump-up yields first, and the demonstration presented on 20 August is the corner balance and commanded slew. The brake mechanism build and its associated electrical work are sequenced after the edge-balance gate for the same reason.

Appendix G lists the risks behind this plan, with their mitigation and owner, and fixes the order scope is cut if the schedule runs short.

Table 7: Milestone gates, from kick-off to the final presentation, with the criterion that closes each one.

Gate	Pass criterion	Date
G1	Long-lead order placed; requirements and plan locked	24 Jul
G2	Dynamics, linearisation and wheel sizing complete	31 Jul
M1	1-DoF rig bring-up: 5 V rail trimmed under load, driver commutating, encoder verified at ≤ 1.5 mm gap	2 Aug
M2	3D frame geometry frozen at 150 mm; mass, inertia and CoM budget closed against the sizing analysis	2 Aug
G4	TDD v1 delivered for assessment	3 Aug
M3	Procurement round 2 consolidated and ordered	5 Aug
M4	1-DoF closed loop live; disarm path and watchdog verified	7 Aug
M5	Manufacturing package released for fabrication	7 Aug
M6	Plant parameters measured on the rig and validated against simulation; measured torque, inertia and friction reconciled with the sizing analysis, or a re-sizing decision recorded	9 Aug
M7	1-DoF control design complete: plant, regulator, estimator and safety logic	9 Aug
G3	1-DoF rig balances $\geq$60 s unassisted	12 Aug
M8	TDD v2 corrected and finally submitted	14 Aug
G5	Cube integrated and powered; all axes addressable	15 Aug
G6	Edge balance (S2a) for $\geq$60 s with disturbance recovery	17 Aug
G7	Corner balance and commanded slew (S2b/c)	19 Aug
G8	Face-to-edge jump-up (S3) captured on video (sacrificial scope)	20 Aug
G9	Final presentation and demonstration	20 Aug

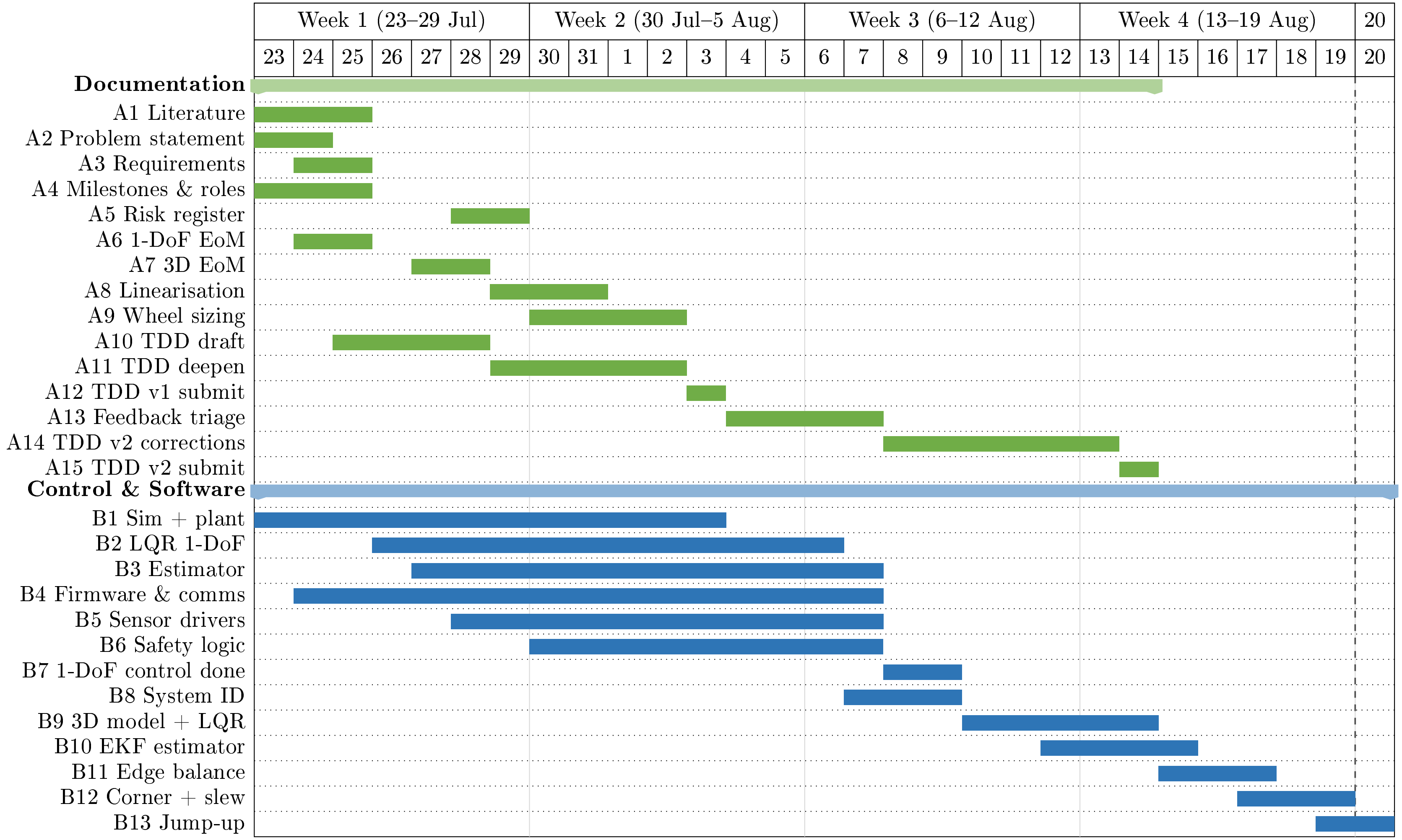

Figure 2: Project schedule, 23 July–20 August 2026 (sheet 1 of 3): documentation and control/software streams. The documentation stream (green) carries two assessed submissions, on 3 and 14 August. Control and software (blue) runs from the simulation environment through to the balancing demonstrations, with the three-dimensional extension concentrated in the final two weeks. The build streams continue in Figure 3 and the milestone gates in Figure 4.

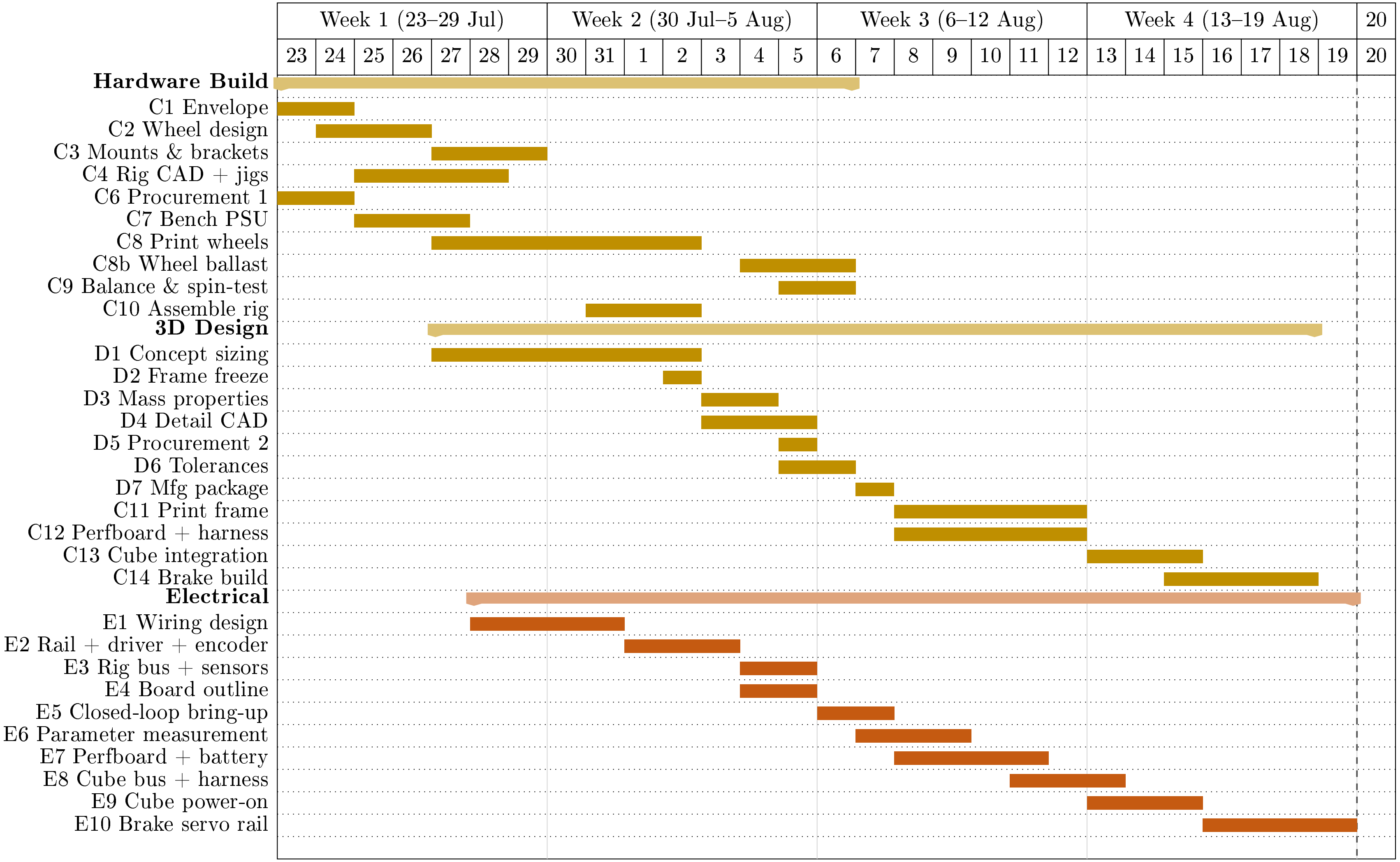

Figure 3: Project schedule, 23 July–20 August 2026 (sheet 2 of 3): hardware, three-dimensional design, electrical and deliverable streams, continued from Figure 2. Hardware (gold) separates the single-axis build from the three-dimensional design, and the electrical stream (orange) proceeds from the single-axis bring-up to the integrated cube. The milestone gates that close each phase are shown in Figure 4.

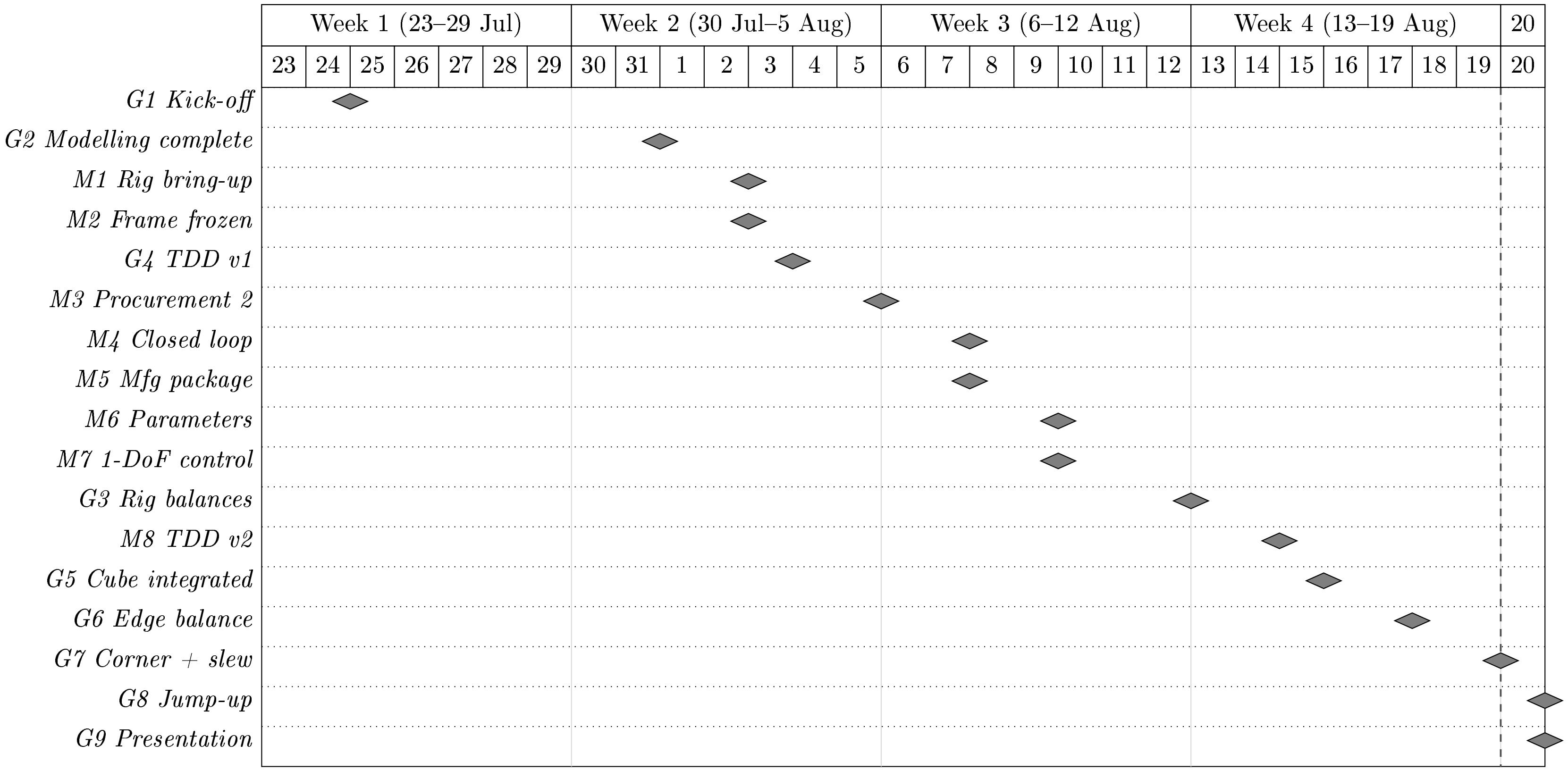

Figure 4: Project schedule, 23 July–20 August 2026 (sheet 3 of 3): the milestone gates that close each phase, continued from Figures 2 and 3. Grey diamonds mark the gates listed in Table 45, placed on the day each gate falls due.

4 Methodology

Section 3.2 assigns every task an owner and a date. This section states the reasoning that assignment is built on: the procedure the team follows to take Armonia from a sizing target to a validated prototype, why its six stages run in this order and not another, and why the result is an iteration rather than a single pass through them. Armonia is an interdisciplinary system whose subsystems cannot be sized, modelled or validated in isolation, which is why the methodology below is organised as a sequence of stages and validation gates rather than a single design pass.

4.1 Development and Iteration Workflow

Figure 5 is that procedure: six stages, each committing an output the next stage cannot start without, and three feedback paths that reopen an earlier stage when a downstream finding invalidates an upstream assumption.

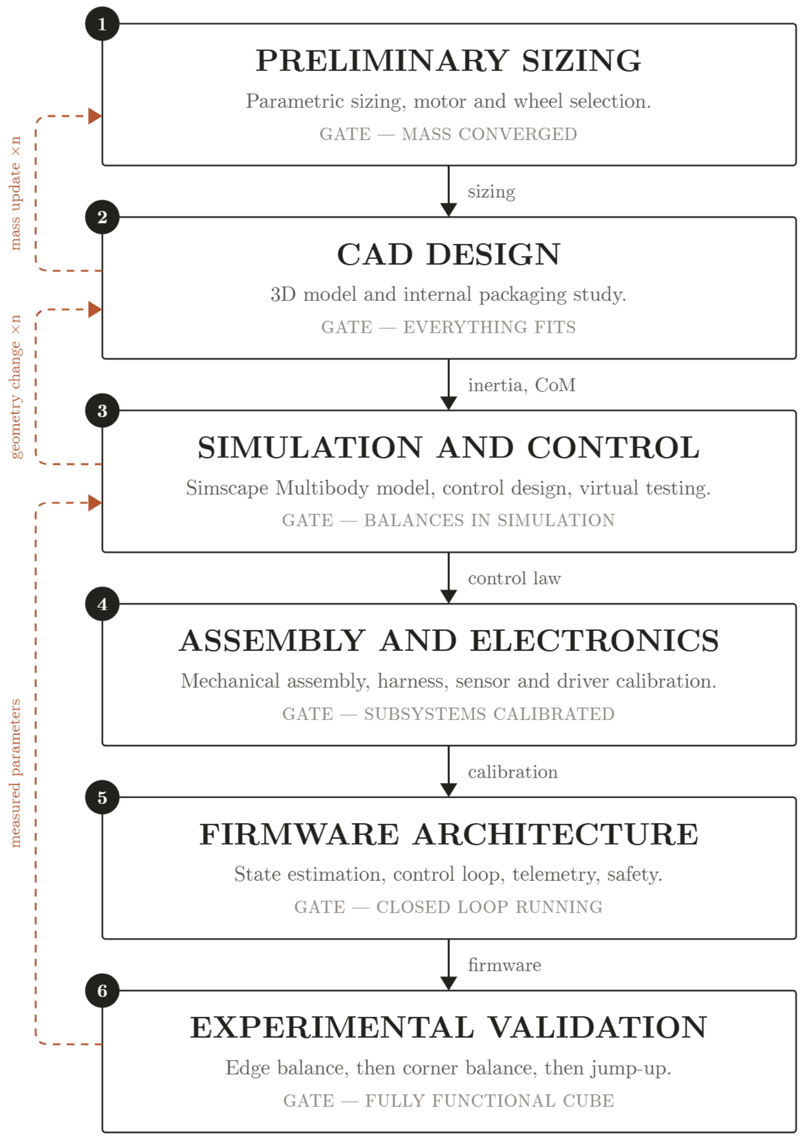

Figure 5: Development and iteration workflow, with the three feedback loops that keep sizing, CAD and control consistent with each other and with measured hardware.

Workflow steps Each stage sits where it does because the stage after it consumes an output only it produces:

1. **Preliminary Sizing** is the first stage of the project: it derives the physics behind Armonia and justifies the design parameters and performance targets the rest of the design must meet.
2. **CAD Design** turns those targets into a 3D model and an internal packaging study, returning a measured inertia tensor and centre of mass where Stage 1 could only estimate them (Section 6).
3. **Simulation and Control** tests control feasibility against the CAD-derived inertia, mass and geometry rather than the Stage 1 estimate (Sections 4.2–4.4, 8).
4. **Assembly and Electronics** begins once the design is closed: the cube is assembled, the electronics are placed and wired, and every axis is tested before it is commanded (Section 7).
5. **Firmware Architecture** is developed in parallel with the previous stage: it integrates the electronics around the MCU and implements the control-dynamics loop that converts IMU orientation data into wheel-torque commands. Alongside the flight code, sub-programs are developed to test and calibrate each electronic module individually (Section **??**).
6. **Experimental Validation** is the last stages: once integration concludes, results are tested and validated. The workflow accommodates iteration at every stage rather than assuming a single pass succeeds.

Iterative loop A single pass through that order is not sufficient, because three of its outputs correct an assumption made earlier in the same pass rather than only feeding forward:

- *Mass update.* CAD's as-modelled mass and CoM close the loop into Stage 1's assumed M (Section 5.2): the sizing reported there is the fixed point of this loop, not its first iteration.
- *Geometry change.* A control or simulation finding — inertia short of target, an infeasible slew, an interference the 2D model could not see — returns to CAD rather than being absorbed by re-tuning gains, since no gain fixes a wheel that does not fit or a body the controller cannot actuate against.
- *Measured parameters.* Once hardware exists, the torque constant, friction and latency measured on it (Section 4.3) replace the CAD/datasheet estimates Stage 3 was designed on, and the control gains are re-derived against the measured plant rather than flown on assumed values.

This is also why the schedule of Section 3.3 gates CAD on the sizing numbers rather than running the two in parallel: a frame frozen ahead of the inertia tensor is a frame that may have to be reprinted, which costs more than the days spent waiting on the number.

Fully assembled, Armonia is underactuated and simultaneously unstable about all three axes when balanced on a corner (Section 1.3). Stage 3 of the procedure above is therefore itself incremental: a first, single-face 1-DoF prototype is built and modelled in two dimensions, so that the dynamics, the parameter identification procedure and the estimator can be validated on a plant simple enough to reason about directly; the same modelling, identification and verification workflow is then generalised to the second and final prototype, the fully three-dimensional cube, in Section 4.4. Each subsection below states the equations that define the problem at that stage and the method used to determine every parameter that appears in them; the control law itself — how those equations are turned into a feedback command — is derived separately in Section 8, since sizing, dynamics and control are three passes over the same plant rather than three independent problems.

4.2 2D Reaction-Wheel Pendulum

The 1-DoF prototype is a 3D-printed plate sized to one cube face, with a motor-driven momentum wheel at its centre and a braking mechanism at one corner. The plate pivots on a bearing at that corner, constraining motion to a single rotational degree of freedom in a vertical plane.

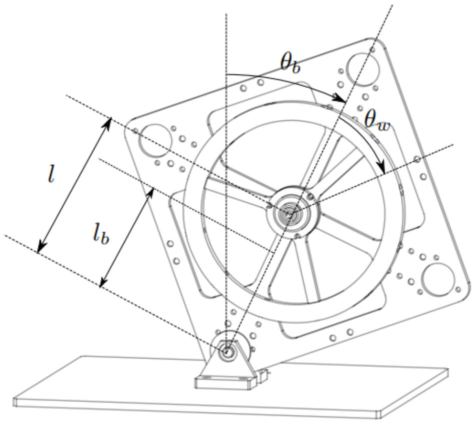

Figure 6: Frames and angles of the 2D reaction-wheel pendulum. θ_b is the plate's tilt from vertical about the pivot; θ_w is the wheel's rotation relative to the plate; l and l_b are, respectively, the pivot-to-motor-axis and pivot-to-plate-CoM distances along the plate's diagonal. Symbols per Table 8.

4.2.1 Equations of Motion

The nonlinear equations of motion are

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l)\, g \sin\theta_b - T_m - C_b\dot{\theta}_b + C_w\dot{\theta}_w}{I_b + m_w l^2}, \tag{1}$$

$$\ddot{\theta}_w = \frac{(I_b + I_w + m_w l^2)(T_m - C_w\dot{\theta}_w)}{I_w(I_b + m_w l^2)} - \frac{(m_b l_b + m_w l)\, g \sin\theta_b - C_b\dot{\theta}_b}{I_b + m_w l^2}, \tag{2}$$

with motor current-torque relation $T_m = K_m u$. Symbols are defined in Table 8 and used throughout this section.

The input u here is a *current*, so K_m enters B directly. The corner-balancing design of Section 8 takes the wheel *torque* itself as its input instead, and its B (Equation (33)) accordingly carries no torque constant at all: there K_m is the conversion applied at the actuator boundary, between the commanded torque and the current handed to the driver, and it sets the torque–current pairs of Table 43 rather than any entry of the plant matrices. Both forms use one figure, $K_m = 0.025{,}13\,\mathrm{N\,m/A}$, the machine constant implied by the motor's 380 rpm/V rating ($K_m = 60/2\pi K_V$); Section 7 gives the bench cross-check and the reason the fitted slope there is not used in its place.

Table 8: 2D pendulum model variables.

Symbol	Description	Unit
θ_b	Body tilt angle	rad
θ_w	Wheel rotation relative to body	rad
m_b, m_w	Body, wheel mass	kg
I_b, I_w	Body (about pivot), wheel (about motor axis) inertia	$\mathrm{kg\,m^2}$
l	Motor axis to pivot distance	m
l_b	Body CoM to pivot distance	m
g	Gravitational acceleration	$\mathrm{m/s^2}$
T_m	Motor torque	N m
C_b, C_w	Body, wheel damping coefficient	$\mathrm{kg\,m^2/s}$
K_m	Motor torque constant	N m/A
u	Motor current input	A

4.3 Parameter Identification & the 2D Testbed

CAD mass-properties simulation, accounting for print material, infill, and mounted hardware (driver, encoder, MCU, gyroscope/accelerometer, wiring), gives m_b, m_w, I_b, I_w directly, while l and l_b follow from the plate's design geometry. The damping coefficients C_b and C_w cannot be obtained from CAD, since they depend on bearing friction, wiring drag, and motor cogging; nominal values are used for the initial control design and replaced by the measured values below once the prototype is fabricated.

Table 9: Parameter determination method (symbols per Table 8).

Symbol	Determination method
l	Calculated during design
l_b	Calculated during design
m_b	CAD mass properties
m_w	CAD mass properties
I_b	CAD mass properties
I_w	CAD mass properties
C_b	Estimated (measured on the 1-DoF rig, below)
C_w	Estimated (measured on the 1-DoF rig, below)
K_m	Datasheet K_V (bench cross-check, Sec. 7)

1-DoF plant identification. Four parameters are measured on the single-axis rig once it is built, each by a method chosen so that the quantity of interest dominates the measurement rather than appearing as a small correction to something else. These measured values replace the CAD/datasheet estimates of Table 9 before the control gains used on hardware are finalised (Section 8).

Tolerances are stated with the measured values, since a parameter without a tolerance cannot support the robustness sweep of Section 8 that consumes it. Loop latency is measured through the complete path rather than estimated from the 1 kHz loop rate, because the scheduling and bus-transport contributions are the ones a rate figure omits.

4.4 Generalization to the 3D Cube

The dynamics, identification workflow, and verification approach validated on the 2D model of the first prototype extend to the second and final prototype: the full three-dimensional cube, where three orthogonal reaction wheels must stabilise the body about its edge and corner equilibria.

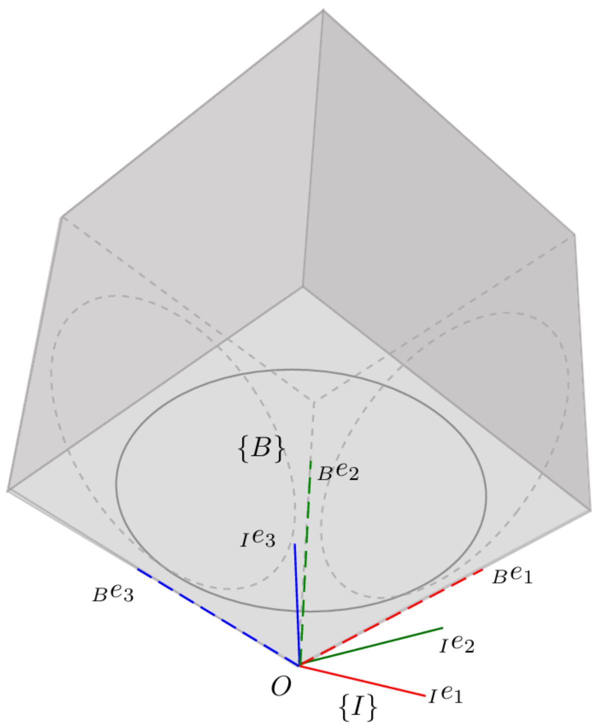

Figure 7: Body frame $\{B\}$ and inertial frame $\{I\}$ of the three-axis cube, both origined at the corner contact point O. The body axes ${}_Be_1, {}_Be_2, {}_Be_3$ coincide with the wheel spin axes e_i of Table 10, each directed outward from O along a cube edge; $\{I\}$ is fixed in space and coincides with $\{B\}$ at the corner-up equilibrium, so $R = \mathbb{1}$ there.

4.4.1 Full-Body Equations of Motion

With $R \in SO(3)$ the cube's orientation and $\omega_b \in \mathbb{R}^3$ its body-frame angular velocity, the attitude kinematics are

$$\dot{R} = R\,[\omega_b]_\times, \tag{3}$$

where $[\cdot]_\times$ is the skew-symmetric cross-product matrix. The assembly's angular momentum about the pivot, expressed in the body frame, generalizes the single-axis momentum term of the 2D model:

$$H = I\omega_b + \sum_{i=1}^{3} I_{w,i}\,\omega_{w,i}\,e_i. \tag{4}$$

Euler's equation gives

$$\dot{H} + \omega_b \times H = \tau_g - C_b\,\omega_b, \tag{5}$$

and each wheel obeys

$$I_{w,i}\big(\dot{\omega}_{b,i} + \dot{\omega}_{w,i}\big) = T_{m,i} - C_{w,i}\,\omega_{w,i}, \qquad i = 1, 2, 3. \tag{6}$$

Table 10 defines all symbols.

The spin axis e_i is directed outward from the corner contact point O along the corresponding cube edge (Figure 7). Each wheel's motor/encoder positive-rotation direction is fixed by its

Table 10: Full-body (3D) model variables.

Symbol	Description
R	Cube orientation relative to world frame
ω_b	Body angular velocity, body frame ($\in \mathbb{R}^3$)
$\omega_{b,i} = e_i^{\mathrm{T}}\omega_b$	Body angular velocity about wheel axis i
$\omega_w = (\omega_{w,1}, \omega_{w,2}, \omega_{w,3})$	Wheel angular velocities relative to body
e_i	Spin axis of wheel i (face normal)
I	3×3 inertia tensor, body+wheels, about pivot
H	Total angular momentum about pivot
$\tau_g = m\, r_{cm} \times (R^{\mathrm{T}}\hat{g})$	Gravity torque about pivot
m	Total assembly mass
r_{cm}	Body-frame vector, pivot to assembly CoM
$\hat{g}$	World vertical unit vector
C_b	3×3 body damping matrix
$I_{w,i}, C_{w,i}$	Inertia, damping of wheel i
$T_{m,i} = K_{m,i}u_i$	Motor torque, wheel i
$u_i, K_{m,i}$	Motor current, torque constant, wheel i

mounting inside the inner structure and runs opposite to e_i, so the raw encoder rate and the commanded driver torque must be sign-flipped before they enter the equations above:

$$\omega_{w,i} = -\,\omega_{w,i}^{\mathrm{enc}}, \qquad T_{m,i} = -\,K_{m,i}\,u_i^{\mathrm{cmd}}, \tag{7}$$

so that H, Euler's equation and the per-wheel equation are all expressed consistently in the e_i-positive convention rather than the driver's native one.

Restricting motion to a single vertical plane with two wheel torques set to zero recovers the 2D model exactly, with $I_b, l_b, m_b, m_w, I_w, C_b, C_w$ the corresponding single-axis projections of $I, r_{cm}, m, I_{w,i}, C_{w,i}$.

4.4.2 Edge vs. Corner Equilibria & Linearisation

Jump-up proceeds in two stages: Flat-to-Edge, where one wheel brakes, followed by Edge-to-Corner, where a second wheel brakes. **Edge balancing** constrains the body to a single axis, so the dynamics reduce to the 2D form of Section 4.2.1, with m_b, l_b, I_b replaced by their edge equivalents.

Corner balancing is genuinely 3D: all three wheels contribute across three rotational DOF. Linearising about the corner-up equilibrium $(\theta, \omega_b, \omega_w) = (0, 0, 0)$, with $\theta \in \mathbb{R}^3$ the small orientation error, gives the same state-space form as the 2D case:

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x := (\theta, \omega_b, \omega_w) \in \mathbb{R}^9, \tag{8}$$

with $A \in \mathbb{R}^{9\times9}$ and $B \in \mathbb{R}^{9\times3}$ built from I, r_{cm}, and the per-wheel parameters $I_{w,i}, C_{w,i}, K_{m,i}$ (Table 10), u again being the vector of wheel currents. This linear model, together with its 2D counterpart of Section 4.2.1, is the plant the balancing controller of Section 8 is designed on — there in the torque-input form noted above, so that $K_{m,i}$ leaves B and appears only in the command conversion.

5 Preliminary Sizing

This section confirms, before fabrication, that a cube of a candidate mass and edge length can perform the jump-up manoeuvre while carrying its own components. It covers: the sizing requirements (Section 1.3); the physics converting a target manoeuvre into a required wheel inertia (Section 5.1); the packaging constraint linking cube size to wheel size (Section 5.2); and the wheel design meeting that target (Section 5.3).

Provenance of the numbers in this section. Every figure reported below — the wheel-inertia target, the ballast station count, the wheel mass and inertia, and the cube mass budget — is produced by a single preliminary sizing program, `analysis/SIZING.py`, rather than assembled by hand from separate calculations. The program implements the jump-up dynamics in full (the ideal momentum transfer of Equation (10), the contact-case geometry factors of Table 11, the brake-amplification factor β and the resulting inertia target of Equation (11)), the wheel configuration (ring and spoke geometry, the radial M6 ballast stations with their parallel-axis and own-axis inertia terms, the axial-fit and edge-gap feasibility rules), and the cube mass budget itemised from the bill of materials with the structural allowance carried explicitly. Because the mass budget feeds back into the inertia target, the program is run to convergence rather than evaluated once, and the values quoted here are those of the converged pass. They are therefore reproducible: re-running the program with the inputs of Table 12 regenerates every number in this section, and changing an input regenerates all of them consistently rather than leaving the tables to be reconciled by hand. The figures remain *preliminary* in the sense that the structural line is still an allowance pending CAD closure, not in the sense of being approximate.

Sizing and control are solved together, not sequentially. Wheel inertia sets the angular momentum and torque available to the controller for arresting a fall — a wheel sized without the control law in mind can be mechanically sufficient yet dynamically useless. Conversely, the control law's saturation behaviour (how aggressively it must desaturate stored momentum, how much recovery angle it must cover) sets how much sizing margin is needed. Each depends on the other, so Section 5.2's closure procedure is a fixed-point iteration between them, not a one-pass calculation handed off afterward.

5.1 Jump-Up Feasibility Analysis

The analysis runs in one direction, which is worth stating before any equation appears. Two quantities enter as *inputs*: the cube mass M and edge length L, fixed by the packaging closure of Section 5.2; and the maximum wheel speed $\omega_{\max}$, a design choice bounded by what the drive can sustain in service (*Selection of* $\omega_{\max}$, below). One quantity leaves as the *output*: the per-wheel inertia $I_{w,\text{target}}$ the manoeuvre demands, which the wheel of Section 5.3 must then deliver through its diameter and ballast. Wheel speed is never solved for during sizing — it is set, and the wheel is sized around it.

Origin of the requirement. The physics comes from the jump-up collision itself, modelled as a perfectly inelastic transfer between a spinning wheel and the body. Conservation of angular momentum at impact and of energy from impact to edge-standing give a closed form for the speed a *given* wheel would need:

$$\omega_w^2 = (2-\sqrt{2})\,\frac{I_w + I_b + m_w l^2}{I_w^2}\,(m_b l_b + m_w l)\,g. \tag{9}$$

Equation (9) is the inverse of the sizing question rather than the sizing path: it presumes a wheel whose inertia I_w is already known, which at this stage is precisely the unknown. It serves two other purposes. Before fabrication it is a sanity check — a candidate wheel can be substituted to confirm that the speed it implies lies inside the motor's operating range at the available bus voltage, without excessively reducing the balancing recovery angle. After fabrication it becomes a verification relation, evaluated on the CAD- and test-derived $I_b, I_w, m_b, l_b, m_w, l$ of the built prototype (Section 4.3), with ω_w refined experimentally to absorb deviations from the ideal-collision assumption. The ω_w of Equation (9) and the $\omega_{\max}$ of Section 5.1.1 are the same physical wheel speed on opposite sides of one relation: here it is solved for; there it is fixed so that inertia can be solved for instead.

The braking mechanism uses a servo-driven barrier that strikes a bolt head on the wheel to stop it abruptly. Servo response time bounds the validity of the "instantaneous stop" assumption; a slower stop yields a lower effective post-impact body velocity than predicted, compensated through the same tuning process.

5.1.1 Torque-Limited Wheel-Inertia Target

Sizing follows the forward chain of Figure 8: cube parameters fix an ideal wheel momentum, a geometric/contact factor inflates it to a design momentum, and the design momentum converts to the target wheel inertia that Section 5.3 sizes the wheel geometry against, with $\omega_{\max}$ held fixed at every step.

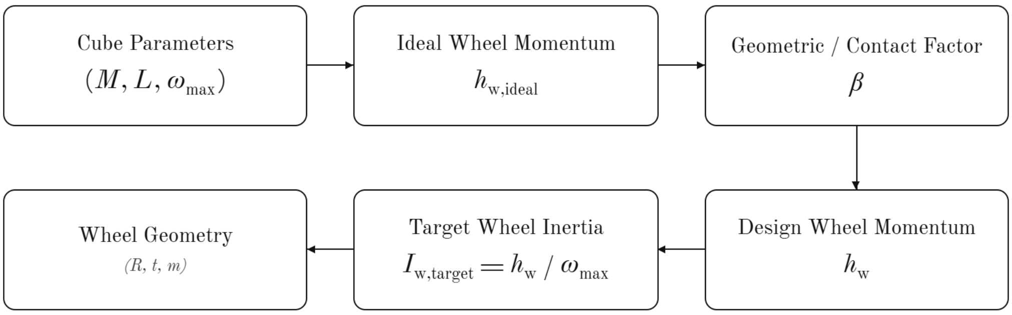

Figure 8: Wheel-inertia sizing chain.

Equation (9) assumes an instantaneous, lossless momentum transfer; the momentum budget below relaxes that assumption by accounting for finite brake torque. Its first step is an ideal per-wheel angular momentum:

$$h_{w,\text{ideal}} = \sqrt{\eta}\,\frac{\sqrt{2g\,\lambda\,\kappa}}{n}\,ML^{3/2}, \tag{10}$$

where M, L, g are cube mass, edge length, and gravity, η is an energy margin, and κ, λ, n are dimensionless geometric factors set by the contact case (Table 11). The corner case is more demanding and governs the design.

Table 11: Contact-case geometric factors.

Case	κ (pivot inertia)	λ (CoM rise)	n (momentum transfer)
Edge	$2/3$	$1/\sqrt{2} - 1/2$	1
Corner (governs design)	$11/12$	$\sqrt{3}/2 - 1/2$	$\sqrt{2}$

A finite brake torque τ_b cannot deliver the impulse instantaneously and must exceed the gravitational tip-over torque $\tau_g = MgL/2$. This loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \qquad h_w = \beta\, h_{w,\text{ideal}}, \qquad I_{w,\text{target}} = \frac{h_w}{\omega_{\max}}, \tag{11}$$

where $\omega_{\max}$ is the maximum practical wheel speed. Equations (10)–(11) are the sizing relations used at step 4 of Section 5.2's closure procedure: given a candidate mass and edge length, they return the inertia the wheel of Section 5.3 must meet. Their inputs, recorded in Table 12, are not fixed in advance — M and L are outputs of the packaging closure, not assumptions preceding it.

Table 12: Torque-limited wheel-inertia target (corner case). Inputs are fixed by the closure procedure of Section 5.2; the table is populated on each iteration. Values are those of the converged iteration.

Symbol	Description	Value
M	Cube mass	1.800 kg
L	Cube edge length	149 mm
η	Energy margin	1.05
τ_b	Brake torque per wheel	5 N m
$\tau_g = MgL/2$	Gravitational tip-over floor	1.3155 N m
β	Brake-amplification factor	1.3570
$\omega_{\max}$	Max wheel speed (design)	4000 rpm*
h_w	Required angular momentum	0.2612 kg m^2/s
$I_{w,\text{target}}$	**Target wheel inertia**	6.2352×10^{-4} kg m^2

* The design speed is set at 57.8 % of the motor's derived end-of-discharge ceiling (PER-08), against 79.5 % at the 5500 rpm of the previous iteration. The reason for the reduction is electrical rather than dynamic: what the lower speed buys is back-EMF headroom at the bottom of the discharge curve — so that the manoeuvre stays available on a depleted pack and not only on a fresh one — together with a lower commutation frequency and a centrifugal ballast load that falls as ω^2. It buys nothing on the control side, and is not claimed to: at a fixed h_w the balancing authority is the motor torque and the time to saturation is h_w/τ, neither of which depends on how the momentum is divided between inertia and speed. The momentum is recovered through inertia instead, and that is where the choice is paid for: holding h_w at 4000 rpm rather than 5500 rpm takes each wheel from 170.37 g to 234.14 g, 191 g over three, making the speed reduction the single largest driver of both the mass budget of Table 15 and the cap stated in CON-02.

Two features of these relations govern how the iteration behaves:

- $h_w \propto ML^{3/2}$ — growing the cube to fit a larger wheel also raises the requirement; convergence needs achievable inertia ($\sim D_w^2$) to grow faster than this.
- β diverges as $\tau_b \to \tau_g$ — a brake only marginally above the tip-over torque cannot deliver the required impulse at any wheel speed. τ_b must therefore be checked against $\tau_g = MgL/2$ every iteration, keeping a comfortable ratio τ_b/τ_g rather than a bare positive margin.

Selection of $\omega_{\max}$. Maximum wheel speed is a design choice, not a datasheet figure: it enters sizing directly through $I_{w,\text{target}} = h_w/\omega_{\max}$, so a higher assumed speed permits a proportionally smaller wheel. It is the primary relief variable when the envelope caps the wheel below the requirement (step 5, Table 13) — but it must reflect what the drive can sustain in service, not the no-load figure $K_v V_{\text{bus}}$. Three effects separate the two:

- winding resistance, which costs speed under the torque needed to spin the wheel up;
- reduced phase voltage available under sinusoidal field-oriented modulation;
- pack voltage sag through the discharge curve, taken at its lower end so the manoeuvre stays available on a partly-depleted pack, not only a fresh one.

The design value sits below the resulting ceiling, leaving margin for motor tolerance, bearing and aerodynamic drag, and residual uncertainty in M. Back-EMF should remain a comfortable fraction of usable bus voltage, and the electrical frequency $\omega_{\max} p/2\pi$ (at p pole pairs) must sit inside the driver's commutation capability — both to be confirmed once motor and pack are selected.

Selection of τ_b. Unlike the other inputs, τ_b cannot be read from a datasheet, because the brake does not generate it directly. The mechanism is a servo-driven barrier struck by a bolt head on the wheel: the servo only *holds* the barrier in the wheel's path, while the wheel's own momentum does the work of the collision. The servo's stall torque is therefore not τ_b and need not approach it — the TowerPro MG92B (Table 62) is rated near 0.29 N m, about one seventeenth of the sizing value. The servo only resists the barrier being deflected; it does not decelerate the wheel.

Since the stop is a collision rather than a friction clutch, τ_b is fixed entirely by how quickly angular momentum is destroyed,

$$\tau_b = \frac{h_w}{\Delta t}, \tag{12}$$

so quoting a brake torque is equivalent to asserting a stopping time: at the present operating point, $\tau_b = 5\,\mathrm{N\,m}$ implies $\Delta t \approx 52\,\mathrm{ms}$, whereas a bolt head striking a printed barrier should arrest the wheel in a few milliseconds. The design value is therefore expected to be conservative, which is the intended direction — since $\beta = \tau_b/(\tau_b - \tau_g)$ falls toward unity as τ_b grows, underestimating τ_b inflates the inertia requirement while overestimating it leaves the wheel undersized.

One consequence carries into the structural design: τ_b is also a *load*, and the sizing value should not be reused to size the barrier, bolt, servo bracket, or wheel spokes — a genuine few-millisecond stop puts an order of magnitude more torque through that load path. Until measurement settles the question, the structure is checked against a short-Δt case while the dynamics use $\tau_b = 5\,\mathrm{N\,m}$. The spin-down test (WBS task E10) closes this gap, reporting both the impulse-equivalent mean $\bar{\tau}_b = h_w/\Delta t$ (the sizing quantity) and the peak torque (the structural one); the two may differ several-fold, and the logging rate must resolve a few-millisecond event without smoothing the peak away.

Spin-up time, and why it is bounded from both sides. The manoeuvre has a second time scale, longer than the braking event and less visible in the equations above: the wheel has to be brought to $\omega_{\max}$ before the brake can do anything, and nothing in Equations (10)–(11) says how long that takes. It follows from the same momentum figure divided by the *drive* torque rather than the brake torque,

$$t_{\text{spin}} = \frac{h_w}{\tau_m} = \frac{I_w\,\omega_{\max}}{k_t\,i}, \tag{13}$$

with k_t the motor torque constant and i the phase current. At $k_t = 0.025{,}13\,\mathrm{N\,m/A}$, the machine constant implied by the motor's 380 rpm/V rating (Section 7), and the driver's 9 A continuous rating (PER-09), $\tau_m = 0.226\,\mathrm{N\,m}$, and the selected wheel — which stores $I_w\omega_{\max} = 0.2627\,\mathrm{kg\,m^2/s}$ — reaches design speed in 1.2 s. At the motor's 16 A short-duration rating the same relation gives $\tau_m = 0.402\,\mathrm{N\,m}$ and 0.65 s, so the figure is set by how the drive is operated rather than by what the motor can do. PER-12 is stated at 2 s to leave room for bearing and aerodynamic drag, for current derating as the winding warms, and for a shaped current ramp rather than a step to full current.

The lower bound is the more interesting one, because it is physical rather than electrical. Spin-up torque is reacted by the body, which at that moment is resting on a face: were it to exceed the tip-over torque $\tau_g = MgL/2$, the cube would begin to rotate during spin-up rather than at the brake event, and would arrive at the edge with no stored impulse behind it. That fixes a floor

$$t_{\text{spin}} > \frac{h_w}{\tau_g} = 0.20\,\mathrm{s},$$

below which the manoeuvre cannot be run at all, however capable the drive. The drive is nowhere near it: 0.226 N m is 17.2 % of the 1.3155 N m floor of Table 12, and even at the 16 A short-duration rating the 0.402 N m it produces reaches under a third of it. The cube therefore cannot tip itself with the motors alone at any current the drive can deliver — which is the quantitative form of the claim that the impulse comes from the collision and not from the drive, and the reason the brake is a separate mechanism. PER-13 fixes the margin at a factor of two, which the actuator meets with the current limit removed.

Neither bound is set by energy. One wheel at design speed holds $\frac{1}{2}I_w\omega_{\max}^2 = 55\,\mathrm{J}$, so all three together draw 165 J from a pack carrying about 35 W h — roughly 0.13 % per jump-up, negligible even across a walking sequence. What the spin-up does cost is *current*: at the end of a three-axis spin-up the bus supplies of order 300 W, some 14 A at pack voltage, which is the transient the XT60 and the anti-spark path of Section 7.3 have to carry. That figure is an estimate from $\tau\omega$ plus copper loss, and is an acceptance measurement at first power-on rather than a closed analysis. For the walking objective the same number sets the cadence: with 2 s of spin-up, consecutive jump-ups (FUN-11) cannot follow each other faster than roughly one every three seconds once the landing transient is allowed to settle.

5.2 Sizing Constraints and Closure Procedure

Cube edge length L and reaction-wheel inertia I_w are not independent choices — they are two ends of a single constraint chain, beginning with what must physically fit inside the cube. This section states that chain and the order in which it closes. No dimensions are committed here: component selection remains open (Sections 6, 7), and the numbers follow from the procedure rather than preceding it.

The chain runs in the order of Table 13. Internal volume is dominated by the LiPo pack and the subframe carrying it (locating the three motors on mutually perpendicular axes through a common centre, together with drivers, flight controller, IMU, and brake servos). The pack is incompressible and sits near the geometric centre to keep the centre of mass on the body diagonal, so L is the smallest edge length containing that volume plus wall thickness and contact features. L then caps wheel diameter D_w, and with it the achievable inertia $I_w \sim m_w D_w^2/4$ — ballast cannot rescue a wheel short of its maximum diameter, since it necessarily sits inboard of the ring it displaces.

Against that stands the jump-up requirement (Section 5.1.1),

$$I_{w,\text{target}} = \frac{h_w(M,L)}{\omega_{\max}}, \qquad h_w \propto ML^{3/2},$$

which rises with both M and L. Two consequences close the loop: enlarging the cube to gain wheel diameter also raises the requirement, but achievable inertia grows faster than the requirement (D_w^2 against $L^{3/2}$), so a short wheel at a given L can be closed by a modest increase in L; and any mass added this way feeds back through $h_w \propto M$ and through the tip-over torque $\tau_g = MgL/2$ the brake must exceed — which is why the procedure terminates at a fixed point rather than in one pass.

This procedure is implemented formally in `analysis/SIZING.py` and run to convergence by hand.

The two contact cases are not equally demanding: the corner equilibrium needs a larger centre-of-mass rise and a less favourable momentum-transfer geometry than the edge equilibrium, so at a common $\omega_{\max}$ it demands the greater wheel inertia (Table 11). Step 4 is therefore always evaluated for the corner case — a wheel meeting the corner requirement meets the edge requirement with margin. Table 14 records both once the procedure closes.

Table 13: Sizing closure procedure. Steps are executed in order; a failure at step 4 returns to step 3 or step 1 as indicated.

Step	Determines	Governing consideration
1	Battery pack geometry	Capacity and discharge rating for the jump-up duty cycle
2	Inner clear volume $V_{\min}$, hence minimum L	Pack + motor subframe + driver/controller stack, non-intersecting; pack near centre
3	Maximum wheel diameter D_w, hence achievable I_w	Clearance to subframe, motor body and the two orthogonal modules
4	Requirement $I_{w,\text{target}} = h_w(M, L)/\omega_{\max}$	Corner jump-up, Eqs. (10)–(11)
5	Accept or iterate	If $I_w < I_{w,\text{target}}$: raise L (step 2), raise $\omega_{\max}$, or trim M; re-enter at step 3
6	Mass reconciliation	Updated M feeds back into step 4; iterate to a fixed point

Table 14: Required per-wheel inertia by contact case, evaluated at $\omega_{\max} = 4000\,\text{rpm}$ for the converged $M = 1.8000\,\text{kg}$. Only the geometry factor differs between the cases: τ_g and β are properties of the cube and the brake, not of the feature it stands on.

Contact case	h_w ($\text{kg}\,\text{m}^2/\text{s}$)	$I_{w,\text{target}}$ ($\text{kg}\,\text{m}^2$)	Relative
Edge	0.2369	5.6566×10^{-4}	90.7 %
Corner (critical)	0.2612	6.2352×10^{-4}	100 %

The corner case demands 9.3 % more inertia than the edge case, so sizing on it covers the edge case with margin. The selected wheel delivers $6.2715 \times 10^{-4}\,\text{kg}\,\text{m}^2$ (Table 16) — 100.6 % of the corner requirement and 110.9 % of the edge requirement.

Table 15 is the bookkeeping side of steps 5 and 6: populated as components are selected and re-totalled each iteration, since the total is both an input to $I_{w,\text{target}}$ and an output of the design.

5.3 Reaction-Wheel Sizing

The wheel must supply the inertia $I_{w,\text{target}}$ demanded by the corner jump-up (Section 5.1.1), fit within the envelope's outer diameter (step 3, Table 13), and stay light enough not to dominate cube mass, since its own mass feeds back into $I_{w,\text{target}}$ through M. These three demands pull against each other, so the wheel architecture is chosen to be trimmable after fabrication rather than needing to be right the first time.

Architecture: ballasted ring. Rather than scaling a solid disc, the wheel is a printed *ring plus spokes* tuned by *ballast*: standard M6 steel bolt-and-nut sets fitted through round clearance holes drilled *radially* through the ring's cross-section, from the inner bore surface to the outer rim surface — a true ID-to-OD through-hole, not one drilled through the wheel's face. Each hole's depth is fixed at the ring's radial width, and its centroid sits at the mean radius $R_{\text{mean}} = (D_w + D_i)/4$ by construction, with no free pitch-circle radius to choose. Hole diameter is instead capped by the wheel's axial thickness, with a printed wall retained on both faces. Each station removes plastic and adds steel at very nearly the largest available radius, so the net inertia gain per station is large, and the target is reached simply by choosing how many stations to populate.

The bolt is also the retention feature: the nut threads on and clamps the ring, so nothing depends on a plastic pocket floor. Each station carries the centrifugal load $F = m_{\text{hw}}\,\omega_{\max}^2\,R_{\text{mean}}$ of its

Table 15: Mass budget (per cube), rolled up from the bill of materials by `analysis/SIZING.py` (Stage 6). The total is an input to the inertia requirement, not only a result: the value below is the *converged* one, which the closure reproduces exactly ($\delta = 0.0\,\mathrm{g}$) against the M assumed in Table 12.

Subsystem	Mass (g)	Share	Basis
Reaction wheels ($3 \times 234.14\,\mathrm{g}$)	702.4	39.0 %	Sized (Sec. 5.3)
Motors and brake servos (3× each)	245.4	13.6 %	Datasheet
Power (battery, XT60, regulator)	279.0	15.5 %	Drives envelope (step 1)
Electronics (drivers, MCU, encoders, IMU)	95.2	5.3 %	Datasheet, as-weighed
Wiring, perfboard and passives	45.5	2.5 %	Estimated
Subtotal — specified hardware	**1367.5**	76.0 %	Measured or datasheet
Structure (subframe, frame, fasteners)	432.5	24.0 %	**Allowance, pending CAD**
Total M	**1800.0**		**Converged fixed point**

The structure line is an *allowance*, not a measurement, and at roughly 24 % of the total it remains the dominant uncertainty in the budget. It has been revised *downward* from the 487.6 g of the previous iteration to 432.5 g (subframe 176.9 g, frame 212.8 g, fasteners/harness 42.8 g, held at the same relative split). Its share of the total falls further than that figure alone suggests, because the wheels grew over the same revision: the ballast needed at 4000 rpm raises each wheel from 170.37 g to 234.14 g, so the specified-hardware subtotal now dominates the budget in a way it did not before. The allowance is reported separately from the specified hardware for that reason: the subtotal is what the design currently knows, and the total is what it currently projects. Closing the CAD model replaces the allowance with a real figure and re-enters the loop at step 6.

This design point is *tighter* than the one it replaces, and the difference is worth stating rather than inheriting the previous reassurance. The fifteen-station wheel holds anywhere between 300.0 g and 440.0 g of structure allowance — a 140 g band, but not a symmetric one. Set at 432.5 g, the budget can absorb a 132.5 g *underrun* without changing the wheel, and only a 7.5 g *overrun* before the station count steps from fifteen to eighteen. The previous iteration could absorb roughly 77 g in either direction. Any growth in the frame therefore forces a wheel change, where before it did not, and step 6 must be re-entered on the first CAD revision that moves the structure line upward at all.

own hardware, reacted by the bolt in tension and by bearing of the head and nut faces — now against the ring's curved ID and OD surfaces rather than a flat face, so both are spot-faced flat (or fitted with a curved washer) to seat squarely. A nyloc or threadlocked nut is specified so vibration cannot back it off.

The geometry is defined by the quantities of Table 16, all of which follow from the envelope closure of Section 5.2 and are recorded there once fixed.

Table 16: Reaction-wheel geometry and ballast parameters. Values are the current design point of the iteration described below; they are provisional while the structural mass of Table 15 remains an allowance.

Symbol	Parameter	Value
D_w	Ring outer diameter (fixed by envelope)	120 mm
D_i	Ring inner diameter ($D_w - 2w_r$)	100 mm
w_r	Ring radial width (= ballast hole depth)	10 mm
t	Wheel axial thickness (caps hole diameter)	20 mm
n_s, w_s	Spoke count and width	3, 10 mm
ρ_p	Printed-material density (PET-CF)	1290 kg/m^3
d_h	Ballast hole diameter (M6 clearance)	6.4 mm
R_{mean}	Ballast centroid radius (forced, $= (D_w{+}D_i)/4$)	55.0 mm
m_{hw}	Hardware mass per station (bolt + nut)	7.5 g
m_b	*Net* mass added per station	7.085 g
ΔI_b	*Net* inertia added per station	2.151×10^{-5} kg m^2
N	Station count reaching $I_{w,\text{target}}$ (of $N_{\max} = 37$)	15
m_w	Resulting wheel mass	234.14 g
I_w	Resulting wheel inertia	6.2715×10^{-4} kg m^2
	as a fraction of $I_{w,\text{target}}$	100.6 %

The ballast is characterised *net* of the plastic each hole displaces: m_b and ΔI_b are the increments the installed bolt-and-nut adds over the radial cylinder of material it replaces, evaluated at the forced centroid radius R_{mean}. Both corrections use the parallel-axis theorem about the spin axis, retaining each feature's own-axis term rather than treating it as a point mass — here the *transverse* solid-cylinder term $r_h^2/4 + w_r^2/12$, since the hole's axis is radial (perpendicular to the spin axis), unlike a face-drilled hole. Sizing then reduces to solving

$$I_w(D_w, N) = I_{w,\text{ring}}(D_w) + N\,\Delta I_b \geq I_{w,\text{target}}$$

for the smallest admissible N at each candidate D_w, subject to the fit and mass checks below.

Sizing calculator and iteration procedure. These relations are implemented as a parametric calculator mapping the geometry of Table 16 onto the mass and inertia of one wheel: it sweeps candidate outer diameters and, for each, returns the required station count, the count the gap rule admits, resulting mass and inertia, and the pass/fail state of every constraint. It is not an optimiser, and the reported design point was not found by one.

Outer diameter D_w is not free to choose — it is imposed by the packaging envelope, since three orthogonal wheel modules and the battery must fit within L — so the sweep is retained for what it reveals, not what it selects. Left free to minimise wheel mass, the calculator prefers a larger ring, since $I_{zz} \sim mR^2$ makes mass at large radius buy inertia more cheaply: the unconstrained optimum at the present target is 150 mm (163.78 g/wheel, no ballast needed), so the envelope constraint at 120 mm costs about 70.4 g per wheel (211.1 g over three). The penalty has roughly quadrupled since the previous iteration, because the higher inertia target now has to be met with steel at R_{mean} rather than with ring diameter, and steel bought at a fixed radius scales linearly where diameter scales quadratically. The intermediate case is worth recording for the CAD review, since it is the one that would recover most of the penalty for the least envelope: at 130 mm the wheel closes at 100.4 % on nine stations and 203.60 g, recovering 30.5 g per wheel (91.6 g over three) for 10 mm of diameter. This is the price of a wheel that fits, and a wheel that does not fit is not a candidate.

The procedure followed is the fixed-point iteration of Table 13, run by hand: a cube mass is estimated and a mass budget assembled; the jump-up analysis converts that mass into an inertia requirement; the wheel is sized against it subject to volume and hole-pattern constraints; and the resulting wheel mass feeds back into the budget that started the loop. The cycle repeats until the geometry simultaneously fits the internal volume, meets the inertia requirement, and reproduces the cube mass it assumed — convergence is what closes the design, with step 6 of Table 13 returning approximately zero.

Selection among admissible diameters is made on minimum *wheel mass*, not minimum diameter.

6 CAD Design

This design is preliminary: final actuator and sensor specifications are not yet confirmed, so the mechanical architecture is kept modular and largely parametric (Section 1.1). This section states the mechanical concept and the CAD conventions the build follows, for both the first prototype (the single-face 1-DoF rig) and the second and final prototype (the three-axis cube), and the checks that confirm the model and the fabricated parts agree. It is deliberately kept to the level of decisions and interfaces: the part-by-part model tree, the manufacturing drawings, the fabrication settings and the assembly sequence are recorded in Appendix C, so that a change to a bracket does not require an edit to the design rationale, and the rationale is not buried in geometry.

6.1 Mechanical Design and CAD Construction

6.1.1 CAD Methodology and Model Conventions

The model is built top-down from the envelope closure of Section 5.2 rather than bottom-up from individual parts, so that the edge length L and the wheel diameter D_w propagate to every dependent feature when an iteration changes them. Three conventions make that propagation reliable, and they are stated here because they constrain how every part in Appendix C is modelled:

1. **A single skeleton drives the assembly.** L, D_w, the motor axis offsets and the wall thickness live in one parameter table referenced by every part; no downstream part carries a hard-coded dimension that duplicates one of them.
2. **The body frame is the CAD origin.** The assembly origin is the cube's geometric centre, with axes on the three motor spin axes, so that CAD-reported mass properties are directly the I and r_{cm} of Table 10 without a change of basis.
3. **Mass properties are an output of the model, not an annotation.** Every part carries its assigned material and print density, so the assembly mass roll-up is the quantity fed back into Table 15 at step 6 of Table 13.

Convention 3 makes the iteration auditable: the mass budget and the inertia requirement are read from the same model, so a design that closes analytically cannot fail to close in the CAD without the discrepancy being detected.

Where the final numbers are. The three conventions above say how the model is driven; they do not say what it is driven *to*. Those values — edge length and wall thickness, wheel diameter, thickness and ballast count, the mass of every item on board, the operating point the wheels were sized for — are collected in one place, Appendix A, and are not restated here or anywhere else. That appendix is a reference sheet rather than an argument: each row quotes a value and names the table or the analysis that derives it, so the reasoning stays in Sections 5 and 6.1 while the numbers themselves have a single address. Table 63 is the model-side view of the same figures, listing only the subset the CAD skeleton drives, and it is subordinate to the specification sheet: where the two differ, the sheet is right and the CAD parameter table is stale. A reader who wants to know what was built, rather than why, should go to Appendix A first.

6.1.2 1-DoF Prototype Build

The single-face rig is the first article built and the bench on which the parameters of Table 9 are measured. Its structure is a 3D-printed plate sized to one cube face, a central motor and wheel mount, a corner bearing defining the single rotational degree of freedom, and encoders and other hardware. Note that the servo (brake system) is not included in the 1-DoF prototype as its purpose is validating the control loop on a more well-understood problem. Detailed geometry and the drawing set are in Appendix C.3.

The rig as built is shown in Figure 9. The plate is mounted diagonally, so that the pivot bearing sits at the lowest corner and the wheel axis passes through the plate centre: this places the centre of mass directly above the contact point at the upright equilibrium, which is the configuration linearised in Section 8. The two printed uprights either side of the frame are hard stops, limiting travel to a few degrees past the point of no return so that a failed balancing attempt ends against a stop rather than on the bench.

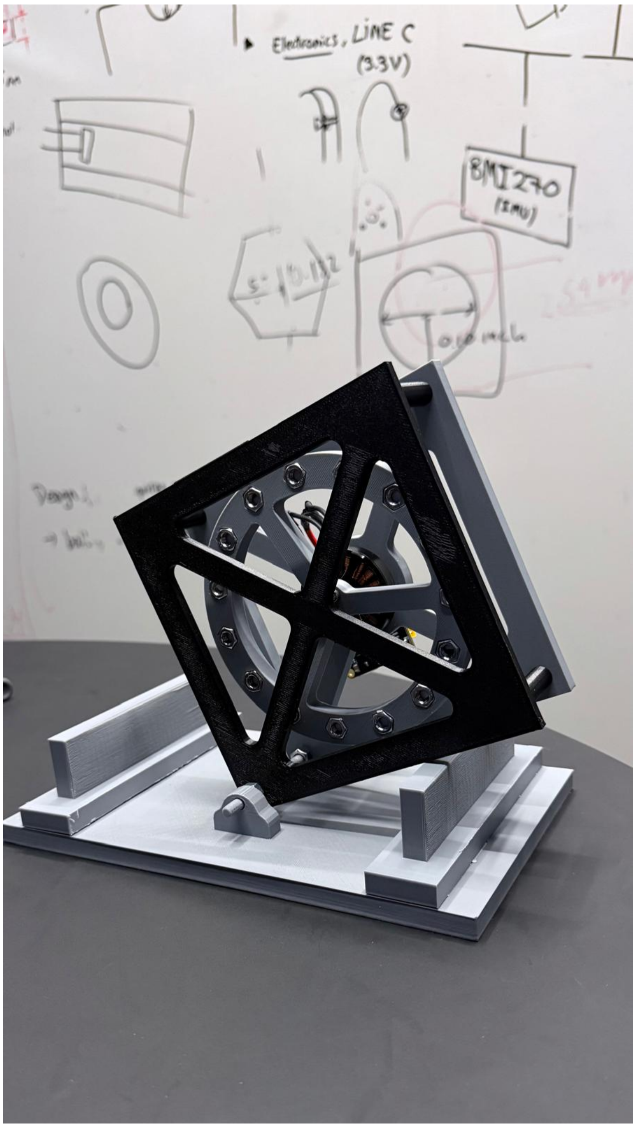

Figure 9: The single-face prototype rig as built: one reaction wheel driven by a brushless motor on a diagonally-mounted printed plate, pivoting on a corner bearing against a printed base with hard stops. This is the article on which the plant parameters of Table 9 are measured and the single-axis controller of Section 8 is validated before the three-axis build. The drawing set is given in Appendix C.3.

6.1.3 3D Cube Frame

The frame is conceived in two parts, reflecting the constraint chain of Section 5.2. An *inner structure* locates the three motors on mutually perpendicular axes through a common centre and carries the battery pack, the three drivers, the flight controller, inertial measurement unit, and the brake servos; it is the element that fixes the minimum internal clear volume and therefore the edge length L. An *outer frame* carries the edge and corner contact features, protects the wheels, and reacts the motor mounting loads from the inner structure into the ground contact.

A first physical prototype of this frame has been printed and is shown in Figure 10. It is explicitly a *concept prototype*, not the flight article: it demonstrates the two-part architecture, the X-braced face pattern and the packaging of three orthogonal wheel modules within the cube envelope, and it serves as a physical check on clearances that are awkward to judge on screen. The final design is not frozen at the time this photograph was taken. Detailed geometry follows the envelope closure of Section 5.2, and the wheel dimensions in particular remain open pending the sizing iteration of Section 5.3 and confirmation of the final motor; the mass properties of the printed prototype are therefore not the mass properties assumed in the budget. The frozen general-assembly drawing is given in Figure 20 of Appendix C.4.

The split is also the main mechanical interface in the machine, and it is managed as one: the inner structure is designed against a fixed bolt pattern and keep-out envelope published to the outer frame, so the two can be revised independently as long as that interface holds. The controlled interfaces are listed in Table 17; the parts realising them, with their drawings, are in Appendix C.4.

Table 17: Controlled mechanical interfaces. Each is frozen once agreed, so that the parts on either side can be revised independently.

Interface	Between	Controlled quantities
Motor mount	Motor ↔ inner structure	Bolt circle, shaft height, concentricity to the wheel axis
Wheel hub	Wheel ↔ motor rotor	Bore, bolt pattern, axial location, balance datum
Frame joint	Inner structure ↔ outer frame	Bolt pattern, keep-out envelope, load path to the contact features
Electronics mount	Board ↔ inner structure	Board outline, mounting-hole pattern, keep-out heights, connector exit directions (Sec. 7.6)
Brake mount	Servo ↔ inner structure	Servo location, barrier throw
Battery retention	Pack ↔ inner structure	Pack envelope, retention method, centring tolerance to the cube centre
Contact features	Outer frame ↔ ground	Corner and edge geometry, material, replaceability

6.2 CAD and Dimensional Verification

The checks below split into those that run against the model, before anything is fabricated, and those that run against the physical part afterwards. The driving dimensions themselves are outputs of the packaging closure of Section 5.2 and are not verified here; what is verified is that the model realises them consistently and that the fabricated part matches the model.

Model checks (pre-fabrication). Interference and clearance verification on the wheel-to-subframe and inter-module clearances c_w and c_o of Table 63, through a full rotation of each wheel and across all three axes simultaneously; orthogonality of the three motor axes; position of the centre of mass relative to the geometric centre, since the modelling of Section 4 assumes a known offset; and the mass roll-up against Table 15, which is the input to the sizing verification and must be re-run whenever it changes.

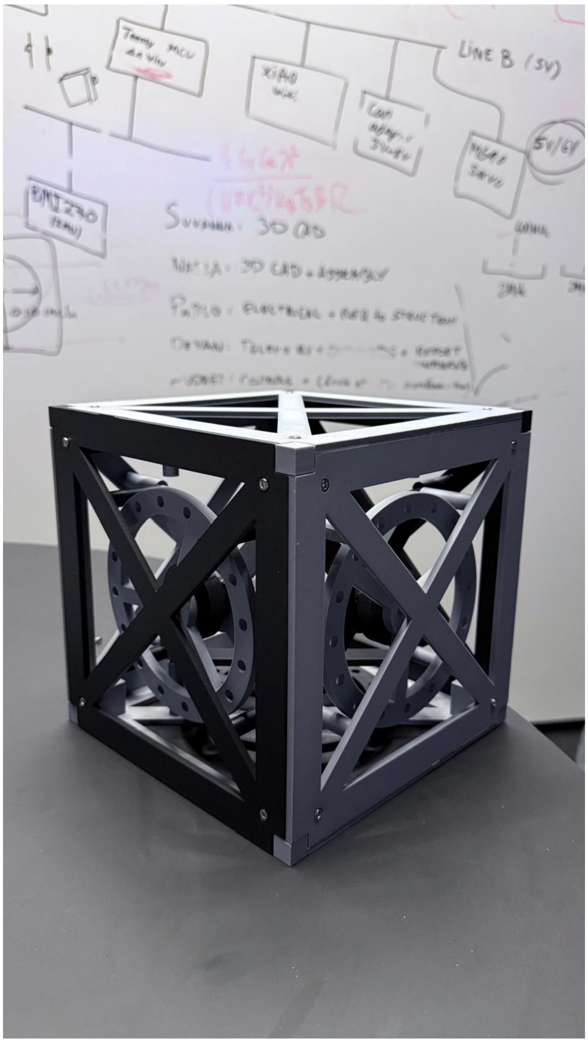

Figure 10: Printed concept prototype of the three-axis cube frame, showing the X-braced outer faces and the three orthogonal reaction-wheel modules carried by the inner structure. This article validates the two-part architecture and the internal packaging only: it is *not* the final design. Wheel geometry and the frame edge length L remain subject to the sizing closure of Sections 5.2 and 5.3, so its mass properties do not represent the budgeted values. The frozen design's general-assembly drawing is given in Figure 20 of Appendix C.4.

Part checks (post-fabrication). Dimensional acceptance of the ballast holes — diameter, pitch circle radius and the radial wall left on each side — since these are structural features rather than cosmetic ones and a print that is dimensionally out of tolerance at a hole is a wheel that cannot be trusted at speed; verification that each bolt is torqued and its nyloc nut engaged, the through-bolt being the retention feature; measured wheel mass against the CAD prediction; and static balance of the assembled wheel, which is the first opportunity to detect a ballast placement error before it appears as vibration at speed.

The modelled-versus-measured comparison is recorded in Table 66 of Appendix C.6 and is not duplicated here. A discrepancy found at that step is carried back into $I_{w,\mathrm{target}}$ and absorbed by ballast trim where the authority allows; where it does not, the finding returns to the closure procedure rather than being accepted.

7 Electronics

This section is presented at the level of the decisions that shape the electrical architecture: what the components are, how the whole system loop fits together, how power is distributed, how the buses are routed and terminated, and how the assembly is made safe to power on. The sheet-level schematics, the connector and pin tables, the harness drawings and the perfboard layout are collected in Appendix D, for the same reason the CAD detail is separated from the mechanical rationale of Section 6 — a connector re-pin should not require an edit here. The acceptance criteria and checks run against this architecture during bring-up (CAN-bus termination, rail voltages, sensor and driver calibration) are tracked against the electrical bring-up stage of the validation strategy, not repeated here.

7.1 System Architecture

Figure 11 shows the closed control loop that ties the four subsystems together. The Teensy 4.1 runs estimation and control at 1 kHz and issues torque commands over CAN-FD to three moteus-n1 drivers, each closing a 30 kHz field-oriented current loop on its MN4006 motor. The resulting reaction torque acts on the cube body; the body's attitude is sensed by the BMI270 IMU and the rotor angle by the MA600 absolute encoder, closing the loop back to the controller. The two loops run at different rates by design: the fast current loop is delegated entirely to the drivers, so the flight controller need only meet the 1 ms outer-loop deadline of Section 8.

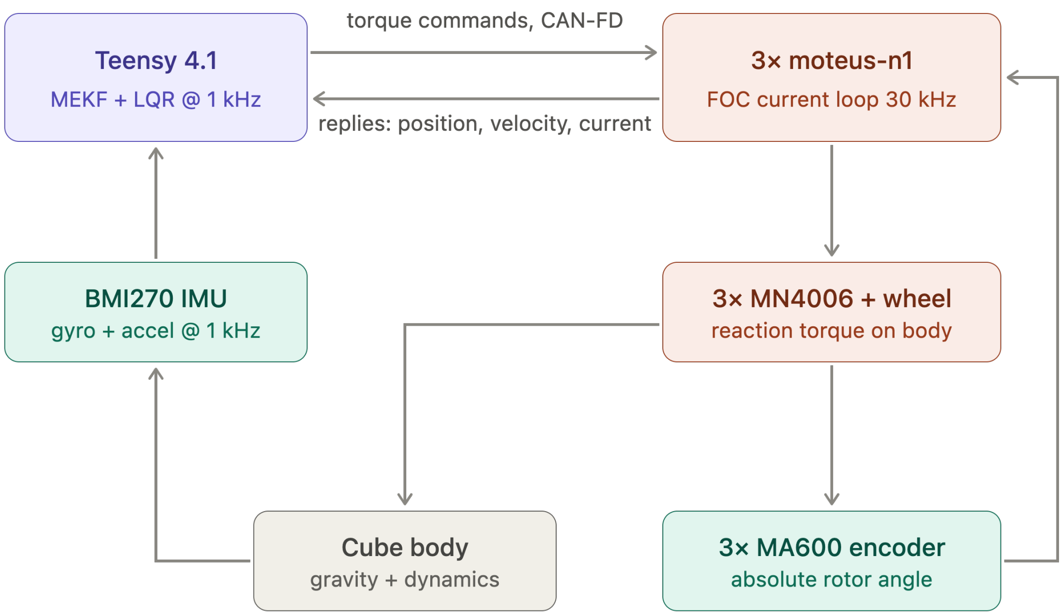

Figure 11: System architecture and control loop. Blue: processing; green: sensing; orange: actuation. Rates shown are the design targets of Section 8.

7.2 Components

The electronics and software are built around the Teensy 4.1 microcontroller, which communicates over the CAN-FD bus with the mjbots moteus-n1 driver that closes the current loop for the MN4006 brushless motor. Wheel speed is measured with the MA600 magnetic encoder on the motor shaft, and body tilt is estimated from the BMI270 IMU. Braking is actuated by a

TowerPro MG92B digital servo driven directly from the microcontroller, and the XIAO ESP32-C6 wireless module provides a telemetry and debugging link. Power is supplied by the 6S LiPo battery, stepped down through an LM2596 DC-DC converter for the logic-level electronics, with the motor drivers powered directly from the battery bus. Table 18 lists each component against the schematic block it belongs to (Appendix D.1); current ratings are given in Appendix D.2 rather than here, so that a part substitution does not require an edit to this section.

Table 18: Component overview: each part against the schematic block that carries it (Figure 21, Appendix D.1).

Component	Function	Schematic block
Teensy 4.1	Flight computer: estimator + control law	Flight computer and IMU
BMI270 IMU	Body attitude reference (tilt + rates)	Flight computer and IMU
mjbots moteus-n1 (3×)	FOC current/torque driver, inner loop	Power entry and protection; CAN-FD bus
T-Motor MN4006 (3×)	Reaction-wheel torque source, via driver	Power entry and protection
MA600 encoder (3×)	Wheel angle: commutation + state feedback	Encoder interfaces
TowerPro MG92B servo (3×)	Wheel brake actuator (jump-up)	Brake servo drive
XIAO ESP32-C6	Wi-Fi telemetry / command link	Telemetry link
LM2596 buck converter	5 V logic rail source	5 V logic rail
6S LiPo pack	Primary energy store	Power entry and protection

The full parts list is catalogued in Appendix B (Table 62). The supplied set is an electronics-only BOM; the reaction wheels, the frame and all printed parts are the team's design responsibility, and their fabrication is the leading schedule risk.

Table 19 states, block by block, the layout, mounting and connection constraints each part is checked against before bring-up: what governs its placement, and how the assembled machine confirms it. These are assembly and layout restrictions on the physical prototype, not schematic content, which is why they are stated here rather than repeated against each schematic block of Section 7.4.

Motor characterisation and operating envelope. Figure 13 shows measured torque and speed against phase current for the MN4006, taken from manufacturer bench data. The measurements were made with the motor driving propellers rather than reaction wheels, so the *load* is not representative of this application — an aerodynamic load rises with the square of speed, whereas a reaction wheel is an inertia with only bearing friction opposing it at constant speed. What does transfer is the motor's own electromechanical behaviour, since torque per ampere and back-EMF per unit speed are properties of the machine and not of what it is attached to. The curves are used here only in that sense: to bound the torque and current the actuator can be expected to deliver.

Three figures relevant to the design follow from the data. First, the torque–current relation in Figure 13(a) admits a straight-line fit ($r = 0.97$) of slope $0.0227\,\mathrm{N\,m/A}$ with a torque-axis intercept of $+0.078\,\mathrm{N\,m}$ — but that intercept is an artefact of fitting a line to a curve rather than the signature of a physical offset, and the slope is correspondingly not the machine constant. A

Table 19: Electrical layout, mounting and connection constraints, by block. Checked against the assembled prototype before and during bring-up.

Component / block	Layout, mounting & connection rule	Verification
Power entry and protection	Pack connector, anti-spark inrush path, E-stop and fuse in series, upstream of the split to the motor bus and logic rail	E-stop functional; fuse rated to worst-case bus current; anti-spark path intact
5 V logic rail	LM2596 buck converter off the motor bus, back-feed diode, bulk capacitance and per-device decoupling at each load	Converter rating vs. measured worst-case rail current (Appendix D.2); capacitor voltage ratings confirmed by position
CAN-FD bus	Linear chain across the three moteus drivers and the flight computer, short stubs, termination at the two physical ends only (Sec. 7.5)	Differential-pair scope capture; sustained error-frame count at full data rate with all nodes addressed
Encoder interfaces (3×)	Short SPI run from each driver to its own encoder; encoder positions fix the driver positions, not the reverse	Air gap ≤ 1.5 mm confirmed; read-back with no error flags, re-checked after mechanical assembly
Flight computer and IMU	Teensy 4.1 and BMI270 mounted at or as close as possible to the pivot; orientation set by packaging, corrected in software by the mounting rotation of Sec. **??**	Static orientation check; noise floor confirmed with wheels spinning
Telemetry link	ESP32-C6 module fed off the logic rail, sized for its transmit-burst current	Transmit-burst return path verified off the logic rail
Brake servo drive	Servo PWM routed clear of the encoder SPI runs, one instance per axis	Stall-current headroom confirmed on the logic rail
Rail separation	Motor bus and logic rail kept separate downstream of the E-stop	Ripple measurement on the logic rail under motor load
Grounding	Single return topology; no motor return current through the sensing ground	Continuity and isolation check before first power-on

PLACEHOLDER — artwork pending

Assembled electronics of Armonia: flight computer, driver stack, distribution board and harness in place inside the frame.

Figure 12: Assembled electronics of Armonia. To be replaced with a photograph once the perfboard and harness of Section 7.6 are populated.

positive torque-axis intercept would assert shaft torque at zero current. What the points actually show is a torque per ampere that *falls* with current, monotonically within every load block, from about 0.050 N m/A at the lowest measured currents (2 A) to 0.026 N m/A at the highest (17.5 A): resistive loss grows as I^2 while torque grows as I, so the characteristic is concave, and any straight line through it must return a positive intercept together with a slope that is a chord across the whole 2–17.5 A span rather than a tangent anywhere inside it.

The design therefore takes the torque constant from the motor's 380 rpm/V rating, $k_t = 60/2\pi K_V = 0.025{,}13\,\mathrm{N\,m/A}$, a property of the machine rather than of a fit, and uses that one figure throughout — Table 9, the spin-up estimate of Section 5.1.1, and the torque limits of Table 43. The bench data then serve as the cross-check they can support, and they pass it in the conservative direction: every measured point returns more torque per ampere than 0.025,13, the closest being the 17.5 A point just quoted, and across the 9 A continuous band of the driver the margin is at least 20 %. Two cautions stay attached to the torque column whatever constant is adopted — it is not mutually consistent with the published electrical power, and the load is a propeller — so a residual error in k_t should be expected rather than assumed away. Its constant part reaches the controller as a torque bias at the plant input, which is the same class of disturbance as the centre-of-mass offset and the constant friction term of Section 8.6, and the disturbance-torque augmentation described there removes it through integral action: the balancing design needs k_t bounded, not exact.

In absolute terms the motor produces at most 0.45 N m, an order of magnitude below the 5 N m brake torque assumed in Section 5.1.1. The two are not alternatives: the motor stores angular momentum in the wheel over about a second, and the brake destroys it in tens of milliseconds, so the impulsive torque comes from the collision and not from the drive. This is why braking is a separate mechanism rather than a motor function. Second, the highest measured point reaches 0.45 N m at 17.5 A, above the 16 A peak rating in Table 62, and that run terminated on the motor's thermal limit — so the top of this envelope is a short-duration capability, not a continuous one. Third, every load tested crosses the 4000 rpm design ceiling of Section 5.2, and does so at or below 60 % throttle — the heaviest propeller, which is the one that ran to the thermal limit, passes it at 4079 rpm on 5.9 A. The ceiling is therefore set by bus voltage and derating, not by any inability of the motor to reach the speed. This margin has widened with the revision of $\omega_{\max}$ from 5500 rpm to 4000 rpm (Section 5.1.1): at the previous value only

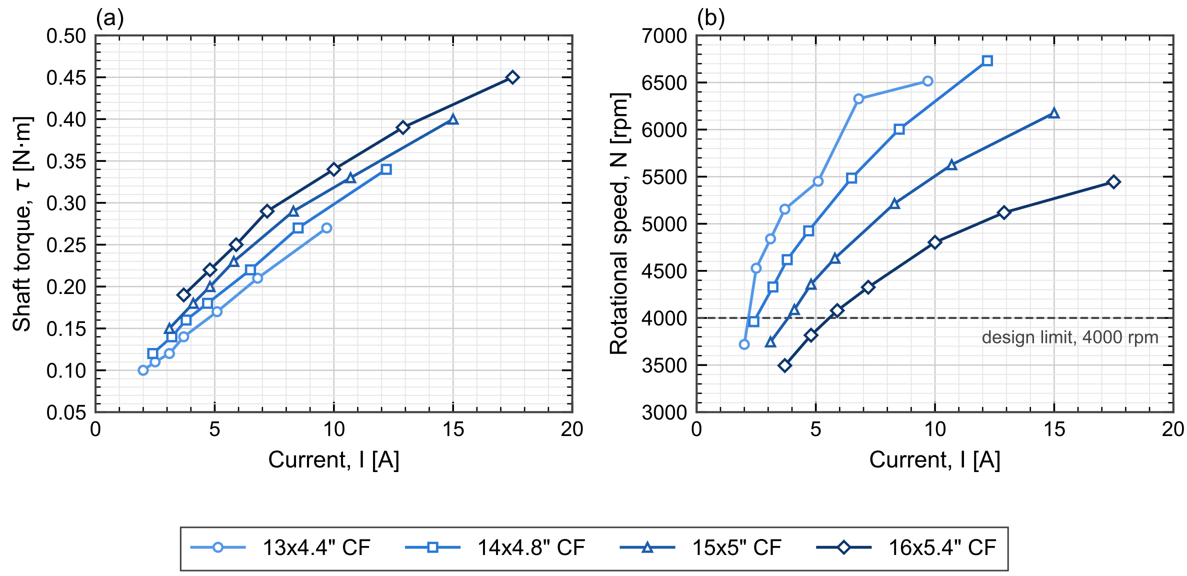

Figure 13: Manufacturer static bench data for the MN4006: (a) shaft torque and (b) rotational speed against phase current, for one motor driving four different loads. Symbols are measured points at throttle settings of 50, 55, 60, 65, 75, 85 and 100 %; lines join successive points and are not fits. The dashed line in (b) is the 4000 rpm reaction-wheel design ceiling of Section 5.2. Loads shown are propellers, not reaction wheels: the curves characterise the *motor*, and the load-dependent spread between them does not carry over to this application. The heaviest load reached the motor thermal limit at 100 % throttle.

the lighter loads cleared the ceiling, whereas the drive now sustains the design speed across the whole of the measured envelope.

7.3 Power Architecture

The power architecture is shown in Figure 14. A single 6S LiPo pack (22.2 V nominal, 1550 mA h) feeds everything through one XT60 connector, behind a combined E-stop and fuse, with an anti-spark path to limit inrush into the driver bulk capacitance. The XT60 is fixed by the pack rather than chosen: it is rated at ≈ 60 A, which covers the three-axis spin-up transient but with a smaller margin than a 90 A housing would give, so the transient is a measured acceptance item at first power-on rather than an assumed one. The bus then splits into two rails. The *motor bus* runs at pack voltage (22–25 V across the discharge curve) directly into the three moteus drivers, with bulk capacitance rated to at least 50 V to absorb regenerative braking transients—which are not incidental here, since the jump-up manoeuvre of Section 5.1 dumps the full wheel kinetic energy back into the bus. The *logic rail* is derived by an LM2596 buck converter stepping 22 V down to 5 V for the Teensy, the ESP32 telemetry link, the three brake servos and the IMU. Keeping the two rails separate downstream of the E-stop means motor-current ripple does not reach the sensing and processing electronics.

The rail-by-rail current budget that sizes the buck converter and the harness conductors is given in Appendix D.2 (Table 67), populated from datasheet ratings and completed from measurement during bench bring-up rather than from typical values alone, because the figure that matters for the 5 V rail is the coincident worst case — telemetry transmit burst, servo stall and full sensor load at once. The 5 V total is checked against the LM2596 rating, which derates without a heatsink.

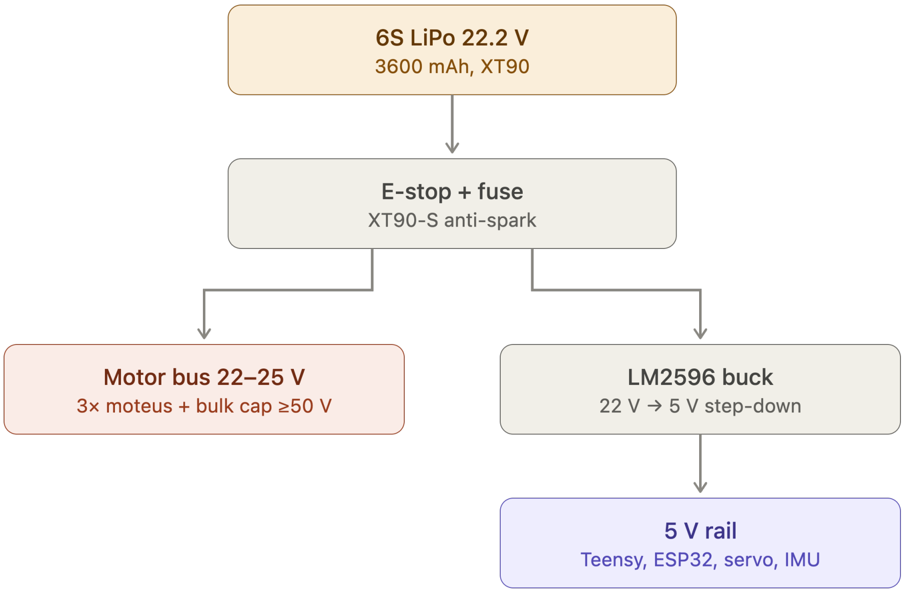

Figure 14: Power distribution tree. A single 6S pack feeds the motor bus directly and the 5 V logic rail through a buck converter, both behind a common E-stop and fuse.

7.4 Schematic and Interconnect Diagrams

The signal-level design is captured as one flat schematic (Figure 21, Appendix D.1), together with the connector and pin-assignment tables. The layout, mounting and connection constraints each block is checked against are given in Table 19 of Section 7.2, ahead of the schematic itself, and are not repeated here.

7.5 Bus Topology, Termination and Signal Integrity

The CAN-FD bus is the one electrical subsystem whose physical routing is a correctness requirement, and it is therefore fixed before the board outline is drawn. At the design data rate the bus must be a linear chain with short stubs and termination at exactly the two physical ends; a star topology or a mid-chain termination will pass a bench continuity check and then fail intermittently under wheel vibration, which is the hardest class of fault to diagnose once the cube is closed. The design rules and their verification are listed in Table 19 of Section 7.2, together with every other block's layout and connection constraints.

The encoder interfaces impose the complementary constraint. Each absolute encoder requires a short SPI run to its own driver and a tightly held air gap to its magnet, so the driver positions are dictated by the encoder positions, and those in turn by the motor axes. The physical layout order is therefore: motor axes fix the encoders, encoders fix the drivers, drivers fix the bus route, and only then is the board outline of Section 7.6 drawn around what remains.

7.6 Board Layout and Harness

The electronics are built on a double-sided perfboard rather than a fabricated PCB. A PCB would have a long lead time, and a perfboard allows for iteration and modification throughout the process. Its consequence is that layout discipline — rail separation, decoupling at each device, deliberate routing of the bus pair — must be enforced by hand and checked explicitly, since no design-rule check will do it.

The board outline, mounting-hole pattern, keep-out heights and connector exit directions constitute the electronics-mount interface of Table 17 and are frozen once delivered to the mechanical design: later electrical changes must fit inside that outline rather than move it. The harness — battery lead, driver drops, bus chain and servo drive — is fabricated to the drawings in Appendix D.4 and is labelled and strain-relieved before installation, because every connector in the machine sits on a structure that vibrates by design.

IMU mounting and axis alignment. The BMI270 is carried on Perfboard 1 alongside the Teensy (Figure 20), placed by wiring and packaging convenience rather than by axis alignment to the body frame $\{B\}$ of Figure 7. Its native sensor axes therefore do not coincide with $\{B\}$; every sample is corrected by the fixed mounting rotation of Section **??** before the estimator uses it, rather than by re-orienting the board to match the model.

7.7 Electrical Safety and Bring-Up Protocol

Two rules govern all electrical bring-up and are stated here because they are design constraints, not merely lab practice. First, every first-time power application is made on a current-limited bench supply, never on the battery; the battery is introduced only after the assembly has passed a full bench power-on. Second, the pack is never connected without the E-stop and fuse in circuit and the anti-spark path intact, since the driver bulk capacitance draws an inrush that will weld a bare connector.

The staged sequence — rails, then buses, then sensors, then open-loop motion, then closed-loop — and the acceptance criteria that close each step are tracked against the project's validation stage gates. The isolation checks that precede first power-on (rail-to-rail separation, bus-to-chassis, and confirmation that no low-voltage capacitor sits on the pack bus) are recorded in Appendix D as a checklist, because they are performed on every re-assembly and not only once.

8 Control Algorithms & Approach

8.1 What has to be regulated

Balanced on a corner, the cube is an inverted pendulum in every direction perpendicular to the vertical. Linearised about the balancing attitude it has two unstable eigenvalues,

$$\lambda_{1,2} = 8.2572,\ 8.2343\ \mathrm{s}^{-1}, \qquad \lambda = \sqrt{\frac{S_g}{\bar{\mathbf{\Theta}}}}, \tag{14}$$

a divergence time constant of 121 ms. Nothing in the open-loop plant resists displacement: the gravitational moment grows with tilt rather than opposing it. The controller supplies all of the stability there is.

Three properties of the plant set the shape of the control problem, and each is addressed by a specific part of what follows.

1. **The attitude state is an element of $SO(3)$, not a vector.** No three-parameter global parameterisation of $SO(3)$ exists without a singularity, and the operating point sits 53.94° from the orientation in which the geometry was authored, so small-angle reasoning about the parameterisation itself is not available (§8.3).

2. **One rotational degree of freedom is degenerate at both ends.** Gravity produces no moment about the vertical, and a single vector measurement cannot observe rotation about it. The problem is eight-dimensional, not nine (§8.5).

3. **Authority is bounded by stored momentum, not by torque.** The wheels exchange momentum with the body rather than pushing against the world, so recovery consumes wheel momentum and the envelope is set by what the wheels can store before saturating (§8.8).

8.2 Control architecture

The loop runs at 400 Hz in four layers. Each layer is separable, testable in isolation, and was brought up in the order listed.

Layer	Input → output	Function
1. Sensing	IMU, encoders → calibrated body-frame quantities	Accelerometer and gyro rotated into cube body axes by a six-face-verified $\boldsymbol{R}_{IB}$; encoder rates from the three drivers over CAN-FD
2. Estimation	$\boldsymbol{a}, \boldsymbol{\omega} \to \hat{\boldsymbol{\gamma}}, \boldsymbol{\omega}$	Explicit complementary filter with PI correction on the gravity direction; the integral term estimates gyro bias online (§8.6)
3. Control	$\boldsymbol{x} \to \boldsymbol{u}_{\mathrm{LQR}}$	Reduced attitude $\boldsymbol{\phi} = -\hat{\boldsymbol{g}}_B \times \hat{\boldsymbol{\gamma}}$, then a constant 3×9 gain matrix (§8.7)
4. Actuation	$\boldsymbol{u}_{\mathrm{LQR}} \to \boldsymbol{u}$	Friction feedforward, torque clamp, wheel-speed cap; torque-mode command to three moteus drivers (§8.8)

Table 20: The four layers of the control loop.

The complete per-cycle cost is three cross products, one 3×9 matrix–vector product, one reciprocal square root and three `tanhf` calls — roughly 60 multiply–accumulates, with no `atan2` and no trigonometric evaluation. On a 600 MHz Teensy 4.1 the 2.5 ms budget is consumed by CAN traffic, not by the control computation.

8.3 Attitude representation

This is the load-bearing decision of the control design. Three representations were implemented and instrumented before the loop closed; two failed in ways that resembled tuning problems and were not.

8.3.1 Euler angles: three measured failures

The first implementation used a Z–Y–X Euler triple. It failed three times, in three distinct ways.

Failure 1 — the parameterisation went singular, after passing four gates. The Simscape plant was built with a Gimbal joint on defensible grounds: it linearises to a clean 9×9 system comparable directly against the hand-derived $\boldsymbol{A}$ and $\boldsymbol{B}$, the equilibrium gimbal angles are $[0, 0, 0]$ because the mount transform pre-applies the corner tilt, and lock sits roughly 75° away at corner-balance amplitudes. It passed validation gates 1 to 4 on that basis, including the linearisation gate at $\|\Delta \boldsymbol{A}\| = 1.742 \times 10^{-7}$, $\|\Delta \boldsymbol{B}\| = 2.157 \times 10^{-16}$ (measured on the $(+1, +1, +1)$ configuration; see §8.9).

It failed later, in closed-loop simulation, with a diagnostic signature: q_x and q_z smooth throughout, while q_y — the middle gimbal primitive, the one carrying the singularity — spiked to $\pm 20000°$ and returned within milliseconds. A rigid body cannot rotate that far in that interval, and the confinement of the excursion to the middle primitive identifies the cause exactly: the Jacobian relating gimbal rates to body rates diverging, with the solver integrating the divergence.

It also retrospectively explained a persistent 19.8 % error in the gravity block of $\boldsymbol{A}$ that never resolved while $\boldsymbol{B}$ stayed exact at 3×10^{-16}. That asymmetry is the tell: $\boldsymbol{B}$ does not involve the attitude parameterisation and $\boldsymbol{A}$ does, so a discrepancy appearing in one and not the other is a coordinates error rather than a mass-properties or wiring error.

The remedy is a Spherical joint, which carries no chart at the cost of a rank-deficient linearisation.

Failure 2 — an Euler triple is not the reduced attitude. Even far from lock, the triple $[q_1, q_2, q_3]$ is not a rotation vector, so projecting out the vertical component removes yaw only to first order.

On this cube yaw is not small, for a mechanical reason. The inertia about the vertical is

$$\hat{\boldsymbol{g}}_B^\top \boldsymbol{\Theta}\, \hat{\boldsymbol{g}}_B = 5.374 \times 10^{-3}\ \mathrm{kg\,m^2}, \tag{15}$$

against a mean tilt inertia of $2.825 \times 10^{-2}\ \mathrm{kg\,m^2}$ in the plane perpendicular to $\hat{\boldsymbol{g}}_B$ — the cube is 5.3× lighter in yaw than in tilt. Wheel momentum along $\hat{\boldsymbol{g}}_B$ therefore spins the body at $I_s/(\hat{\boldsymbol{g}}_B^\top \boldsymbol{\Theta} \hat{\boldsymbol{g}}_B) = 0.0906\ \mathrm{rad\,s^{-1}}$ per $\mathrm{rad\,s^{-1}}$ of $\boldsymbol{\rho} \cdot \hat{\boldsymbol{g}}_B$. Released from 3.6°, yaw reaches 23.6° within 0.3 s — entirely physical and momentum-conserving, and far outside the range where a first-order yaw removal is valid. Feeding the Euler triple to the controller cost 19.5 % of the recovery angle, and the loop diverged.

Failure 3 — Euler rates are not body rates. Found only after the first two were fixed. The Euler-rate vector $[\dot{q}_1, \dot{q}_2, \dot{q}_3]$ equals the body angular velocity $\boldsymbol{\omega}$ only at $\boldsymbol{q} = \boldsymbol{0}$; over the recovery trajectory the discrepancy reached 73 %. Correcting both the attitude and the rate feedback moved the Simscape recovery from 3.33° to 4.14°, at which point it matched an independently written nonlinear simulator to 0.35 %.

All three are the same mistake in different clothes: a signal correct at the equilibrium was treated as correct everywhere. That is precisely the assumption a balancing controller is not entitled to make, because the controller only acts when the system is away from equilibrium.

8.3.2 The options

Representation	Params	DoF	Disposition
Euler angles	3	3	Rejected. Singular; carries yaw; the triple is not a rotation vector and its rates are not body rates
Quaternion	4 (+1)	3	Rejected for this stage. Global and singularity-free, but carries yaw explicitly, and with one vector measurement that direction is unobservable
Rotation matrix	9 (+6)	3	Rejected. Singularity-free, but three DoF where two are wanted, and nine states to renormalise every cycle
Reduced attitude on S^2	**3 (+1)**	**2**	**Adopted.** Exactly the observable and controllable subspace, and nothing else

Table 21: Attitude representations considered.

8.3.3 Reduced attitude

Let $\hat{\boldsymbol{\gamma}}$ be the unit vector along gravity expressed in the cube body frame, and $\hat{\boldsymbol{g}}_B$ its value at the balanced attitude — a per-corner constant. Then

$$\hat{\boldsymbol{\gamma}} = \mathrm{normalise}(-\boldsymbol{a}_{\mathrm{cal}}) \qquad \text{directly what the accelerometer gives} \tag{16}$$

$$\dot{\hat{\boldsymbol{\gamma}}} = -\boldsymbol{\omega} \times \hat{\boldsymbol{\gamma}} \qquad \text{kinematics, one line, no chart} \tag{17}$$

$$\boldsymbol{\phi} = -\hat{\boldsymbol{g}}_B \times \hat{\boldsymbol{\gamma}} \qquad \text{the entire attitude computation} \tag{18}$$

Three numbers, one norm constraint, two degrees of freedom; $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{g}}_B$ is balanced.

The property that does the work is that $\boldsymbol{\phi}$ in (18) is perpendicular to $\hat{\boldsymbol{g}}_B$ *by construction* — a cross product with $\hat{\boldsymbol{g}}_B$ always is. Yaw therefore cannot enter the tilt channel. Not approximately, not to first order: exactly, at any attitude, at any tilt magnitude. This is the direct repair of Failure 2, where the projection had to be applied as a correction valid only near equilibrium.

Each property was checked against the nonlinear model rather than assumed:

Property	Test	Result
$\boldsymbol{\phi} \perp \hat{\boldsymbol{g}}_B$ exactly	$\lVert \boldsymbol{P}\boldsymbol{\phi} - \boldsymbol{\phi} \rVert$	6.0×10^{-18}
Yaw-immune	$30°$ pure yaw, motion of $\hat{\boldsymbol{\gamma}}$	1.6×10^{-16}
Consistent with the nonlinear plant	$mg(\boldsymbol{r}_c \times \hat{\boldsymbol{\gamma}})$ vs $S_g\boldsymbol{\phi}$	$9.9 \times 10^{-17}\,\mathrm{N\,m}$

Table 22: Verification of the reduced-attitude formulation, all at machine precision. Row 3 is the consistency check between the nonlinear plant used for envelope work and the linear plant used for gain synthesis.

8.3.4 The limits of this choice

Stating them is part of the argument. The correction term $\boldsymbol{e} = \hat{\boldsymbol{g}} \times \hat{\boldsymbol{\gamma}}$ has magnitude sin(error): well conditioned below $90°$, but zero again at $180°$, so an inverted cube can lock onto the antipode. The mitigation is a reset when $\hat{\boldsymbol{g}}\cdot\hat{\boldsymbol{\gamma}} < 0$ for N consecutive cycles. Reduced attitude is also the right choice *because* yaw is irrelevant to corner balancing; jump-up and corner-to-corner locomotion, where $\hat{\boldsymbol{g}}_B$ moves through 67–107°, require large-attitude tracking against a full reference and a quaternion MEKF becomes the appropriate structure. Finally, a magnetometer would not help: it would make yaw observable while leaving it uncontrollable, and the degeneracy is two-sided (§8.5).

8.3.5 Loop rate

The outer loop runs at 400 Hz. The figure is bounded from below by two independent requirements and from above by the CAN-FD budget, and the binding lower bound is not the one usually quoted.

Lower bound 1: discrete stability. Discretising the open-loop plant with a zero-order hold and closing the loop with the gain of §8.7, including the one-cycle computational delay between sampling and actuation, the closed loop leaves the unit circle below

$$f_{\min}^{\mathrm{stab}} = 38.3\ \mathrm{Hz}, \tag{19}$$

which is 29× the open-loop divergence rate $\lambda/2\pi = 1.31$ Hz. The margin above this floor is large but the floor itself is not the design driver.

The delay term matters more than the sampling. Without it the discretised loop remains stable down to well under 20 Hz; with it the floor rises to 38.3 Hz. Sample-and-actuate latency, not sample rate as such, is what consumes phase.

Lower bound 2: anti-aliasing. The accelerometer sees structural and wheel-borne content well above the control bandwidth. Any of it above $f/2$ folds down into the estimator passband, where it is indistinguishable from tilt and cannot be removed by any subsequent filtering. Requiring the folding frequency to sit clear of the sensor's useful band gives

$$f_{\min}^{\mathrm{alias}} = 281\ \mathrm{Hz}, \tag{20}$$

which exceeds (19) by a factor of 7.3 and is therefore the binding constraint. *The loop rate is set by the sensor, not by the controller.* This is worth stating explicitly because the stability floor is the quantity usually computed and it is, here, an order of magnitude too permissive.

Upper bound: cycle budget and rate quantisation. At 400 Hz the cycle budget is 2.5 ms, which must accommodate an IMU read over SPI, three encoder and torque exchanges over CAN-FD, and the control computation. The control computation is negligible — roughly 60 multiply–accumulates (§8.2) — so the budget is dominated by bus traffic, and this is the constraint that closes the interval from above.

Encoder quantisation contributes a second, weaker upper bound. Differencing a 16-bit absolute encoder gives a rate noise floor of LSB × f = $9.587 \times 10^{-5} \times 400 = 0.038\,\mathrm{rad\,s^{-1}}$, 0.10 % of the wheel-speed cap, and it scales linearly with f. At 400 Hz this is immaterial; it is recorded because it is the term that would eventually penalise a faster loop.

What 400 Hz buys. Discretisation costs closed-loop damping, and the cost is a smooth function of rate rather than a threshold:

At 400 Hz the loop retains 95 % of the continuous design's damping. Doubling to 800 Hz recovers a further three points and halves the cycle budget: a poor trade against the bus. Below 100 Hz the loss becomes steep, and by 40 Hz — still nominally stable — only 13 % of the damping survives, which is the practical rather than the nominal floor.

400 Hz therefore sits 1.4× above the binding anti-aliasing bound, 10× above the stability floor, and comfortably inside the CAN-FD budget, and it is verified against the discretised plant rather

f [Hz]	$\max\lvert z\rvert$	equivalent $\mathrm{Re}(s)$ [rad s^{-1}]	% of continuous
40	0.97503	-1.01	13 %
50	0.88308	-6.22	80 %
100	0.93478	-6.74	86 %
200	0.96480	-7.17	92 %
281	0.97427	-7.33	94 %
400	**0.98153**	$\mathbf{-7.46}$	**95 %**
800	0.99051	-7.63	98 %

Table 23: Discrete closed-loop performance against loop rate, one-cycle computational delay included, marginal modes excluded. The continuous design places the slowest controlled mode at $-7.82\,\mathrm{rad\,s^{-1}}$.

than assumed from the continuous design. *The return-difference identity $M_s = 1$ of §8.9 does not survive discretisation*, so the discrete poles are checked directly at the design rate and are not inferred from the continuous margin.

8.4 Plant model and linearisation

8.4.1 Modelling assumptions

Four assumptions define the model. Each is stated with the reason it is admissible and the circumstance under which it would fail.

1. **Fixed point contact, no slip and no bounce.** The cube rotates about a stationary point at the balancing corner. The corner truncation on this build was restored to a point, so the contact is the exact geometric corner and no contact offset is required. Even the original 0.45 mm edge cut amounted to only 0.26 mm along the diagonal — an angular contribution of 0.081°, below the estimator noise floor. The assumption fails at jump-up, where contact is impulsive and intermittent.
2. **Rigid bodies throughout.** Frame and wheels are rigid, with no structural compliance. Admissible because the lowest structural mode is well above the control bandwidth of §8.3.5; it is the assumption that would be challenged first if the printed frame were made lighter.
3. **Wheels axisymmetric about their spin axes.** This is stronger than it appears: it makes $\boldsymbol{\Theta}$ *body-constant*. The inertia tensor does not depend on wheel *angle*, only on wheel *rate* through the momentum term. Without it the plant would be time-varying in the wheel angles and no constant-gain design would be available.
4. **Pivot friction neglected in the design plant.** Retained in the nonlinear model of §8.8 but dropped from the linearisation, where it is second order about the equilibrium. The coefficient is a placeholder pending a breakaway test.

8.4.2 Nonlinear equations of motion

Let O be the contact corner, $\boldsymbol{\omega}$ the body angular velocity about O, $\boldsymbol{\rho}$ the three wheel rates *relative to the body*, $\boldsymbol{A}_w$ the matrix whose columns are the wheel spin axes, I_s the wheel inertia about its own spin axis, $\boldsymbol{\Theta}$ the inertia of the whole assembly about O with all bodies rigid, $\boldsymbol{r}_c$ the vector from O to the centre of mass, and $\hat{\boldsymbol{\gamma}}$ the gravity direction in body coordinates (§8.3.3).

The angular momentum about O, expressed in body axes, is

$$\boldsymbol{L} = \boldsymbol{\Theta}\boldsymbol{\omega} + \boldsymbol{A}_w I_s \boldsymbol{\rho}. \tag{21}$$

The first term is the momentum the assembly would have if the wheels were locked; the second accounts for their relative spin. Euler's equation in the rotating body frame gives

$$\dot{\boldsymbol{L}} + \boldsymbol{\omega} \times \boldsymbol{L} = \boldsymbol{M}_{\text{ext}}, \qquad \boldsymbol{M}_{\text{ext}} = mg\,(\boldsymbol{r}_c \times \hat{\boldsymbol{\gamma}}) + \boldsymbol{\tau}_{\text{pivot}}. \tag{22}$$

Each wheel obeys

$$I_s(\hat{\boldsymbol{a}}_k \cdot \dot{\boldsymbol{\omega}} + \dot{\rho}_k) = u_k, \tag{23}$$

where u_k is the motor torque applied to wheel k. Solving (23) for $I_s\dot{\boldsymbol{\rho}}$ and substituting into the time derivative of (21) eliminates $\dot{\boldsymbol{\rho}}$ from the body equation:

$$\boldsymbol{\Theta}\dot{\boldsymbol{\omega}} + \boldsymbol{A}_w\boldsymbol{u} - \boldsymbol{A}_w I_s \boldsymbol{A}_w^\top\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \boldsymbol{L} = \boldsymbol{M}_{\text{ext}}, \tag{24}$$

and defining the *reduced inertia*

$$\bar{\boldsymbol{\Theta}} \;=\; \boldsymbol{\Theta} - \boldsymbol{A}_w I_s \boldsymbol{A}_w^\top \tag{25}$$

the equations of motion close as

$$\bar{\boldsymbol{\Theta}}\,\dot{\boldsymbol{\omega}} = mg\,(\boldsymbol{r}_c \times \hat{\boldsymbol{\gamma}}) \;-\; \boldsymbol{u} \;-\; \boldsymbol{\omega} \times \boldsymbol{L} \;-\; \boldsymbol{\tau}_{\text{pivot}} \tag{26}$$

$$I_s\dot{\boldsymbol{\rho}} = \boldsymbol{u} - I_s\boldsymbol{A}_w^\top\dot{\boldsymbol{\omega}} \tag{27}$$

$$\dot{\hat{\boldsymbol{\gamma}}} = -\,\boldsymbol{\omega} \times \hat{\boldsymbol{\gamma}} \tag{28}$$

Three features of (26)–(28) govern everything that follows.

Momentum Exchange. The wheels exchange momentum with the body; they do not generate it. Torque does not push against the world, and $\boldsymbol{L}$ is conserved under $\boldsymbol{u}$ alone. This is the origin of the momentum-limited envelope (§8.8).

Reduced Inertia. The wheel inertia about its own axis is subtracted because that inertia is already accounted for in (27); leaving it in $\boldsymbol{\Theta}$ would double-count it. On this build $\bar{\boldsymbol{\Theta}}$ differs from $\boldsymbol{\Theta}$ by 4.87×10^{-4} on each diagonal entry, about 2.4 % — small, but it shifts λ by 1.2 % and it is not optional.

Gyroscopic Term. $\boldsymbol{\omega} \times \boldsymbol{L}$ is quadratic and vanishes at the equilibrium, so it plays no part in the gain synthesis. It is retained in the nonlinear simulator used for the envelope work, and it is the reason the envelope results of §8.10 are simulated rather than computed in closed form.

8.4.3 The state vector

Throughout the remainder of this chapter the state is

$$\boxed{\boldsymbol{x} = \begin{bmatrix}\boldsymbol{\phi}\\ \boldsymbol{\omega}\\ \boldsymbol{\rho}\end{bmatrix} \in \mathbb{R}^9, \qquad \boldsymbol{\phi} = -\,\hat{\boldsymbol{g}}_B \times \hat{\boldsymbol{\gamma}}} \tag{29}$$

— reduced attitude, body rate, wheel rate, each in body axes X, Y, Z in that order. $\boldsymbol{\phi}$ and $\boldsymbol{\omega}$ come from the estimator of §8.6; $\boldsymbol{\rho}$ comes directly from the encoders and needs no estimation.

The interpretation of $\boldsymbol{\phi}$ deserves a line, because it is what makes the projector $\boldsymbol{P}$ appear later. For a small rotation $\delta\boldsymbol{\theta}$ away from the balanced attitude, $\hat{\boldsymbol{\gamma}} \approx \hat{\boldsymbol{g}}_B - \delta\boldsymbol{\theta} \times \hat{\boldsymbol{g}}_B$, and therefore

$$\begin{aligned}\boldsymbol{\phi} &= -\hat{\boldsymbol{g}}_B \times (\hat{\boldsymbol{g}}_B - \delta\boldsymbol{\theta} \times \hat{\boldsymbol{g}}_B) = \hat{\boldsymbol{g}}_B \times (\delta\boldsymbol{\theta} \times \hat{\boldsymbol{g}}_B)\\ &= \delta\boldsymbol{\theta}\,(\hat{\boldsymbol{g}}_B\cdot\hat{\boldsymbol{g}}_B) - \hat{\boldsymbol{g}}_B\,(\hat{\boldsymbol{g}}_B\cdot\delta\boldsymbol{\theta}) = \boldsymbol{P}\,\delta\boldsymbol{\theta}, \qquad \boldsymbol{P} = \boldsymbol{I} - \hat{\boldsymbol{g}}_B\hat{\boldsymbol{g}}_B^\top.\end{aligned} \tag{30}$$

$\boldsymbol{\phi}$ is the rotation vector with its vertical component removed. It is not an approximation of the attitude error; it is exactly the part of the attitude error that gravity can act on and that an accelerometer can see.

8.4.4 Linearisation

Gravitational Term. Because $\boldsymbol{r}_c = -\ell\,\hat{\boldsymbol{g}}_B$ is body-fixed,

$$mg\,(\boldsymbol{r}_c \times \hat{\boldsymbol{\gamma}}) = -mg\ell\,(\hat{\boldsymbol{g}}_B \times \hat{\boldsymbol{\gamma}}) = S_g\,\boldsymbol{\phi}, \qquad S_g = mg\ell, \tag{31}$$

with no small-angle step anywhere. Substituting (30) gives the form used in the design plant, $S_g\boldsymbol{P}\,\delta\boldsymbol{\theta}$, and since $\boldsymbol{P}$ is idempotent and $\boldsymbol{\phi}$ already lies in its range, $S_g\boldsymbol{\phi} = S_g\boldsymbol{P}\boldsymbol{\phi}$.

This identity is the reason the linear plant used for gain synthesis and the nonlinear plant used for envelope work can be compared at all. It was checked numerically rather than assumed: the two agree to 9.9×10^{-17} N m (§8.3.3). The controller is not approximating the plant's attitude behaviour — the two are the same object.

The design plant. Dropping the gyroscopic and pivot-friction terms, which are second order about the equilibrium, and taking $\boldsymbol{A}_w = \boldsymbol{I}_3$ since the wheel axes are the cube axes:

$$\bar{\boldsymbol{\Theta}}\,\dot{\boldsymbol{\omega}} = S_g\boldsymbol{P}\boldsymbol{\phi} - \boldsymbol{u}, \qquad I_s\dot{\boldsymbol{\rho}} = \boldsymbol{u} - I_s\dot{\boldsymbol{\omega}}. \tag{32}$$

In the ordering (29) this is $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$ with

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{G} & \boldsymbol{0} & \boldsymbol{0} \\ -\boldsymbol{G} & \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}, \qquad \boldsymbol{B} = \begin{bmatrix} \boldsymbol{0} \\ -\bar{\boldsymbol{\Theta}}^{-1} \\ I_s^{-1}\boldsymbol{I} + \bar{\boldsymbol{\Theta}}^{-1} \end{bmatrix}, \qquad \boldsymbol{G} = \bar{\boldsymbol{\Theta}}^{-1}S_g\boldsymbol{P}. \tag{33}$$

Three structural remarks.

First, the third block row is $-\boldsymbol{G}$, an exact copy of the second with the sign reversed. That is momentum exchange written in state-space form: whatever angular acceleration gravity imparts to the body appears with opposite sign on the wheels.

Second, the first block row is $\dot{\boldsymbol{\phi}} = \boldsymbol{\omega}$, not $\boldsymbol{P}\boldsymbol{\omega}$. The component of $\boldsymbol{\phi}$ along $\hat{\boldsymbol{g}}_B$ therefore accumulates with yaw rate. It is harmless, because $\boldsymbol{G}$ annihilates it — $\boldsymbol{P}\hat{\boldsymbol{g}}_B = \boldsymbol{0}$ exactly — so it never feeds back into the dynamics. It is nevertheless removed from the control law by the projection of §8.5, because a state that drifts without bound should not multiply a gain.

Third, $\boldsymbol{P}$ has rank 2. That single fact generates both structural degeneracies of §8.5 and is the reason the Riccati problem must be posed on eight states rather than nine.

Numerical values. With the measured parameters of §8.4.5, in SI units:

$$\bar{\boldsymbol{\Theta}} = \begin{bmatrix} 0.020384 & -0.0076938 & -0.0074549 \\ -0.0076938 & 0.019648 & -0.0077143 \\ -0.0074549 & -0.0077143 & 0.020379 \end{bmatrix} \text{kg m}^2, \tag{34}$$

$$\hat{\boldsymbol{g}}_B = \begin{bmatrix} -0.571549 \\ -0.588646 \\ -0.571688 \end{bmatrix}, \qquad \boldsymbol{P} = \begin{bmatrix} 0.67333 & -0.33644 & -0.32675 \\ -0.33644 & 0.65350 & -0.33652 \\ -0.32675 & -0.33652 & 0.67317 \end{bmatrix}, \tag{35}$$

$$\boldsymbol{G} = \begin{bmatrix} 45.202 & -22.008 & -22.530 \\ -23.502 & 45.528 & -23.382 \\ -22.623 & -21.984 & 45.254 \end{bmatrix} \text{s}^{-2}. \tag{36}$$

$\boldsymbol{G}$ is not symmetric — it is $\bar{\boldsymbol{\Theta}}^{-1}$ times a symmetric matrix — and its eigenvalues are

$$\text{eig}(\boldsymbol{G}) = \{68.1814,\ 67.8029,\ 0\}\ \text{s}^{-2}, \tag{37}$$

whose square roots are the unstable poles of (14),

$$\lambda_{1,2} = \sqrt{68.1814},\ \sqrt{67.8029} = 8.2572,\ 8.2343\ \mathrm{s}^{-1}, \tag{38}$$

and whose third, exactly zero, eigenvalue is the degenerate vertical direction. The two unstable values differ by 0.28 %: the cube is very nearly isotropic about its balancing diagonal, and for comparison a uniform cube of the same size gives $0.99\sqrt{g/a} = 8.01\ \mathrm{s}^{-1}$.

The lower blocks of $\boldsymbol{B}$ are

$$-\bar{\boldsymbol{\Theta}}^{-1} = \begin{bmatrix} -90.525 & -56.908 & -54.656 \\ -56.908 & -95.556 & -56.988 \\ -54.656 & -56.988 & -90.635 \end{bmatrix}, \quad I_s^{-1}\boldsymbol{I} + \bar{\boldsymbol{\Theta}}^{-1} = \begin{bmatrix} 2144.4 & 56.908 & 54.656 \\ 56.908 & 2149.5 & 56.988 \\ 54.656 & 56.988 & 2144.5 \end{bmatrix}. \tag{39}$$

The ratio of the diagonal entries, $2144/90.5 \approx 24$, is the inertia ratio $\bar{\boldsymbol{\Theta}}/I_s$ and is the quantitative statement of why the envelope is momentum-limited: a torque that accelerates the body by $1\ \mathrm{rad\,s^{-2}}$ accelerates the wheel by 24.

8.4.5 The measured plant

Every entry below is a measurement or an explicit arithmetic consequence of measurements. The two assumed quantities are named as such.

Quantity	Symbol	Value	Origin
Total mass	m	1.633 kg	weighed; CAD 1.6185, −3.2 %
Wheel mass (each)	–	0.216 kg	weighed
Ballast (each)	–	0.130 kg	weighed
Centre of mass	–	$[-4.790,\ -2.690,\ -4.773]$ mm	from weighed groups
Balancing corner	–	$(-1,-1,-1)$	§8.4.6
Lever arm	ℓ	122.84 mm	$\lVert \boldsymbol{r}_{\mathrm{com}} - \boldsymbol{r}_{\mathrm{corner}} \rVert$
Gravitational moment	S_g	1.8875 N m	$mg\ell$
Wheel spin inertia	I_s	$4.8688 \times 10^{-4}\ \mathrm{kg\,m^2}$	CAD −8.7 % after mass correction
Wheel transverse	I_t	$2.4804 \times 10^{-4}\ \mathrm{kg\,m^2}$	
Unstable poles	λ	$8.2572,\ 8.2343\ \mathrm{s^{-1}}$	$\sqrt{\mathrm{eig}(\boldsymbol{G})}$
Yaw inertia	$\hat{\boldsymbol{g}}_B^\top \boldsymbol{\Theta} \hat{\boldsymbol{g}}_B$	$5.374 \times 10^{-3}\ \mathrm{kg\,m^2}$	
Tilt-plane inertia	–	$2.769,\ 2.784 \times 10^{-2}\ \mathrm{kg\,m^2}$	eigenvalues of $\bar{\boldsymbol{\Theta}}$ on range($\boldsymbol{P}$)
Wheel axis offset	–	$(+7,+7)$ mm from the cube axes	CAD, battery clearance
Axial station	d_{ax}	−61.6 mm	CAD
Momentum cap	h_{cap}	0.019475 N m s	$I_s\omega_{\mathrm{cap}}$ at $40\ \mathrm{rad\,s^{-1}}$
Mount rotation	–	53.9390° about $[-0.7072,\ 0,\ +0.7070]$	from $\hat{\boldsymbol{g}}_B$
Equilibrium tilt	–	0.797°	COM offset ⊥ the diagonal
Assumed, not measured:			
Continuous torque	τ_{cont}	0.12 N m	estimate; thermal test outstanding
Print density split	ρ_{scale}	0.84	assumed; λ moves 0.5 % across the band

Table 24: Measured cube plant, balancing corner $(-1,-1,-1)$.

Two entries in table 24 carry more weight than their size suggests.

Wheel Axis Location. Each wheel's spin axis is offset $(+7,+7)$ mm from the corresponding cube axis — battery clearance along the body diagonal. The axis position is *not* $[d, 0, 0]$, and modelling it that way misplaces the joint. The parameter file keeps the axis location and the wheel centre of mass as separate named quantities even though they coincide numerically to 10^{-8} mm, precisely so that an unbalanced wheel would move one and not the other rather than silently corrupting the joint position.

Centre-of-mass Offset. The offset from the geometric centre is mm, but the quantity that sets the equilibrium tilt is its projection perpendicular to the balancing diagonal, 1.708 mm. That figure also sits below the ∼2 mm threshold at which the cube would cease to rest on its corner truncation, so multi-corner operation remains geometrically viable.

8.4.6 Choice of balancing corner

The two ends of the balancing diagonal are dynamically near-equivalent:

Corner	ℓ	S_g	λ	θ_{eq}
$(-1,-1,-1)$ **(balancing)**	122.84 mm	1.8875 N m	8.2572 s^{-1}	0.797°
$(+1,+1,+1)$	136.99 mm	2.1048 N m	7.9341 s^{-1}	0.714°
the other six	126.1–134.0 mm	—	—	2.61–3.17°

Table 25: All eight corners. The six off-diagonal corners carry three to four times the equilibrium tilt because the centre-of-mass offset is not aligned with their diagonals.

$(-1,-1,-1)$ was selected on mechanical grounds: it is the corner *between* the wheels, and its truncation has been restored to a point so that the contact is the exact geometric corner.

The choice also happens to help, and the reason is worth stating because it runs opposite to intuition. $(-1,-1,-1)$ has the *shorter* lever arm and therefore the *smaller* destabilising moment S_g, which gives it the larger recoverable envelope — 3.88° against 3.42° on $(+1,+1,+1)$ (§8.10). The corner with the marginally lower equilibrium tilt is the corner with the smaller envelope, and the envelope is worth more.

8.4.7 From one degree of freedom to three

The Stage-1 planar panel is the same plant with every matrix collapsed to a scalar, and the correspondence is close enough to be useful as a check.

What generalises. The block structure of (33) survives intact with scalars promoted to matrices; the unstable-pole relation $\lambda = \sqrt{S_g/\bar{\Theta}}$ survives with $\bar{\boldsymbol{\Theta}}$ read on the tilt plane; the wheel remains a pure integrator, so a wheel-rate gain remains necessary; and the Bryson weights were carried from panel to cube *unchanged*. That last was deliberate and is a debugging strategy rather than an economy — same physics, same weights, so any surprise in the cube result is located in the plant and not in the tuning.

What breaks. Four things, in descending order of consequence.

1. **Gravity acquires a projector.** The scalar S_g becomes $S_g\boldsymbol{P}$ of rank 2, and with it come the two structural degeneracies of §8.5. There is no one-dimensional analogue; it must be handled by an exact state-space reduction, not by tuning.
2. **Attitude stops being a number**, with the consequences set out in §8.3.
3. **The wheel axes acquire an offset** that the obvious model does not have.
4. **One accelerometer error term stops cancelling.** The centripetal term is quadratic in $\boldsymbol{\omega}$ and always directed inward, so unlike the tangential term it has a DC component and does not average away over a completed recovery (§8.6).

And the envelope shrinks. The panel recovers from roughly 11.5°, the cube from roughly 3.9°. The cube is the larger and more capable machine and has the smaller envelope, which invites the question if left unaddressed. Taking the momentum-limited bound $h_{\text{cap}}^2/(2\bar{\Theta}S_g)$, which is approximately $1-\cos\theta_{\max}$ for a release from rest:

	Panel	**Cube**	**Ratio**
Stored momentum h_{cap}	7.432×10^{-3}	$1.9475\times10^{-2}\,\mathrm{N\,m\,s}$	$\times2.62$
Gravitational moment S_g	0.2624	$1.8875\,\mathrm{N\,m}$	$\times7.19$
Tilt inertia $\bar{\Theta}$	3.213×10^{-3}	$2.776\times10^{-2}\,\mathrm{kg\,m^2}$	$\times8.64$
$h_{\text{cap}}^2/(2\bar{\Theta}S_g)$	3.275×10^{-2}	3.618×10^{-3}	$\div\,9.05$
Implied bound $\theta_{\max}$	14.7°	4.88°	

Table 26: Momentum capacity grew by 2.6× between the two stages while the load it must arrest grew by 7 to 9×.

The bound sits above both simulated figures, as it must, since it accounts for neither the torque limit nor friction. **The wheels did not scale with the machine**, and no amount of gain tuning recovers that.

8.5 Structural properties: what cannot be controlled and what cannot be seen

The plant of (33) has nine states, but the control problem is eight-dimensional. One rotational degree of freedom — rotation about the vertical — is degenerate, and it is degenerate at *both* ends: it cannot be driven, and it cannot be observed. The two facts have different causes and different consequences, and both must be handled explicitly.

This is not a numerical near-degeneracy to be regularised away. Both statements hold exactly, to machine precision, and both follow from a single line of the model.

8.5.1 Yaw momentum is conserved: the uncontrollable direction

Derivation. Let $\boldsymbol{L}=\boldsymbol{\Theta}\boldsymbol{\omega}+I_s\boldsymbol{\rho}$ be the angular momentum about the contact point. Its component along the vertical evolves as

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(\hat{\boldsymbol{g}}_B\cdot\boldsymbol{L}) &= \hat{\boldsymbol{g}}_B^\top(\boldsymbol{\Theta}\dot{\boldsymbol{\omega}}+I_s\dot{\boldsymbol{\rho}})=\hat{\boldsymbol{g}}_B^\top(\boldsymbol{\Theta}\dot{\boldsymbol{\omega}}+\boldsymbol{u}-I_s\dot{\boldsymbol{\omega}})\\ &= \hat{\boldsymbol{g}}_B^\top\big(\bar{\boldsymbol{\Theta}}\dot{\boldsymbol{\omega}}+\boldsymbol{u}\big)=\hat{\boldsymbol{g}}_B^\top(S_g\boldsymbol{P}\boldsymbol{\phi}-\boldsymbol{u}+\boldsymbol{u})=S_g\,\hat{\boldsymbol{g}}_B^\top\boldsymbol{P}\,\boldsymbol{\phi}=0,\end{aligned}\tag{40}$$

because $\boldsymbol{P}\hat{\boldsymbol{g}}_B=\boldsymbol{0}$ by construction. The control $\boldsymbol{u}$ cancels identically between the body and wheel equations — which is exactly the statement that the wheels are internal. Verified numerically on random states and inputs: $\hat{\boldsymbol{g}}_B^\top(\boldsymbol{\Theta}\dot{\boldsymbol{\omega}}+I_s\dot{\boldsymbol{\rho}})=6.7\times10^{-16}$, and $\hat{\boldsymbol{g}}_B^\top S_g\boldsymbol{P}=3.4\times10^{-16}$.

Physically: gravity produces no moment about the vertical, and the wheels can only exchange momentum with the body, not with the world. Nothing in the plant can change $\hat{\boldsymbol{g}}_B\cdot\boldsymbol{L}$. A yaw kick therefore persists indefinitely in the frictionless model. On hardware, contact friction at the corner supplies an external yaw torque and it bleeds away, so the model is the pessimistic case.

The numerical evidence. The controllability matrix has

so $\operatorname{rank}\mathcal{C}=8$ of 9. The left singular vector of the vanishing direction aligns with the predicted conserved quantity to

$$\left|\cos\angle\left(\boldsymbol{v}_{\text{unc}},\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{\Theta}\hat{\boldsymbol{g}}_B\\ I_s\hat{\boldsymbol{g}}_B\end{bmatrix}\right)\right|=1.0000000000,\tag{41}$$

σ_i/σ_1	1.00	9.72×10^{-1}	1.04×10^{-2}	1.01×10^{-2}	2.05×10^{-6}	1.31×10^{-6}	1.31×10^{-6}	1.85×10^{-7}	$\mathbf{9.19\times10}$

Table 27: Normalised singular values of $\mathcal{C}(\boldsymbol{A}, \boldsymbol{B})$. The gap between the eighth and ninth spans ten orders of magnitude.

which is what converts a rank deficiency into a physical statement. A rank count alone would only say that *some* direction is uncontrollable; (41) identifies it as the vertical component of angular momentum, and that identification is what licenses removing it rather than regularising it.

The Popov–Belevitch–Hautus test gives the same result in the form ETH used [**gajamohan2013cubli**]: $\operatorname{rank}[\boldsymbol{A} - s\boldsymbol{I},\ \boldsymbol{B}] = 8$ at $s = 0$, with the left null vector aligned with the same direction to ten figures, and $\boldsymbol{w}^\top \boldsymbol{A} = 1.9 \times 10^{-13}$, $\boldsymbol{w}^\top \boldsymbol{B} = 3.3 \times 10^{-13}$.

Stabilisability. The system is *stabilisable*, not controllable, and stabilisability is sufficient here for a specific reason: the uncontrollable mode sits at exactly $s = 0$. It is marginally stable, it does not grow, and balancing requires only the two tilt directions. An uncontrollable mode in the right half-plane would end the project; one at the origin is a bookkeeping problem.

8.5.2 Yaw angle is unobservable: the second degeneracy

The second degeneracy is independent of the first and has a different cause. An accelerometer measures the direction of gravity. One vector measurement determines two rotational degrees of freedom, not three: it tells you which way is down, and it never tells you which way you are facing.

With the full state available, $\operatorname{rank}\mathcal{O} = 9$ of 9. With the reduced-attitude measurement actually available on this machine, $\operatorname{rank}\mathcal{O} = 8$ of 9, and the unobservable direction is

$$\boldsymbol{v}_{\text{unobs}} = \begin{bmatrix} -0.5715 & -0.5886 & -0.5717 & 0 & \cdots & 0 \end{bmatrix}^\top, \tag{42}$$

which is $[\hat{\boldsymbol{g}}_B;\ \mathbf{0};\ \mathbf{0}]$ to ten figures — the yaw *angle*, and nothing else. Yaw *rate* is fully observable: the gyroscope measures it directly. It is the integral that cannot be recovered.

	Uncontrollable	**Unobservable**
Quantity	$\hat{\boldsymbol{g}}_B \cdot \boldsymbol{L}$, yaw momentum	yaw angle
Direction	$[\mathbf{0};\ \boldsymbol{\Theta}\hat{\boldsymbol{g}}_B;\ I_s\hat{\boldsymbol{g}}_B]$	$[\hat{\boldsymbol{g}}_B;\ \mathbf{0};\ \mathbf{0}]$
Cause	wheels internal, $\boldsymbol{P}\hat{\boldsymbol{g}}_B = \mathbf{0}$	one vector measurement gives 2 DoF
Remedy	8-state reduction before the Riccati solve	$\boldsymbol{K}_p = \boldsymbol{K}\,\operatorname{blkdiag}(\boldsymbol{P}, \boldsymbol{I}, \boldsymbol{I})$

Table 28: The two degeneracies are distinct and require distinct treatment.

Magnetometer. It would make yaw observable while leaving it uncontrollable. Removing one side of a two-sided degeneracy buys nothing, and it adds a sensor with a hard-iron calibration problem inside a cube containing three motors.

8.5.3 Exact Eight-state Reduction

Let $\boldsymbol{v} = [\mathbf{0};\ \boldsymbol{\Theta}\hat{\boldsymbol{g}}_B;\ I_s\hat{\boldsymbol{g}}_B]$ and let $\boldsymbol{V}_2 \in \mathbb{R}^{9\times8}$ be an orthonormal basis for $\boldsymbol{v}^\perp$. The design is performed on

$$\boldsymbol{A}_r = \boldsymbol{V}_2^\top \boldsymbol{A} \boldsymbol{V}_2, \quad \boldsymbol{B}_r = \boldsymbol{V}_2^\top \boldsymbol{B}, \quad \boldsymbol{Q}_r = \boldsymbol{V}_2^\top \boldsymbol{Q} \boldsymbol{V}_2, \qquad \boldsymbol{K} = \boldsymbol{K}_r \boldsymbol{V}_2^\top. \tag{43}$$

The reduced pair is fully controllable, $\operatorname{rank}\mathcal{C}(\boldsymbol{A}_r, \boldsymbol{B}_r) = 8$ of 8, and the resulting gain places no authority on the conserved direction: $\|\boldsymbol{K}\boldsymbol{v}\|/\|\boldsymbol{v}\| = 5.5 \times 10^{-16}$.

Nine-state Formulation. Posing the same problem on all nine states:

	9 states	**8 states**
`icare` report	2 — returns **empty**	0 — success
Hamiltonian eigenvalues on the imaginary axis	**2**	**0**
$\|\boldsymbol{K}\boldsymbol{v}\|/\|\boldsymbol{v}\|$	1.68×10^{-1}	5.5×10^{-16}
$\text{cond}(\boldsymbol{X})$	—	2.00×10^{6}
`lqr`	**succeeds silently**	succeeds

Table 29: The nine-state Riccati problem is ill-posed. `icare` says so; `lqr` does not.

The Hamiltonian row is the diagnostic. A well-posed algebraic Riccati problem has no eigenvalues on the imaginary axis; each one is a mode the stable and anti-stable invariant subspaces cannot separate. At nine states there are two — the conserved direction and its adjoint. At eight there are none.

Silent Failure in LQR. `lqr` returns a gain matrix without warning, and that gain places $\|\boldsymbol{K}\boldsymbol{v}\| = 0.168$ on a direction the actuators cannot move. The controller then reads a real, drifting, measurable quantity — accumulated yaw momentum — and commands torque in response to it. That torque cannot change the quantity it is responding to, so its entire effect is a disturbance to the two directions that *do* respond. Authority is spent to no purpose, and the loss is invisible in any check that only asks whether the closed loop is stable.

A design that runs without error is not the same as a design that is well posed. This is the clearest instance in the project of a check mattering more than a result.

8.5.4 The yaw-angle projection

The reduction of (43) removes the uncontrollable direction. It does not remove the unobservable one, because the two are different directions — compare the rows of table 28. A second, separate step is required.

Recall from §8.4.4 that the first block row of $\boldsymbol{A}$ is $\dot{\boldsymbol{\phi}} = \boldsymbol{\omega}$ rather than $\boldsymbol{P}\boldsymbol{\omega}$, so the component of $\boldsymbol{\phi}$ along $\hat{\boldsymbol{g}}_B$ accumulates with yaw rate without bound. It is harmless to the *dynamics*, because $\boldsymbol{G}$ annihilates it. It is not harmless in the *control law*, because the reduced gain places $\|\boldsymbol{K}[\hat{\boldsymbol{g}}_B; \mathbf{0}; \mathbf{0}]\| = 0.346$ on it — and that component of the estimate is not a measurement of anything. It is integrated gyro drift.

The remedy is to project it out of the gain itself:

$$\boxed{\boldsymbol{K}_p = \boldsymbol{K}\,\text{blkdiag}(\boldsymbol{P}, \boldsymbol{I}, \boldsymbol{I})} \tag{44}$$

after which the gain on that direction is 4.97×10^{-16}. **Use $\boldsymbol{K}_p$ in firmware, not $\boldsymbol{K}$.**

Gain	**Marginal modes**	**Slowest controlled**	**Interpretation**
$\boldsymbol{K}$	1	$-4.2594\,\text{s}^{-1}$	conserved momentum only
$\boldsymbol{K}_p$	2	$\mathbf{-7.8167}\,\text{s}^{-1}$	momentum, plus yaw angle drifting to a constant

Table 30: Both counts are physically correct. Two marginal modes under $\boldsymbol{K}_p$ is the expected and desired outcome.

Marginal mode Influence. The second column is the part worth pausing on. Under $\boldsymbol{K}$ the yaw angle is driven to zero at $4.26\,\text{s}^{-1}$; under $\boldsymbol{K}_p$ it is left to drift. Removing that control action makes the *slowest controlled mode faster*, from -4.26 to $-7.82\,\mathrm{s}^{-1}$ — a factor of 1.8. The projection is not a cosmetic tidying of a harmless term: authority previously spent chasing an unobservable state is returned to the states that matter, and the loop is measurably better for it.

8.5.5 A tolerance trap in the rank test

The entire argument above rests on a rank count, so it is worth recording how that count can be got wrong.

Call		Result
`rank(Co)`	(default, relative tolerance)	**8** — correct
`rank(Co, 1e-9)`	(absolute tolerance)	9 — **wrong**

$\|\mathcal{C}\| = 1.105 \times 10^9$, so an absolute tolerance of 10^{-9} is 9×10^{-19} of the matrix scale — far below the double precision floor, and therefore no threshold at all. Every singular value, including the one at 9.19×10^{-17} that *is* numerically zero, counts as nonzero.

The failure is silent and directional: it reports the system as fully controllable, which is the answer that requires no further work. Had it been believed, the reduction of §8.5.3 would never have been written, `lqr` would have returned a gain without complaint, and the first symptom would have appeared on hardware as authority quietly leaking into a direction that does not move.

A tolerance must be chosen relative to the scale of the matrix it is applied to. MATLAB's default does this; a hand-supplied absolute figure carried over from another problem does not.

8.6 State estimation

8.6.1 The accelerometer measurement problem

An accelerometer measures specific force, not gravity: the difference between the acceleration of its case and free fall. Stationary, the case is supported at $1\,g$ and the reading corresponds to the gravity direction. Under any other acceleration it does not.

For a body rotating about a fixed pivot with the sensor at distance d, the case experiences a tangential acceleration $\ddot{\theta}\,d$ and a centripetal acceleration $\dot{\theta}^2 d$, giving in body axes

$$f_x = \ddot{\theta}\,d + g\sin\theta, \qquad f_y = g\cos\theta - \dot{\theta}^2 d, \tag{45}$$

so the direct readout $\hat{\theta} = \mathrm{atan2}(f_x, f_y)$ is corrupted by both terms.

The corruption is correlated with the manoeuvre rather than random. At rest $\ddot{\theta} = 0$ and the measurement is exact; at maximum control effort $\ddot{\theta}$ peaks and the error is largest. Table 31 gives the magnitude for this cube with the IMU at the geometric centre, $d = 130\,\mathrm{mm}$ from any corner.

Tilt	$\ddot{\theta}$ free	$\ddot{\theta}$ saturated	Apparent tilt error
1.00°	1.19	$7.30\,\mathrm{rad\,s^{-2}}$	0.90° / 5.54°
2.00°	2.37	$8.49\,\mathrm{rad\,s^{-2}}$	1.80° / 6.44°
3.00°	3.56	$9.67\,\mathrm{rad\,s^{-2}}$	2.70° / 7.35°
3.88° *(envelope edge)*	4.60	$10.71\,\mathrm{rad\,s^{-2}}$	3.49° / 8.14°

Table 31: Angular acceleration during recovery and the resulting lever-arm error at $d = 130\,\mathrm{mm}$. "Free" is gravity alone; "saturated" adds the wheel torque $\sqrt{2}\,\tau_{\mathrm{cont}}$ opposing it.

At the envelope edge the apparent tilt error is 8.14°, against a recoverable envelope of 3.88°. The measurement error exceeds the disturbance being corrected by a factor of 2.1.

The equivalent panel configuration gives a 25.9° apparent error at its envelope edge. Measured in closed loop, the unfiltered accelerometer produced a 32.44° peak estimate error and a terminal attitude of 88.83° — the rig driven onto its side by its own controller.

8.6.2 Sensitivity to standing tilt error

The lever-arm term is transient. A second requirement, arising from constant estimate error, binds more tightly on this machine.

A constant tilt-estimate error e causes the controller to drive the estimate to zero, so the cube balances e from true vertical. Gravity then applies a constant moment which the wheels absorb indefinitely, and the steady state gives a standing wheel speed

$$\dot{\varphi}_{ss} = \frac{|K_1|}{|K_3|}\, e, \qquad \frac{|K_1|}{|K_3|} = 4365, \tag{46}$$

against 635 for the planar panel. The cube is 6.9× more sensitive to estimate bias, and panel tolerances do not transfer.

Tilt-estimate error	Standing wheel speed	Fraction of the $40\,\mathrm{rad\,s^{-1}}$ cap
0.05°	$3.8\,\mathrm{rad\,s^{-1}}$	10 %
0.10°	$7.6\,\mathrm{rad\,s^{-1}}$	19 %
0.25°	$19.0\,\mathrm{rad\,s^{-1}}$	48 %
0.50°	$38.1\,\mathrm{rad\,s^{-1}}$	95 %
1.00°	$76.2\,\mathrm{rad\,s^{-1}}$	190 % (saturated at rest)

Table 32: Momentum consumed by a standing tilt error before any disturbance.

Allocating one fifth of the momentum budget to standing error gives a tolerance of $e_{\max} = 0.105°$, against 0.722° for the panel. This figure governs the estimator requirements of §8.6.4 and the calibration requirements of §8.6.8.

8.6.3 Candidate filter structures

8.6.4 Selected structure

An explicit complementary filter with PI correction, applied directly to the gravity direction. The filter estimates $\hat{\boldsymbol{g}}$, the gravity direction in body coordinates, and a gyro bias $\hat{\boldsymbol{b}}$:

$$\boldsymbol{e} = \hat{\boldsymbol{g}} \times \hat{\boldsymbol{\gamma}}_{\text{meas}} \tag{47}$$

$$\dot{\hat{\boldsymbol{g}}} = -\left(\boldsymbol{\omega} - \hat{\boldsymbol{b}}\right) \times \hat{\boldsymbol{g}} \;+\; k_P\, \boldsymbol{e} \times \hat{\boldsymbol{g}} \tag{48}$$

$$\dot{\hat{\boldsymbol{b}}} = +\,k_I\, \boldsymbol{e} \tag{49}$$

with $\hat{\boldsymbol{g}}$ renormalised each cycle. The controller state follows as $\boldsymbol{\phi} = -\hat{\boldsymbol{g}}_B \times \hat{\boldsymbol{g}}$ (§8.3.3).

Equation (48) combines two sensors over complementary frequency bands. The gyro term is accurate over short intervals and drifts over long ones; the accelerometer term has no drift but carries the lever-arm error of (45) at high frequency. The gain k_P sets the crossover at $\tau = 1/k_P$.

Filter	Structure	Assumes	Cost
Direct nonlinear complementary	correction from the measured vector directly	reconstructed attitude available	requires an attitude reconstruction step, reintroducing a parameterisation
Explicit complementary	correction from vector measurements, no reconstruction	one reference vector	P-only leaves a standing offset proportional to gyro bias
Passive complementary	correction rotated into the estimate frame	one reference vector	better conditioned at large error; no advantage within the operating range
Linear Kalman	optimal for a linear plant and Gaussian noise	linearity	see §8.6.10
MEKF	error-state Kalman on the tangent space, reset each step	small error between resets	explicit bias states; six additional states and a covariance to tune

Table 33: Estimator structures considered. The taxonomy of the first three follows Mahony, Hamel and Pflimlin [**mahony2008complementary**].

Gyro bias	Tilt error at $k_P = 4$	Standing wheel	Fraction of cap
$1.00°\,\mathrm{s}^{-1}$ (uncalibrated MEMS)	0.250°	$19.0\,\mathrm{rad\,s}^{-1}$	47.6 %
$0.05°\,\mathrm{s}^{-1}$ (startup calibration)	0.013°	$1.0\,\mathrm{rad\,s}^{-1}$	2.4 %
$0.01°\,\mathrm{s}^{-1}$ (online estimation)	0.003°	$0.2\,\mathrm{rad\,s}^{-1}$	0.5 %

Table 34: Gyro bias propagated to standing wheel speed through a proportional-only filter.

Bias estimation. A proportional-only filter leaves a steady-state tilt offset $e = b/k_P$ proportional to gyro bias. Propagated through (46):

An uncalibrated consumer MEMS gyro typically exhibits 1–2° s^{-1} of zero-rate offset, consuming approximately half the momentum budget at rest. Startup calibration is therefore required, and the integral term (49) removes the residual, including thermal drift of order 0.1–0.3° s^{-1} over a ten-minute session for a sensor mounted near three motors without airflow.

The sign in (49) is $+k_I\boldsymbol{e}$, established by measurement: the opposite sign converges smoothly to five times the true bias.

Yaw-rate bias observability. The component of gyro bias along $\hat{\boldsymbol{g}}_B$ is unobservable, gravity providing no information about rotation about gravity (§8.5.2). Against a true bias of 0.149° s^{-1} in that direction the estimator returns exactly 0.000, while both perpendicular components converge to truth. This is the correct response to an unconstrained direction. A quaternion MEKF represents that direction explicitly and its covariance grows without bound unless constrained by hand [**markley2003attitude**]; the reduced-attitude form does not admit the state.

Gain selection. $k_P = 4$, $k_I = 0.5$. The response to k_I is flat across 0.1–2.0. The response to k_P is not flat and is bounded above, as set out in §8.6.7.

k_P: the firmware note and the estimator specification give 4; the cube envelope sweep gives 5 as the optimum (2.75° recovery against 2.49–2.58° at $k_P = 2$), with a hard edge above. If 5 is current, this paragraph and §8.6.7 both change, and 4 should be presented as deliberate headroom rather than as the optimum.

8.6.5 Rejection of the lever-arm term

Two mechanisms act on the tangential error term, and their combined effect exceeds what either provides alone.

The first is attenuation. The lever-arm error is a transient of duration $\sim 0.2\,\mathrm{s}$ and the accelerometer path is low-passed at $\tau = 1/k_P$, so most of the transient does not reach the estimate.

The second is cancellation. The error term is $\ddot{\theta}\,d$ and low-pass filtering approximates integration, but $\int \ddot{\theta}\,\mathrm{d}t = \Delta\dot{\theta}$, which is zero across a completed recovery: the rig is at rest before and after. The sign of the error reverses between the accelerating and decelerating phases, and the two halves cancel.

Measured on the panel, a 25.9° raw peak becomes a 0.29° filtered error, a reduction of 300×. The residual is not lever arm but the gyro-bias floor $b\tau = 0.005\ \mathrm{rad\,s^{-1}} \times 1.0\ \mathrm{s} = 0.286°$, which matches the observed value to two decimals at every mounting distance.

Excluded correction terms. An explicit $\dot{\boldsymbol{\omega}} \times \boldsymbol{r}$ correction is not applied. It requires a differentiated gyro signal and the filter already rejects the term it would remove, so it would contribute noise without benefit.

Accelerometer rejection gate. The gate discarding samples whose magnitude departs from $1\,g$ is set at 10–20 %. A tighter gate fires during the acceleration phase but not the deceleration phase, which removes one half of the cancelling pair and leaves the other. The width is chosen to pass both halves of the transient while still rejecting an impact.

8.6.6 Differences from the planar case

The panel study established the rejection mechanism. Three of its quantitative conclusions do not extend to three dimensions and were re-measured on the cube.

Centripetal acceleration. The tangential term cancels because $\ddot{\theta}$ changes sign. The centripetal term $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$ is quadratic in $\boldsymbol{\omega}$ and directed inward at all times, so it carries a DC component.

ω [$\mathrm{rad\,s^{-1}}$]	$\omega^2 r$ at $r = 130\,\mathrm{mm}$	Fraction of g	Apparent tilt
1	$0.130\,\mathrm{m\,s^{-2}}$	1.3 %	0.76°
2	$0.520\,\mathrm{m\,s^{-2}}$	5.3 %	3.04°
3	$1.170\,\mathrm{m\,s^{-2}}$	11.9 %	6.80°
5	$3.250\,\mathrm{m\,s^{-2}}$	33.1 %	18.34°

Table 35: Centripetal contribution to specific force. Negligible on the panel; subtracted explicitly on the cube.

The correction uses the measured $\boldsymbol{\omega}$ and the known mounting position $\boldsymbol{r}$, requiring no differentiation.

IMU mounting position. On the panel the estimate error was 0.29° at every distance from 0 to 106 mm, and mounting position was unconstrained. On the cube the rejection has a range limit within the available geometry.

The change is geometric. The panel's largest available lever arm was 106 mm; on the cube, opposite corners are 259.8 mm apart, so an IMU placed near one corner lies beyond the limit for its antipode. Mounting at the geometric centre places the sensor 129.9 mm from all eight

Estimator	IMU at the corner	at 137 mm	at 206 mm
Raw accelerometer	3.18°	0.00 (fails)	—
Complementary filter with bias	2.85°	2.75°	0.00 (fails)

Table 36: Recovery angle against IMU distance from the balancing corner.

corners, within the ∼150 mm limit for every one simultaneously. No other position satisfies the constraint for all corners. Thermal isolation and cable routing remain secondary criteria.

Envelope penalty. Closing the loop through the filter costs 11.59° → 11.53° on the panel, a penalty of 0.5 %; on the cube it costs 3.42° → 2.75°, a penalty of 20 %.

The cause is neither sensor noise nor lever arm. With noise and bias set to zero the recovery is 2.76°, essentially unchanged, and §8.6.5 accounts for the lever-arm term. The residual is phase lag: the complementary filter is model-free and can only integrate the gyro forward with slow correction toward the accelerometer, so the estimate trails the truth during a fast transient. On a plant with a 121 ms divergence time constant and a 3.88° envelope there is no margin to absorb the delay. Section 8.6.10 addresses this deficiency.

8.6.7 Proportional gain limit

The response to k_P has a hard upper bound rather than the shallow optimum observed on the panel.

k_P	1	2	5	10	20
Recovery	2.22–2.41°	2.49–2.58°	2.75°	0.00	0.00

Table 37: Recovery angle against complementary-filter proportional gain.

Below the optimum, phase lag dominates. Above it, sufficient lever-arm error reaches the estimate to close a positive feedback loop: accelerometer error tilts the estimate, the controller commands torque, the torque produces angular acceleration, and the angular acceleration produces further accelerometer error of the same sign. The estimator and plant form a loop whose gain is k_P. The panel did not exhibit this mechanism because its loop gain was below the threshold.

Consequently k_P cannot be increased to reduce the phase lag identified in §8.6.6. Removing that lag requires a model-based estimator.

8.6.8 Calibration requirements

Startup gyro-bias calibration. Three seconds stationary at boot, averaged and subtracted from subsequent samples. Required by table 34.

Twelve-parameter accelerometer calibration. Three biases, three scale factors and six cross-axis terms, obtained from a set of static orientations. Normalising $\hat{\gamma}$ to unit length removes common-mode scale error but not differential scale error between axes. In one plane a differential error appears as a gain change and is largely absorbed; in three dimensions it appears as a direction-dependent tilt bias, against the 0.105° total budget of table 32.

Mounting rotation. $\boldsymbol{R}_{IB}$, the rotation from IMU axes to cube body axes, is verified by a six-face test rather than taken from the datasheet drawing. A mounting misalignment produces a constant tilt offset and enters table 32 identically to gyro bias.

8.6.9 Standing wheel speed as a diagnostic

A non-zero standing wheel speed has two possible causes, gyro bias or a centre-of-mass offset, and they are separable without instrumentation: rotating the cube 180° about the vertical and repeating the run reverses the sign of a COM offset and leaves a gyro bias unchanged. The test distinguishes recalibration from shimming.

By table 32, standing wheel speed is the most informative single health indicator available and is logged on every run.

8.6.10 Upgrade path

The complementary filter is sufficient for corner balancing and is the implemented configuration. The 0.67° of envelope identified in §8.6.6 is recoverable through a four-stage progression.

1. Explicit complementary filter with PI bias correction. Implemented.
2. Steady-state linear Kalman filter on $[\boldsymbol{\phi}; \boldsymbol{\omega}; \boldsymbol{b}_{\text{gyro}}]$. The standard objection to a linear Kalman filter for attitude — that any parameterisation of $SO(3)$ carries a singularity — does not apply, because $\boldsymbol{\phi}$ is not a parameterisation of $SO(3)$ but a vector in the tangent plane at the balanced attitude, already linear in the plant (§8.4.4). Possessing the model, the filter reconstructs the lever-arm term analytically rather than rejecting it, removing the phase lag at source. The runtime cost is a constant gain matrix, identical to the complementary filter, and the separation principle allows composition with $\boldsymbol{K}_p$ unchanged. This stage accounts for the outstanding 0.67°.
3. Augmentation with a disturbance-torque state, providing integral action against COM offset and constant friction without additional structure.
4. MEKF. Deferred: it provides explicit bias states in place of (49), at the cost of six additional states and a covariance to tune. It becomes the appropriate structure at Stage 3, where large-attitude tracking against a full reference is required and the reduced-attitude formulation no longer applies (§8.3).

8.7 Control design

8.7.1 Choice of method

The plant of (33) is linear, time-invariant after the reduction of §8.5.3, fully actuated in the two tilt directions, and equipped with a full state estimate. Under these conditions linear quadratic regulation is the natural formulation, for three reasons specific to this machine.

Multivariable coupling. The two unstable tilt directions are not aligned with any wheel axis, so each wheel contributes to both. Three independent single-axis loops would require the coupling terms to be tuned by hand against each other. LQR produces the coupled gain matrix directly, and the resulting structure is visible in $\boldsymbol{K}_p$ below: negative diagonal terms opposing each wheel's own tilt component, positive off-diagonal terms assisting the other two.

Momentum regulation as a stated objective. The wheel-rate state enters the cost function on equal footing with attitude. Wheel speed is not a secondary quantity to be limited after the fact; it is the constraint that determines whether the machine balances indefinitely or for several seconds (§8.7.2).

Verifiable correctness. Full-state LQR satisfies the return-difference identity of Kalman [**kalman1964optimal**], which forces $M_s = 1$ exactly. This provides an independent check on the implementation that no manually tuned design offers (§8.9).

8.7.2 Weight selection

Weights are assigned by Bryson's rule [**bryson1969applied**]: each state is weighted by the inverse square of its maximum acceptable excursion,

$$\boldsymbol{Q} = \mathrm{diag}\left(\tfrac{1}{\theta_{\max}^2}\boldsymbol{I}_3,\ \tfrac{1}{\dot\theta_{\max}^2}\boldsymbol{I}_3,\ \tfrac{1}{\omega_{\mathrm{des}}^2}\boldsymbol{I}_3\right), \qquad \boldsymbol{R} = \frac{\rho}{\tau_{\mathrm{ref}}^2}\boldsymbol{I}_3. \tag{50}$$

The advantage of this parameterisation is that every entry is a physical commitment rather than a tuning constant, and each can be checked against an independently computed quantity.

Parameter	Value	Physical quantity it is checked against	Ratio
$\theta_{\max}$	0.20 rad (11.5°)	recoverable envelope, 3.88°	3.0×
$\dot\theta_{\max}$	2.0 rad s^{-1}	peak body rate from the envelope edge, 0.558 rad s^{-1}	3.6×
ω_{des}	10 rad s^{-1}	firmware wheel cap, 40 rad s^{-1}	0.25×
τ_{ref}	0.24 N m	continuous torque rating, 0.12 N m	2.0×
ρ	12	—	—

Table 38: Bryson weights and the measured quantities against which they are checked. Peak body rate is computed as $\dot\theta = \sqrt{2S_g(1-\cos\theta)/\bar\Theta}$.

Three of the four entries carry margin above the physical quantity, which is intentional: a weight set at the exact envelope produces a controller that treats the boundary as acceptable rather than as a limit to be avoided.

Wheel-speed weight. The entry ω_{des} requires justification because it is the one weight that materially affects the outcome. A recovery from the full 3.88° envelope requires

$$h = \sqrt{2\,\bar\Theta\,S_g\,(1-\cos\theta)} = 0.0155\ \mathrm{N\,m\,s} = 31.8\ \mathrm{rad\,s^{-1}}\ \text{of wheel speed}, \tag{51}$$

which is 80 % of the available cap. There is no margin to spend on standing wheel speed, so ω_{des} is set to $0.25\,\omega_{\mathrm{cap}}$ rather than to the motor's free-run speed. The consequence of the alternative is quantified in §8.7.6.

Torque scale and control cost. Only the ratio $\rho/\tau_{\mathrm{ref}}^2$ enters $\boldsymbol{R}$, so ρ and τ_{ref} are not independent parameters. The effective control weight is $R = 208.3$.

Weight transfer from Stage 1. The weights were carried unchanged from the planar panel to the cube. Retaining them isolates the source of any discrepancy between predicted and measured cube behaviour in the plant model rather than in the tuning.

8.7.3 Gain matrix

Solving the reduced Riccati problem of (43) and applying the projection (44):

$$\boldsymbol{K}_p = \left[\begin{array}{ccc|ccc|ccc} -6.9157 & 3.3923 & 3.4212 & -0.8348 & 0.4207 & 0.4270 & -0.000700 & 0.006235 & 0.006218 \\ 3.5194 & -6.8296 & 3.5136 & 0.4121 & -0.8481 & 0.4117 & 0.003826 & -0.003165 & 0.003816 \\ 3.4291 & 3.3947 & -6.9237 & 0.4265 & 0.4198 & -0.8377 & 0.006068 & 0.006070 & -0.000870 \end{array}\right] \tag{52}$$

with blocks acting on $\boldsymbol{\phi}$, $\boldsymbol{\omega}$ and $\boldsymbol{\rho}$ respectively, and rows corresponding to wheels X, Y, Z. The control law is $\boldsymbol{u} = -\boldsymbol{K}_p\boldsymbol{x}$.

Three properties of (52) serve as structural checks.

Sign pattern. Attitude and rate blocks have negative diagonal and positive off-diagonal entries: each wheel opposes its own tilt component and assists the other two. A row deviating from this pattern indicates an error in the contact point or in the gravity direction.

Diagonal uniformity. The attitude diagonal is 6.9157, 6.8296, 6.9237, a spread of 1.4 %, consistent with a mass distribution near-isotropic about the balancing diagonal. A spread beyond approximately 30 % indicates that the plant is not balancing on a corner.

Wheel-rate block. The entries are three orders of magnitude below the attitude entries. This is the ratio that governs bias sensitivity (§8.6.2): $|K_1|/|K_3| = 4365$, giving a standing tilt tolerance of 0.105° for one fifth of the momentum budget.

8.7.4 Closed-loop response

Pole [s^{-1}]	ω_n [$\mathrm{rad\,s}^{-1}$]	ζ	τ [ms]
0 (double)	—	—	marginal
−7.8167	7.817	1.000	127.9
−7.8352	7.835	1.000	127.6
−8.4230	8.423	1.000	118.7
−8.4513	8.451	1.000	118.3
−14.654	14.654	1.000	68.2
−14.658	14.658	1.000	68.2
−20.898	20.898	1.000	47.9

Table 39: Closed-loop eigenvalues under $\boldsymbol{K}_p$. All controlled modes are real; the slowest has a time constant of 127.9 ms against an open-loop divergence constant of 121 ms.

All controlled modes are real and non-oscillatory. The design is not attempting to place poles far into the left half-plane: the slowest controlled mode at $-7.82\,\mathrm{s}^{-1}$ sits close to the open-loop instability at $+8.26\,\mathrm{s}^{-1}$, which is the appropriate scale. Faster closed-loop poles would demand torque the actuators cannot supply and bandwidth the estimator cannot support (§8.6.7).

The two marginal modes are the conserved yaw momentum and the drifting yaw angle, both expected under $\boldsymbol{K}_p$ and both physically correct (§8.5.4).

Discretisation. At the 400 Hz loop rate, including the one-cycle computational delay, the discrete closed loop retains 95 % of the continuous damping with $\max|z| = 0.98153$. The full loop-rate analysis is given in §8.3.5.

8.7.5 Margins and robustness

Return-difference identity. The reduced-order design gives $M_s = 1.000000$, the value forced exactly by the identity of [**kalman1964optimal**] for full-state LQR. Any deviation indicates an implementation fault rather than a marginal design. The identity does not survive discretisation or the introduction of an estimator [**doyle1978guaranteed**], so the discrete poles are verified directly at the design rate and are not inferred from this figure.

Parametric robustness. The three plant parameters carrying the largest uncertainty are $\bar{\mathbf{\Theta}}$, I_s and S_g. Evaluating the closed loop over a $\pm 30\,\%$ grid on all three, 27 combinations:

$\delta\bar{\mathbf{\Theta}}$	δI_s	δS_g	$\max \mathrm{Re}$ [s^{-1}]
0.7	0.7	0.7	−5.5006
0.7	1.0	1.0	−4.9590
1.0	1.0	1.0	−7.8167
1.3	1.0	1.0	−4.6291
1.3	1.3	1.3	−7.7715
1.3	1.3	0.7	**−1.9283** *(worst)*

Table 40: Closed-loop stability under parameter perturbation, marginal modes excluded. All 27 combinations remain stable.

The worst case is $\delta\bar{\mathbf{\Theta}} = 1.3$, $\delta I_s = 1.3$, $\delta S_g = 0.7$: heavier than modelled with a weaker restoring moment. Stability is retained with a slowest mode at $-1.93\,\mathrm{s}^{-1}$. The $\pm 30\,\%$ range substantially exceeds the residual uncertainty in table 24, where the largest open item moves λ by $0.5\,\%$.

8.7.6 Weight sensitivity

Each weight was varied across an approximately eight-fold span with the others held at nominal.

Varied	$\theta_{\max}$	$\dot{\theta}_{\max}$	ω_{des}	R	**Slowest** [ms]	**Standing wheel** [% cap]
— (nominal)	0.20	2.0	10	208	127.9	13
$\theta_{\max}$	0.05	2.0	10	208	154.6	22
$\theta_{\max}$	0.40	2.0	10	208	123.5	11
$\dot{\theta}_{\max}$	0.20	0.5	10	208	130.9	18
$\dot{\theta}_{\max}$	0.20	8.0	10	208	130.7	11
ω_{des}	0.20	2.0	5	208	124.6	5
ω_{des}	0.20	2.0	20	208	154.2	28
ω_{des}	0.20	2.0	40	208	288.3	21
R	0.20	2.0	10	26	126.7	4
R	0.20	2.0	10	1667	201.6	39

Table 41: Closed-loop response to weight variation. Standing wheel speed is evaluated at the placeholder $\tau_{cw} = 8\,\mathrm{mN\,m}$.

$M_s = 1.000000$ throughout, confirming that every entry represents a well-posed design rather than a numerical artefact.

The attitude and rate weights have limited effect: an eight-fold variation in $\theta_{\max}$ moves the slowest mode by $25\,\%$, and $\dot{\theta}_{\max}$ by $2\,\%$. The wheel-speed weight and the control weight dominate, and both act principally on standing wheel speed rather than on speed of response.

Raising ω_{des} from 10 to $40\,\mathrm{rad\,s^{-1}}$ more than doubles the slowest time constant and leaves the machine consuming a fifth of its momentum budget at rest. Weighting against the motor's free-run speed of $441\,\mathrm{rad\,s^{-1}}$ rather than the firmware cap reduces the wheel-rate gains by approximately two orders of magnitude and removes momentum regulation altogether, producing the failure mode in which the rig balances for several seconds and then saturates.

Gain tuning is therefore not a route to improved envelope. Across the variation above the recoverable envelope moves by approximately $2\,\%$; the envelope is set by the hardware limits of §8.8.

8.7.7 Firmware interface

The control law reduces to one matrix–vector product per cycle:

$$\boldsymbol{u}_{\text{LQR}} = -\boldsymbol{K}_p \begin{bmatrix} \boldsymbol{\phi} \\ \boldsymbol{\omega} \\ \boldsymbol{\rho} \end{bmatrix}, \qquad \boldsymbol{\phi} = -\hat{\boldsymbol{g}}_B \times \hat{\boldsymbol{g}}. \tag{53}$$

- State order is $[\boldsymbol{\phi}(3);\ \boldsymbol{\omega}(3);\ \boldsymbol{\rho}(3)]$ in body axes X, Y, Z.
- $\boldsymbol{\phi}$ and $\boldsymbol{\omega}$ are expressed in the cube body frame. IMU output is rotated by $\boldsymbol{R}_{IB}$, verified by a six-face test.
- $\rho_k > 0$ denotes rotation about the positive body axis k, verified per wheel on the bench.
- $\boldsymbol{K}_p$ is used, not $\boldsymbol{K}$ (§8.5.4).
- $\boldsymbol{K}_p$ is a per-corner constant. Changing balancing corner requires regenerating $\hat{\boldsymbol{g}}_B$, $\boldsymbol{P}$ and $\boldsymbol{K}_p$; no structural change is required.
- The output of this section is $\boldsymbol{u}_{\text{LQR}}$, not the commanded torque. Friction feedforward, torque clamping and wheel-speed capping are applied afterwards (§8.8).

8.8 Nonlinearities

The design plant of §8.4.4 drops every term that vanishes at the equilibrium. The real machine retains them, and each imposes a constraint on the recoverable envelope or introduces a signal the control law must handle without amplifying.

8.8.1 Torque saturation and the recoverable envelope

Two independent bounds limit the tilt from which the cube can recover, and which one binds depends on a single hardware parameter.

Momentum bound. Recovery from a tilt θ at rest requires angular momentum

$$h = \sqrt{2\,\bar{\Theta}\,S_g\,(1 - \cos\theta)}\,, \tag{54}$$

which must not exceed the stored capacity $h_{\text{cap}} = I_s\,\omega_{\text{cap}} = 0.019475\,\mathrm{N\,m\,s}$. Inverting,

$$\theta_{\max}^{\text{mom}} = \arccos\left(1 - \frac{h_{\text{cap}}^2}{2\,\bar{\Theta}\,S_g}\right) = 4.88°. \tag{55}$$

Torque bound. The controller can hold a static tilt only while the gravitational moment does not exceed the available torque. At a corner the worst-case actuator projection is $\sqrt{2}\,\tau_{\text{cont}}$, giving

$$\theta_{\max}^{\text{torq}} = \arcsin\left(\frac{\sqrt{2}\,\tau_{\text{cont}}}{S_g}\right) = 5.16° \quad \text{at } \tau_{\text{cont}} = 0.12\ \mathrm{N\,m}. \tag{56}$$

The recoverable envelope is the smaller of the two. At the current hardware point this is the momentum bound, but the margin is narrow.

The two bounds cross at $\tau_{\text{cont}} = 0.113\,\mathrm{N\,m}$. The current estimate of $0.12\,\mathrm{N\,m}$ places the machine 6 % above that crossover, so the classification as momentum-limited rests on an unmeasured parameter. A thermal test confirming τ_{cont} determines which constraint the design should target, and is listed accordingly in §**??**.

ω_{cap}	τ_{cont}	Momentum bound	Torque bound	Limiting
40	0.08	4.88°	3.44°	torque
40	0.10	4.88°	4.30°	torque
40	**0.12**	**4.88°**	**5.16°**	**momentum**
40	0.20	4.88°	8.62°	momentum
60	0.12	7.32°	5.16°	torque
80	0.12	9.76°	5.16°	torque
80	0.20	9.76°	8.62°	torque

Table 42: Recoverable envelope against wheel cap and continuous torque.

Torque clamp. Commanded torque is clamped to $\pm\tau_{\text{peak}} = \pm 0.40\,\text{N m}$. This figure derives from the MN4006 rated current of 16 A at $K_t = 0.02513\,\text{N m/A}$ — the machine constant implied by the 380 rpm/V rating, §7 — giving 0.4021 N m. Two qualifications apply to its use as a clamp.

First, 16 A is the datasheet's *maximum continuous current over 180 s*, not an instantaneous rating. Second, and more significantly, that figure is specified for the motor in its intended application, driving a propeller which forces air through the ventilation channels. The reaction wheels operate at up to $40\,\text{rad}\,\text{s}^{-1}$ inside a sealed enclosure with no forced airflow, so the datasheet's thermal assumptions do not hold. At 16 A the ohmic dissipation is $1.5 I^2 R_\phi = 74.5\,\text{W}$ per motor with $R_\phi = 194\,\text{m}\Omega$, or 223 W across three motors, into a closed volume.

The binding continuous constraint is therefore neither of these but the driver: the moteus-n1 is rated at 9 A continuous phase current without thermal management, corresponding to 0.226 N m. Its 26 A rating requires a fan or heat spreader, which this build does not carry.

Limit	**Torque**	**Current**	**Basis**
τ_{cont} (design)	0.120 N m	4.78 A	thermal estimate, this enclosure
MN4006 efficiency band	0.126 N m	5.00 A	>92 % efficiency at 2–5 A
moteus-n1, uncooled	**0.226** N m	**9.00** A	**binding continuous limit**
τ_{peak} (clamp)	0.400 N m	15.92 A	MN4006 180 s rating
moteus-n1, cooled	0.653 N m	26.00 A	requires fan or heat spreader
moteus-n1, peak	2.513 N m	100.00 A	instantaneous

Table 43: Torque limits and their sources. $K_t = 0.02513\,\text{N m/A}$ throughout (datasheet K_V, §7).

The clamp at 0.40 N m is retained as a firmware backstop against a runaway command rather than as a thermal limit, since a 0.40 N m command sustained for more than a few seconds would exceed the driver's continuous rating by a factor of 1.8. Thermal protection is the driver's own temperature fault, not the clamp. The design envelope is set by τ_{cont} (table 42), and the gap between 0.120 and the driver's 0.226 N m indicates available headroom should the thermal test support it.

8.8.2 Wheel-speed cap

Each wheel is capped at $\omega_{\text{cap}} = 40\,\text{rad}\,\text{s}^{-1}$, which is 4.5 % of the motor no-load speed of $883\,\text{rad}\,\text{s}^{-1}$. The cap is a firmware policy rather than a hardware limit. Its purpose is to make the momentum-limited failure mode reachable and testable during bring-up, rather than a rare event occurring only at the envelope edge under a particular disturbance sequence.

Implementation is a one-sided clamp:

$$\text{if } (\rho_k \geq +\omega_{\text{cap}} \text{ and } u_k > 0) \text{ or } (\rho_k \leq -\omega_{\text{cap}} \text{ and } u_k < 0): \quad u_k = 0. \tag{57}$$

Torque that would accelerate the wheel further is zeroed; torque decelerating it back toward the cap is permitted. A symmetric clamp would hold the wheel at the cap and remove the actuator from the system.

Budget. A full-envelope recovery consumes 0.0155 N m s, equivalent to 31.8 rad s^{-1} of wheel speed, or 80 % of the cap. The remaining 20 % covers standing wheel speed from estimator bias (§8.6.2) and from friction feedforward error (§8.8.3) combined. Exceeding it at rest means the machine saturates before reaching the envelope edge.

8.8.3 Friction feedforward

Coulomb friction at the wheel bearings applies a constant retarding torque whose sign depends on the direction of rotation. Uncompensated, it enters the plant as a constant disturbance and the regulator responds by holding a standing wheel speed, consuming momentum budget continuously.

Structure. The compensation is a smooth approximation of the sign function applied per wheel:

$$u_k = u_{\mathrm{LQR},k} + \tau_{cw} \tanh\left(\frac{\rho_k}{\varepsilon}\right) + b_w\, \rho_k, \tag{58}$$

with $\varepsilon = 0.05\,\mathrm{rad\,s^{-1}}$ setting the transition width. The tanh reaches 0.9987 of its asymptote at $3\varepsilon = 0.15\,\mathrm{rad\,s^{-1}}$.

tanh rather than sign. All three wheels sit near zero at equilibrium, so a sign function chatters between $\pm\tau_{cw}$ at every crossing, injecting a square wave into the torque command at the loop rate. The smooth form removes the discontinuity while recovering sign behaviour within three transition widths.

Sensitivity. Without feedforward the standing wheel speed is $\tau_{cw}/|K_3|$:

τ_{cw}	Standing wheel speed	Fraction of cap
4 mN m	2.5 rad s^{-1}	6 %
8 mN m (placeholder)	5.1 rad s^{-1}	13 %
12 mN m	7.6 rad s^{-1}	19 %
20 mN m	12.7 rad s^{-1}	32 %

Table 44: Standing wheel speed without feedforward, evaluated at the placeholder friction values.

Over-compensation is worse than under-compensation: excess feedforward drives the wheel rather than retarding it, and the regulator must then oppose its own feedforward term. The starting value is 70 % of the measured friction, adjusted upward against the standing wheel speed observed on the bench.

Parameter status. $\tau_{cw} = 8\,\mathrm{mN\,m}$ and $b_w = 0$ are placeholders pending a spin-down test; the pivot Coulomb torque $\tau_{cp} = 0$ is a placeholder pending a breakaway test. No friction measurement has been made on either stage. The sensitivity study establishing the thresholds above was conducted against these placeholders, so it bounds the tolerance rather than predicting the behaviour. The measurements are listed in §**??**.

8.8.4 Cogging torque

The MN4006 motors have 12 pole pairs, producing torque ripple at 12× the mechanical frequency. At the wheel cap this is 76 Hz, below the Nyquist frequency of the 400 Hz loop and therefore representable in the control signal.

No explicit cogging compensation is applied. The moteus field-oriented control loop runs well above any cogging harmonic and absorbs most of the ripple internally; the residual is a periodic disturbance to the body, attenuated by the body inertia. The cogging frequency reaches $f/2 = 200\,\mathrm{Hz}$ at a wheel speed of $105\,\mathrm{rad\,s^{-1}}$, which is the threshold above which raising ω_{cap} would require the residual amplitude to be characterised first. At the present cap of $40\,\mathrm{rad\,s^{-1}}$ there is a factor of 2.6 in hand.

8.8.5 Vibration and anti-aliasing

Three sources of high-frequency content reach the accelerometer.

1. **Structural resonance** of the printed frame. Frequency unmeasured; expected above 200 Hz given the stiffness of PETG-CF at the 150 mm scale, but this has not been confirmed and is the assumption most likely to be challenged if the frame is lightened.
2. **Cogging torque** at 12× the wheel frequency (§8.8.4).
3. **Wheel unbalance**, producing a once-per-revolution forced vibration at 6.4 Hz at the cap. CAD places each wheel centre of mass on its spin axis to within 10^{-8} mm, so the wheels are balanced by design; the as-built residual has not been measured. The sensitivity is 0.035 N of radial force per 0.1 mm of eccentricity at $40\,\mathrm{rad\,s^{-1}}$, scaling as ω^2.

Content above $f/2 = 200\,\mathrm{Hz}$ folds into the estimator passband and is indistinguishable from tilt. The complementary filter provides some attenuation but is not designed as an anti-aliasing filter, and its rejection above 200 Hz is finite.

This mechanism, not discrete stability, sets the binding lower bound on loop rate: the anti-aliasing floor is 281 Hz against a stability floor of 38.3 Hz (§8.3.5).

8.8.6 Per-cycle torque computation

Collecting §8.7 and the preceding subsections, the torque applied to wheel k on each cycle is

$$\boxed{u_k = \mathrm{clamp}\Big(\underbrace{-\sum_{j=1}^{9} K_{p,kj}\, x_j}_{\text{LQR}} + \underbrace{\tau_{cw} \tanh\Big(\frac{\rho_k}{\varepsilon}\Big) + b_w \rho_k}_{\text{friction feedforward}},\ \pm\tau_{\mathrm{peak}} \Big)} \tag{59}$$

followed by the one-sided wheel cap of (57). The ordering is fixed: feedforward is added before the clamp, and the wheel cap is applied after it. Reversing either produces a different controller.

Listing 1: Per-cycle actuation, applied after the estimator of §8.6 and the control law of §8.7.

```
for (int k = 0; k < 3; k++) {
    u[k] = 0.0f;
    for (int j = 0; j < 9; j++)
        u[k] -= Kp[k][j] * x[j];                    // LQR

    u[k] += tau_cw * tanhf(rho[k] / eps_ff);       // Coulomb
    u[k] += b_w * rho[k];                          // viscous
```

```
    u[k] = clampf(u[k], -tau_peak, tau_peak);      // torque clamp

    if ((rho[k] >= +omega_cap && u[k] > 0.0f)      // wheel cap,
     || (rho[k] <= -omega_cap && u[k] < 0.0f))     // one-sided
        u[k] = 0.0f;
}
```

8.9 Validation

8.9.1 Gate methodology

Each stage of the model was accepted only after passing an independent numerical check with a pre-declared tolerance. The gates are ordered so that each one depends on the results of the previous ones, and a failure at any gate invalidates everything downstream.

Gate	Tests	Criterion	Panel	Cube*
1	Hang test	Gravity direction and mount rotation: released from the balanced attitude, the model falls along $\hat{\boldsymbol{g}}_B$	pass	pass
2	Swing periods	Small-amplitude period about the pivot matches $2\pi/\lambda$, confirming Θ and S_g	$<0.1\,\%$	0.11, 0.70, 0.77 %
3	Linearisation	$\boldsymbol{A}$, $\boldsymbol{B}$ from the Simscape linearisation command match the hand-derived matrices of (33)	$\|\Delta\boldsymbol{A}\|=1.667\times10^{-7}$ $\|\Delta\boldsymbol{B}\|=0$	1.742×10^{-7} 2.157×10^{-16}
4	Poles	Unstable eigenvalues of the linearised plant match $\sqrt{S_g/\Theta}$	±9.037, 0	+7.9341, +7.9152
5	Energy	Open-loop simulation, no damping: total energy drift quantifies solver dissipation	0.0000 %	—
6	Momentum	One torque pulse per wheel: $\|\boldsymbol{L}\|$ constant confirms sign conventions and axis assignment	$\lvert\boldsymbol{L}\rvert = 1.09\times10^{-13}$	subsumed by Gate 3

Table 45: Validation gates.
*Cube results are from the $(+1,+1,+1)$ configuration, before the corner change of §8.4.6. The qualitative conclusions are unaffected; the affected numbers (λ, ℓ, mount angle) differ by the amounts tabulated in table 25.

Gate 3 is the definitive one. Passing it establishes that the Simscape model and the analytic state-space matrices describe the same plant to 1.7×10^{-7}, so every subsequent analysis — the LQR design, the structural properties of §8.5, the discrete-loop check of §8.3.5 — is a statement about the Simscape model rather than a parallel calculation that might or might not correspond to it.

Gate 2 has a methodological subtlety that cost time on the panel and is recorded here for that reason.

8.9.2 Amplitude correction in swing tests

The period of a pendulum is amplitude-dependent. For a swing of half-angle θ_0 the exact period is

$$T(\theta_0) = T_0 \cdot \frac{2}{\pi} K\left(\sin^2\frac{\theta_0}{2}\right), \qquad T_0 = 2\pi\sqrt{\frac{\bar{\Theta}}{S_g}}, \tag{60}$$

where K is the complete elliptic integral of the first kind. Since $\bar{\Theta}$ enters the small-amplitude period as $T_0 \propto \sqrt{\bar{\Theta}}$, an uncorrected period measurement overstates $\bar{\Theta}$ as T^2/T_0^2:

Swing amplitude	T/T_0	$\bar{\Theta}$ overstatement
5°	1.000476	0.10 %
10°	1.001907	0.38 %
20°	1.007669	1.54 %
30°	1.017409	3.51 %
50°	1.049783	10.20 %

Table 46: Apparent inertia error from an uncorrected period measurement. A 50° swing overstates $\bar{\Theta}$ by 10.2 %, which propagates directly into λ.

The error scales as θ_0^2 to leading order, so period errors roughly double the inertia error. On the panel this was caught because the small-amplitude formula was validated against a finite-amplitude simulation and the two disagreed by more than Gate 2's tolerance until the correction was applied. The remedy is either to keep swings below 10° (where the error is under 0.4 %) or to apply (60) and extract $\bar{\Theta}$ from the corrected period.

8.9.3 The return-difference identity

For full-state linear quadratic regulation with the cost $J = \int(\boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x} + \boldsymbol{u}^\top \boldsymbol{R}\boldsymbol{u})\,\mathrm{d}t$, the return-difference matrix satisfies

$$M_s = \bar{\sigma}\left[(\boldsymbol{I} + \boldsymbol{L})^{-1}\right] = 1.000000000\,, \tag{61}$$

where $\boldsymbol{L} = \boldsymbol{K}_r(s\boldsymbol{I} - \boldsymbol{A}_r)^{-1}\boldsymbol{B}_r$ is the loop transfer of the reduced system. This is not a computed margin; it is the identity of Kalman [**kalman1964optimal**], forced to hold exactly by the algebraic structure of the Riccati equation. Any deviation from 1 is an implementation fault — a wrong matrix, a non-positive-semidefinite $\boldsymbol{Q}$, or a Riccati solution that did not converge — rather than a property of the design.

The identity serves as a single-number acceptance test on the gain computation, catching classes of error that would otherwise require inspecting a 3×9 matrix entry by entry. It is evaluated on the reduced system of §8.5.3, where the Riccati problem is well-posed, and not on the full nine-state system, where it is not (table 29).

Scope. Three things the identity does not cover.

1. **Discretisation.** Discretising the reduced-order LQR at 400 Hz gives $M_s = 1.039$, a 3.9 % deviation. The guarantee does not survive the zero-order hold [**doyle1978guaranteed**], and discrete margins must be verified directly against the z-plane eigenvalues (§8.3.5).

2. **An estimator in the loop.** Replacing the full state with the output of §8.6 breaks the loop transfer on which the identity is defined, and Doyle's counter-example [**doyle1978guaranteed**] shows that LQG can have arbitrarily small margins. The return-difference identity is a statement about the *controller*, not about the closed loop with the estimator included.

3. **Gain structure.** $M_s = 1$ confirms that the gain is optimal for the stated cost, but the cost may be wrong. The sign-pattern and uniformity checks of §8.7.3 test the gain's physical plausibility independently of the cost function.

8.9.4 Tolerance in rank computations

The controllability argument of §8.5 rests on a rank count. An incorrect tolerance can invert the conclusion silently.

Call	Result
`rank(Co)` (default, relative tolerance)	8 — correct
`rank(Co, 1e-9)` (absolute tolerance)	9 — incorrect

$\|\mathcal{C}\| = 1.105 \times 10^9$, so an absolute tolerance of 10^{-9} is 9×10^{-19} of the matrix scale — below the double-precision floor. Every singular value, including the one at 9.19×10^{-17} that is numerically zero, exceeds that threshold.

The failure reports the system as fully controllable, which requires no further action. Had the result been accepted, the eight-state reduction (§8.5.3) would not have been written, `lqr` would have returned a gain without complaint (table 29), and the first indication of a problem would have arrived on hardware as unexplained authority loss in a direction that cannot be identified from the symptoms.

A tolerance must be chosen relative to the scale of the matrix it is applied to. MATLAB's default does this; a hand-supplied absolute figure carried from another problem does not. The general form: tolerances that are safe for one matrix may be orders of magnitude too loose or too tight for the next one, and the failure mode in both cases is silence.

8.10 Results

8.10.1 Simulation methodology

Three models serve different roles. The linear design plant (§8.4) is used for gain synthesis and structural analysis; a Simscape Multibody model with imported STEP geometry provides independent verification; and a nonlinear MATLAB simulator ('cubli_nlsim') carries the full sensor and actuator budget and is fast enough for the thousands of runs a bisection sweep requires (∼50 ms per run against ∼20 s for Simscape). The Simscape model is the reference; the MATLAB simulator's only job is to agree with it.

Agreement was verified at a 2° recovery with ideal sensors:

Table 47: Nonlinear simulator against Simscape, 2° release.

t [s]	`cubli_nlsim`	Simscape
0.0	1.6254°	1.6254°
0.1	1.0766°	1.0854°
0.3	0.1135°	0.1108°
0.6	0.3369°	0.3389°
1.0	0.1224°	0.1285°

Agreement is within 1 % through the transient; late-time divergence is Simscape solver tolerance, not a modelling difference.

Recovery metric. The largest initial tilt, released from rest with wheels stationary, from which the cube returns to upright, minimised over tilt azimuth in the plane perpendicular to $\hat{\boldsymbol{g}}_B$. Four to eight azimuths are tested per bisection, with 12 bisection steps per azimuth. Best and worst azimuths differ by approximately 6 %, so a single-axis number overstates capability.

Shipping configuration. Unless otherwise noted, all results below use: $\tau_{\max} = 0.12\,\mathrm{N\,m}$, $\omega_{\mathrm{cap}} = 40\,\mathrm{rad\,s^{-1}}$, bearing friction at $\tau_{cw} = 8\,\mathrm{mN\,m}$ with feedforward, pivot friction at $5\,\mathrm{mN\,m}$, 5 ms loop delay, BMI270-grade sensor noise (accelerometer 3.2 mrad, gyro $2.4\,\mathrm{mrad\,s^{-1}}$), gyro bias $[0.3, -0.2, 0.15]^\circ\,\mathrm{s^{-1}}$, IMU at the geometric centre, and the reduced-attitude complementary filter with bias states at $k_P = 4$, $k_I = 0.5$.

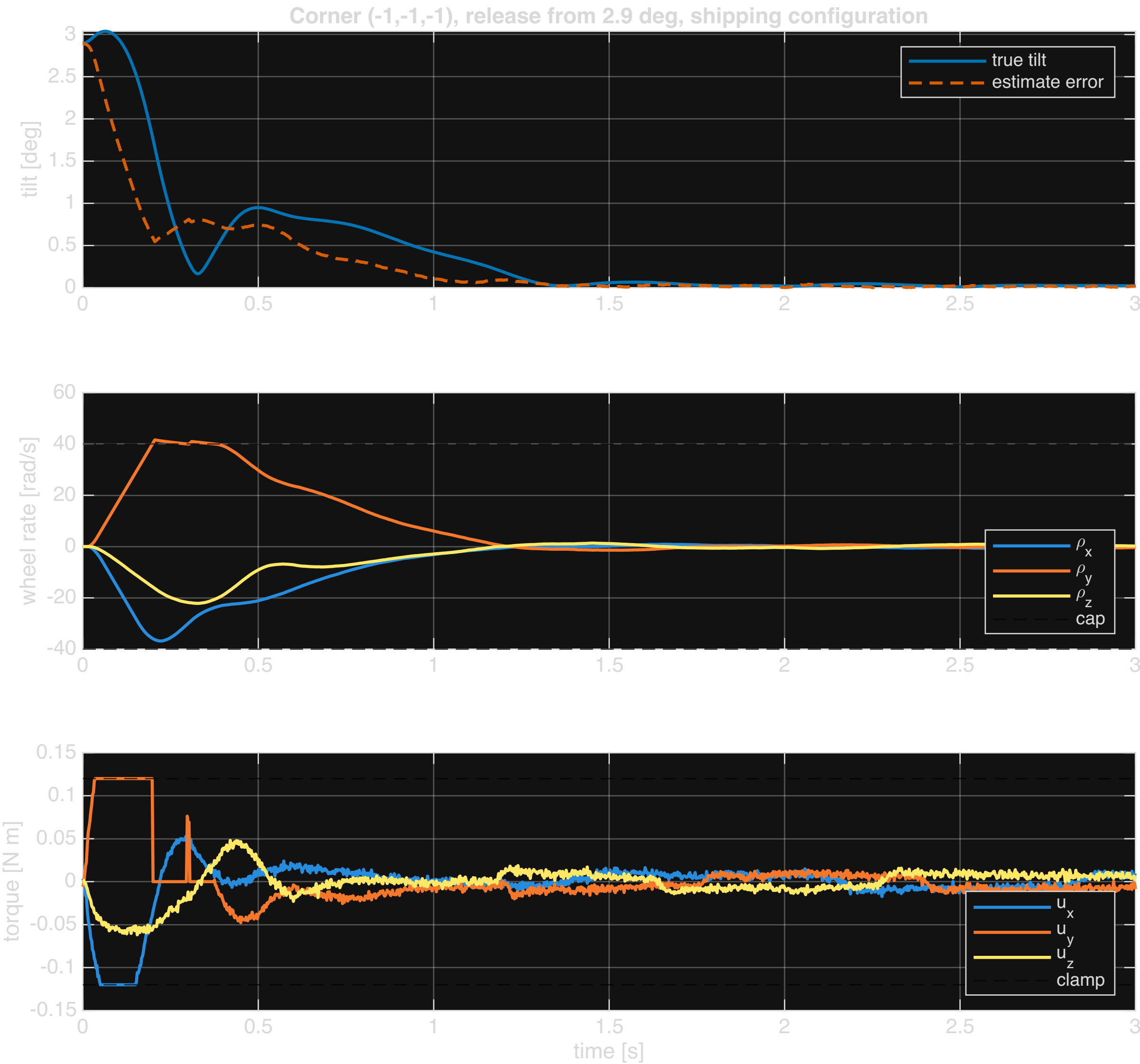

Figure 15: Simulated cube recovery from 3.14° (best corner), shipping configuration. Top: tilt magnitude, true and estimated — the divergence during the first 0.2 s is the estimator phase lag of §8.6.6. Middle: the three wheel speeds against the 40 rad s^{-1} cap. Bottom: commanded torques against the 0.12 N m clamp.

8.10.2 Per-corner recovery envelope

Each corner is designed with its own gain set at the same Bryson weights (§8.7.2).

All eight corners balance. The spread is 14 %, and recovery tracks $1/S_g$ exactly: the longest lever arm produces the largest destabilising moment and therefore the smallest envelope. The primary corner $(-1, -1, -1)$ is the best of the eight.

Every corner is momentum-limited. This is confirmed by the wheel peak: 41.7 rad s^{-1} against the cap at every corner, while the torque stays inside its clamp.

One gain set does not work. With the primary corner's gains applied to all eight, six corners diverge at the open-loop rate (max Re from +7.7 to +7.9 s^{-1}): the controller pushes the wrong way because $\hat{\boldsymbol{g}}_B$ rotates 67–107° between corners, and both $\boldsymbol{P}$ and $\boldsymbol{\Theta}$ follow it. Firmware carries eight gain matrices (216 floats).

Corner	ℓ [mm]	S_g [N m]	λ [s^{-1}]	θ_{eq}	Recovery	Wheel peak
$(-1,-1,-1)$ **(primary)**	122.84	1.8875	8.257	0.797°	**3.14°**	41.7 rad s^{-1}
$(-1,+1,-1)$	126.08	1.9373	8.159	2.773°	3.03°	41.7
$(-1,-1,+1)$	128.54	1.9750	8.121	3.170°	2.93°	41.3
$(+1,-1,-1)$	128.56	1.9753	8.118	3.171°	2.92°	41.7
$(-1,+1,+1)$	131.64	2.0226	8.048	3.097°	2.89°	41.3
$(+1,+1,-1)$	131.66	2.0229	8.050	3.095°	2.83°	41.7
$(+1,-1,+1)$	134.01	2.0591	7.980	2.609°	2.77°	41.7
$(+1,+1,+1)$	136.99	2.1048	7.934	0.714°	2.76°	41.7

Table 48: All eight corners, shipping configuration, own gains. Wheel peak exceeds the 40 rad s^{-1} cap by 4 % because the cap zeroes torque rather than braking.

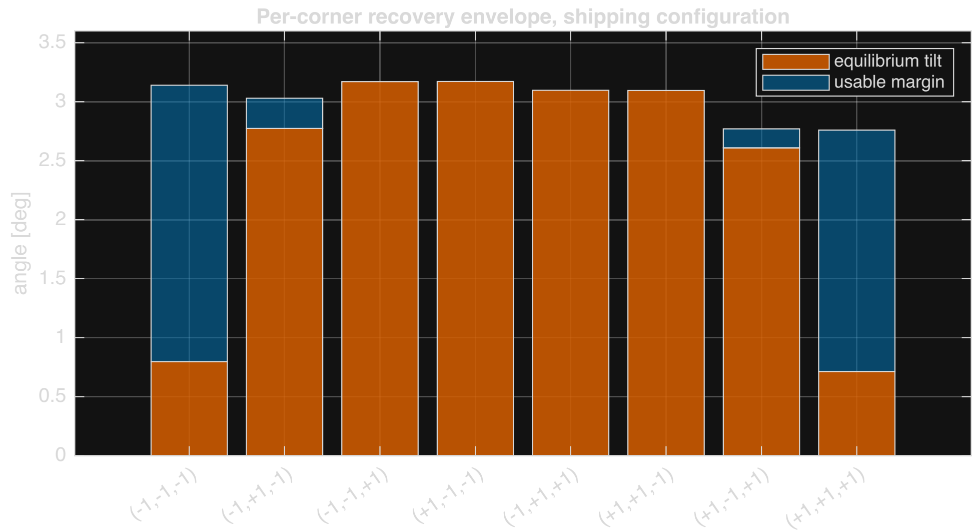

Figure 16: Recoverable envelope and equilibrium tilt for all eight corners. The shaded region is the margin available for a corner-to-corner transition.

The antipodal trap. $\boldsymbol{P} = \boldsymbol{I} - \hat{\boldsymbol{g}}_B\hat{\boldsymbol{g}}_B^\top$ is invariant under $\hat{\boldsymbol{g}}_B \to -\hat{\boldsymbol{g}}_B$, so the primary gains *almost* work at the antipodal corner: $\max \mathrm{Re} = +0.149$, a divergence time of 6.7 s. A 2.5 s scoring window marks this as a pass; it is not. Runs must exceed 15 s or check the eigenvalues directly.

8.10.3 Nonlinearity budget

Each effect was isolated against a 1° tilt, 8 s run, steady state over the last 2 s.

Feedforward removes the wheel drift and torque penalty from bearing friction completely (row 3 vs row 2), confirming it is mandatory rather than beneficial. Accelerometer noise dominates the torque activity and the tilt rms; gyro noise is an order of magnitude below it.

Parametric robustness, nonlinear. Sweeping $\bar{\boldsymbol{\Theta}}$ from ×0.70 to ×1.30 with gains fixed: recovery ranges from 3.41° to 3.00°, stable throughout. The ρ_{scale} assumption does not affect the envelope.

Effect	**Recovery**	**Tilt rms**	**Wheel drift**	**τ rms**
Ideal	3.42°	0.000°	$0.0\,\mathrm{rad\,s^{-1}}$	$0.0000\,\mathrm{N\,m}$
Wheel Coulomb $8\,\mathrm{mN\,m}$, no FF	3.29°	0.001°	$1.8\,\mathrm{rad\,s^{-1}}$	0.0080
Same, with feedforward	3.29°	0.000°	0.0	0.0000
Pivot friction $5\,\mathrm{mN\,m}$	3.36°	0.002°	0.0	0.0001
Accelerometer noise 3.2 mrad	3.42°	0.040°	0.7	0.0308
Gyro noise $2.4\,\mathrm{mrad\,s^{-1}}$	3.42°	0.004°	0.1	0.0030
Delay 2 samples (5 ms)	3.33°	—	—	—
Delay 3 samples (7.5 ms)	3.29°	—	—	—
All, FF on, 5 ms delay	**3.16°**	0.038°	0.5	0.0308

Table 49: Nonlinearity budget. Total realistic degradation −7.6 % from the ideal-sensor baseline.

$\tau_{\max} \setminus \omega_{\text{cap}}$	40	60	80	120	200
0.10	3.09	3.53	3.45	3.45	3.45
0.15	3.80	4.61	5.12	5.16	5.16
0.20	4.25	5.38	6.15	**7.04**	6.86
0.30	4.79	6.36	7.53	9.21	10.83
0.40	5.03	6.92	8.41	10.76	13.39

Table 50: Worst-case recovery angle [deg] across the actuator design space. Non-monotonic cells (0.20/200 worse than 0.20/120): with fixed gains a larger cap lets the wheel run away. Raising ω_{cap} requires re-running the gain grid.

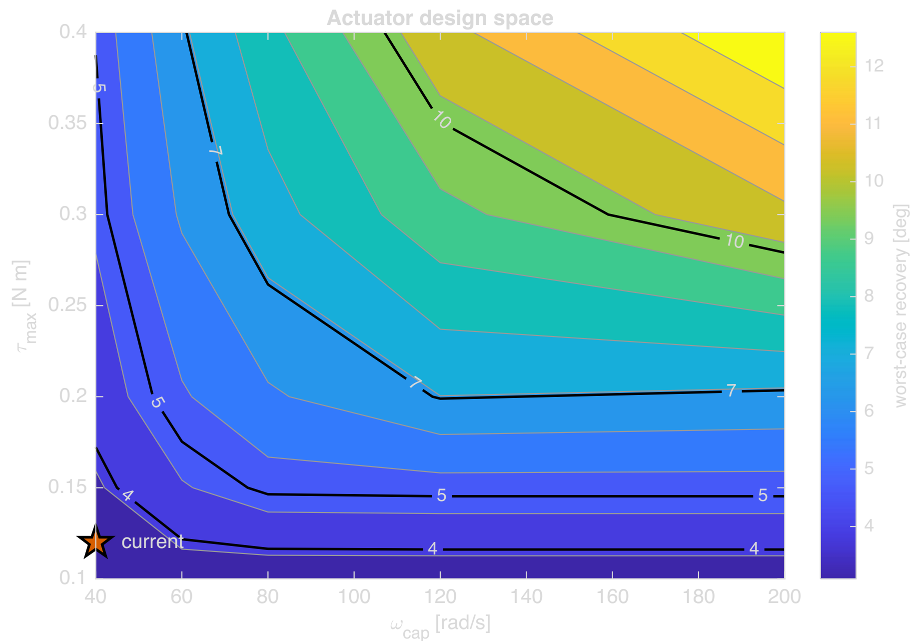

Figure 17: Contour of worst-case recovery [deg] over the actuator design space, with the current operating point ($0.12\,\mathrm{N\,m}$, $40\,\mathrm{rad\,s^{-1}}$) and the momentum/torque crossover line marked.

8.10.4 Actuator design space

The recovery envelope across the ($\tau_{\max}$, ω_{cap}) plane, gains fixed at the Bryson set of §8.7.2:

Below $\tau_{\max} \approx 0.15\,\mathrm{N\,m}$ the cube is torque-limited and the wheel cap is irrelevant; above it, $\tau_{\max}$ and ω_{cap} multiply. Neither is worth relaxing alone.

8.10.5 Weight selection robustness

The wheel-rate weight ω_{des} was selected for robustness against plant uncertainty rather than for peak performance.

Table 51: Recovery angle [deg] at the shipping configuration, sweeping ω_{des} against perturbation in $\bar{\boldsymbol{\Theta}}$.

ω_{des}	$\Theta \times 0.8$	$\times 0.9$	$\times 1.0$	$\times 1.1$	$\times 1.2$	Verdict
5	3.11	3.05	2.99	0.00	0.00	fails at +10 %
6	3.07	3.00	2.94	2.57	0.00	fails at +20 %
8	2.98	2.92	2.86	2.80	0.00	fails at +20 %
10	2.91	2.85	**2.79**	2.73	2.68	survives
12	2.84	2.78	2.72	2.67	2.62	survives, 3 % worse

$\omega_{\text{des}} = 10$ costs 4 % against the fragile optimum at 5 and is the only value that holds across ±20 % on $\bar{\boldsymbol{\Theta}}$ — the residual uncertainty from the print-density assumption (ρ_{scale}). Below $\omega_{\text{des}} = 5$ there is a cliff: $4 \to 2.06°$, $3 \to 1.58°$, $2 \to 0.00°$, as the peak gain magnitude climbs from 13 to 29 and the controller begins fighting the estimator.

8.10.6 Stage 1 — planar panel

The panel is demonstrated on hardware.

Table 52: Stage-1 panel, demonstrated on hardware.

Quantity	**Value**	**Origin**
Demonstrated recovery	10°	measured on hardware
Settling time	0.75 s	measured
Peak wheel speed	39.5 rad s^{-1} (99 % of cap)	measured
Gain vector $\boldsymbol{K}$	$[-1.0998,\ -0.1232,\ -0.001732]$	$M_s = 1.000000$
Closed-loop poles	$-9.72 \pm 0.76j,\ -5.05$	
Estimator penalty	$11.59° \to 11.53°$ (0.5 %)	simulated

8.10.7 Edge balance

No equivalent simulation study exists for edge balance. Edge bring-up went directly from the reduced-attitude estimator design to staged hardware bring-up, validated on hardware rather than through a prior envelope study.

9 Results

9.1 Validation Results to Date

This subsection reports evidence against the acceptance criteria of Table 4, in the order the stages are entered. Where no result exists yet, the criterion is stated so the gap is explicit rather than silent. Appendix F carries the per-requirement verification status that the results below close out.

9.1.1 1-DoF Control Validation

The single-axis rig is the protected experimental item of Section 3.3, and its balance result is the evidence carried into this document. Gate G3 requires the rig to balance for at least 60 s unassisted.

Beyond the duration criterion, three quantities are recorded because they determine whether the result transfers to the integrated cube rather than merely holding on the rig. Recovery from a defined impulse establishes that the balance is a stabilised equilibrium and not a fortunate initial condition. Steady-state wheel-speed drift measures momentum accumulation: a controller that balances while monotonically spinning up is one that will saturate, and the drift rate predicts when. Control effort is compared against the continuous and peak current limits of the driver given in Appendix B, since a balance sustained only by exceeding the continuous rating is not a balance that survives to the demonstration.

Balancing performance examples. Table 53 reports representative runs against Gate G3, each initialised from rest at the upright equilibrium with no applied disturbance.

Table 53: Representative unassisted balance runs on the single-axis rig, against the Gate G3 threshold of 60 s.

Run	Balance duration	Steady-state wheel-speed drift	Peak driver current
1	$> 60\,\mathrm{s}$	–	0.36 A
2	$> 60\,\mathrm{s}$	–	0.38 A
3	$> 60\,\mathrm{s}$	–	0.40 A

Note: No constant drift was appreciated during static zero-position balancing. When the center of mass was uncalibrated, small oscillations were observed instead of steady drift due to torque saturation.

Recovery-angle disturbance tests. Beyond the zero-disturbance case, the rig was displaced by hand to a defined initial angle, measured from the support, and released to test whether the controller recovers to upright rather than merely holding a small perturbation. Table 54 reports the three angles used to characterise this margin.

Table 54: Recovery outcome versus the initial displacement angle, measured from the support.

Initial angle	Recovered?	Recovery time	Peak driver current
7°	Yes	1.2 s	0.86 A
8°	Yes	1.5 s	0.98 A
12°	Yes	2.1 s	1.34 A

The largest initial angle tested to date, again measured from the support, is 24°; recovery from this angle was also satisfactory, so the demonstrated disturbance-rejection envelope currently extends at least this far beyond the three angles of Table 54. This angle represents the physical hard stop of the experimental rig support as well as the maximum tilt evaluated in simulation, so requirement PER-03 (recovery from 12° or more) is satisfied with margin, and the demonstrated envelope is bounded by the rig rather than by the controller.

9.1.2 3-DoF Control Validation

For the 3D model, control-loop validation to date has achieved edge balance on the final prototype using the 1-DoF control law; corner balance has not yet been demonstrated. Table 55 sets out the result template that will be populated once edge-balance testing on the cube is complete, mirroring the 1-DoF quantities of Section 9.1.1 so the two are directly comparable, and Table 56 mirrors the recovery-angle protocol of Table 54 for the cube. TODO: populate both tables from edge-balance testing on the final prototype, run the corner-balance case once edge balance is closed out, and compare against requirements PER-01, PER-02 and PER-04 of Table 4 and the 1-DoF bench results of Section 9.1.1.

Table 55: Result template: unassisted balance on the cube, edge and corner equilibria, against the Gate G3 threshold of 60 s.

Equilibrium	Balance duration	Steady-state wheel-speed drift	Peak driver current
Edge	TBD	TBD	TBD
Corner	TBD	TBD	TBD

Table 56: Result template: recovery from a defined initial angle on the cube, measured from the support, edge equilibrium.

Initial angle	Recovered?	Recovery time	Peak driver current
7°	TBD	TBD	TBD
8°	TBD	TBD	TBD
12°	TBD	TBD	TBD
24°	TBD	TBD	TBD

10 Conclusions & Next Steps

10.1 Conclusions

The team has demonstrated the ability to build and control Armonia across the planned difficulty progression: a working 1-DoF model, and edge balance on the final 3D prototype, both achieved through the staged validation approach described in this document. Several of the originally planned tasks have been rescheduled in response to deadline pressure and issues encountered along the way, but the team has continued to make progress toward the final objective. Corner balance remains the outstanding milestone and is expected to be resolved in the days before final delivery and presentation. Armonia has confirmed itself to be a tightly coupled, multidisciplinary system: reliably building and controlling it depends on close coordination and continuous communication across the team's subsystem interfaces.

Jump-up and walking. The two stretch objectives, jump-up and walking (Section 1.3, Table 1), have not been demonstrated on the delivered cube, and the requirements that depend on them — FUN-10, FUN-11, PER-07, PER-10 and CON-02 (Table 4) — remain open in Appendix F. The cause is not a gap in the underlying analysis: the jump-up sizing chain of Section 5.1.1 closes, the required wheel inertia is verified against it (PER-06, Table 4), and the mass and mechanical-interface budgets carry the brake servos throughout the design rather than as an afterthought — they are costed in the mass budget (Table 15) and given a controlled interface, "Brake mount", alongside the motor and battery interfaces (Table 17). What is missing is the mechanism itself: the servo bracket and the wheel strike feature, parts M5 and W4 of the CAD part list (Table 64, Appendix C), were never carried from a reserved interface to a drawn and fabricated part, so no brake was ever mounted on the integrated cube and the manoeuvre was never attempted.

This gap is the direct consequence of a staging decision, taken at the outset and recorded in Section 3.3: balance the lighter, mechanically simpler cube first, and add the brake in a second design pass once that control problem was solved. The ordering was deliberate, not an oversight. A cube carrying three brake servos and their mounts is heavier and has a different mass and inertia distribution than one without, and an already-marginal balancing problem (Section 1.1) is harder to close, not easier, with that mass present from the first bring-up. Proving the balancing controller on the lighter configuration before committing to the brake's added mass was meant to let the team iterate quickly to a brake-equipped second version once the control loop was trusted, rather than debug an unstable equilibrium and a partially designed actuator at the same time. Consistent with this, the jump-up demonstration and the brake mechanism build were designated sacrificial scope from the start of the plan, sequenced after the edge-balance gate and positioned to yield first if time ran short (gate G8 of Table 45; risk RE12 and cut-order item 2 of Table 72, Appendix G).

That second pass never happened. Schedule pressure and the setbacks recorded in Appendix G consumed the float the plan had reserved for it, and the team reached the final week with edge and corner balance — the committed objectives of Section 1.3 — still the active critical path. At that point, integrating the brake meant reopening the electronics build on a heavier, structurally revised frame, and the drive, sensing and control electronics exist as a single set with no spare (RE11, Appendix G): building a brake-equipped cube meant moving that one electronics set into a new prototype and re-establishing balance on it from scratch, with a real chance of reproducing, on the new build, the same stabilisation difficulties that had only just been solved on the current one. Weighed against the committed corner-balance and slew objectives, that risk was judged unacceptable, and the standing decision was to protect the current cube's balancing result rather than force a late brake integration under time pressure. This is a common and, in the team's judgement, correct trade in engineering practice: a verified baseline is worth more than an unverified stretch capability chased at its expense, and the jump-up and walking objectives are carried forward as the clearly scoped starting point of the next design iteration rather than rushed into this one.

10.2 AI-Driven Control & Dynamic Coefficient Tuning

10.2.1 Motivation: Limitations of a Fixed Gain

The balancing controller of Section **??** computes a single gain K_d offline, from an (A, B) pair built from the parameters of Tables 8 and 10. That gain is optimal for exactly one plant: the one those numbers describe. It is retained here as the baseline precisely because it is simple and verifiable, but its optimality is conditional in three ways that the hardware of this project is expected to violate.

- **Parametric drift.** C_b and C_w are not obtained from CAD and are carried as estimates until the identification tests of Table 9. Bearing friction, wiring drag and motor cogging also change with temperature, wear and reassembly, so the identified values are a snapshot rather than a constant. The inertia tensor I and the centre-of-mass offset r_{cm} are subject to the same objection at the few-percent level, and r_{cm} in particular shifts whenever the pack or harness is disturbed.
- **Payload and configuration changes.** Any added mass changes m, r_{cm} and I simultaneously, and therefore changes A. A gain tuned for one configuration is not merely suboptimal for another; near a fully unstable equilibrium it may be destabilising.
- **Contact and surface variation.** The pivot is modelled as an ideal frictionless point or line. A real corner resting on a real surface slips, deforms and dissipates energy in a way that is neither in the model nor constant across surfaces, and this is the dominant unmodelled effect for the walking objective of Section 1.3.

The standard remedy within the existing design is gain scheduling on $\hat{\theta}_b$ (Section **??**), but scheduling interpolates between gains that were themselves computed from the same assumed parameters. It addresses the weakening of the small-angle linearisation; it does not address a plant that differs from the model. The proposal here is to keep the verified feedback law and adapt its coefficients online.

10.2.2 Proposed Hybrid Architecture

The intended structure is deliberately conservative: a classical inner loop whose stability properties are understood, wrapped by a slow outer loop that adjusts the inner loop's coefficients. Writing ϕ for the tunable coefficient vector,

$$u[k] = -K_d(\phi[k])\,\hat{x}[k], \qquad \phi[k+1] = \phi[k] + \Delta\phi\big(\hat{x}[k],\,\hat{x}[k-1],\ldots\big), \tag{62}$$

where $\hat{x}$ is the estimated state of Section **??** and $\Delta\phi$ is produced by the tuner. Two properties of (62) matter more than the choice of learning algorithm. First, the inner loop remains a linear state feedback at every instant, so the controller that runs on the hardware is the one already implemented and tested. Second, ϕ is updated on a timescale far slower than the control period Δt, which keeps the frozen-gain analysis approximately valid and prevents the tuner from acting as an unmodelled fast feedback path. Three candidate tuners are of interest, and they are not equivalent in cost:

- **Bayesian optimisation** over a low-dimensional parameterization of Q and R, minimising a measured cost from short balancing trials [**shahriari2016bayesopt**]. This is the cheapest option and the one with a direct precedent on this class of hardware: automatic LQR tuning has been demonstrated on an inverted pendulum with on the order of tens of experiments [**marco2016automaticlqr**], and a self-tuning LQR has been run on the Cubli's own ancestor platform at ETH [**trimpe2014learningcubli**]. It tunes between trials, not within them.

- **Adaptive dynamic programming**, which solves the LQR problem online from measured data rather than from (A, B), converging toward the optimal gain for the true plant without an explicit model [**lewis2012adp**]. This is the option with the strongest theoretical guarantees and the strictest excitation requirements.
- **Reinforcement learning**, in which a policy trained in simulation outputs $\Delta\phi$ continuously from the state-error history. This is the most capable and the least verifiable, and is treated below as the exploratory branch rather than the default.

Deployment reservations. The literature supporting learned control is overwhelmingly validated in simulation or on well-instrumented hardware, and the failure mode of an online tuner on an open-loop-unstable plant is a fall, not a degraded score. Any deployment on this cube should therefore be gated behind: a projection of ϕ onto a set for which the frozen-gain closed loop is verified stable across the parameter uncertainty; a fallback to the fixed K_d of Section **??** on any estimator fault or divergence; and bench trials with the cube tethered. This is future work in the literal sense — none of it is a committed deliverable of the present design, and the baseline controller is not contingent on any of it.

10.2.3 Module 1: Dynamic Gain Tuning

The tuner does not emit motor currents. It emits the coefficients from which K_d is computed, which bounds what a bad update can do. Two parameterizations are worth comparing:

- **Weight-space tuning.** The policy outputs the diagonal entries of the cost matrices, $\phi = (\log q_1, \ldots, \log q_9,\ \log r_1, \log r_2, \log r_3)$ with $Q = \mathrm{diag}(q_i) \succ 0$ and $R = \mathrm{diag}(r_i) \succ 0$, and K_d follows from the discrete algebraic Riccati equation. The logarithmic parameterization enforces positive definiteness by construction, so every ϕ the policy can emit corresponds to a well-posed LQR problem and, for a stabilisable (A_d, B_d), to a stabilising gain for the *modelled* plant. The cost is that a Riccati solve must run onboard whenever ϕ changes, which is what confines updates to a slow outer loop.
- **Gain-space tuning.** The policy perturbs the entries of K_d directly, or equivalently the K_p, K_d pair of a PD form, avoiding the Riccati solve at the price of losing the structural guarantee above. This is the practical choice if the outer loop must run fast, and it makes the projection step of the previous paragraph mandatory rather than merely prudent.

For the 3D corner-balancing case the state is $x = (\theta, \omega_b, \omega_w) \in \mathbb{R}^9$ and K_d is 3×9 (Section 4.4), so tuning the 54 free entries directly is not sensible on the available data. Weight-space tuning reduces the problem to 12 coefficients, and symmetry of the three wheel axes reduces it further if the wheels are treated as identical. A reward or cost of the form

$$c[k] = \hat{x}^{\mathrm{T}}[k]\, W_x\, \hat{x}[k] + \rho\, \|u[k]\|^2 + \sigma\, \|\omega_w[k]\|^2 \tag{63}$$

Equation (63) adds an explicit penalty on stored wheel momentum, which the nominal LQR cost $J(u)$ of Section **??** does not distinguish from any other state deviation. This term is the one directly relevant to the saturation problem raised in Section 1.2: it expresses a preference for holding attitude with the wheels near zero speed, leaving momentum headroom for disturbance rejection.

10.2.4 Module 2: Sim-to-Real Transfer via Domain Randomization

A policy trained against a single nominal parameter set learns to exploit that parameter set. Domain randomization instead samples the plant parameters from a distribution at the start of each training episode [**tobin2017domainrand**], so that the policy is optimised against a family of plants rather than one, and the transfer method has a substantial record on physical legged hardware [**hwangbo2019anymal**]. For this cube the randomized quantities follow directly from the identification gaps already documented:

- **Damping:** C_b, C_w over a wide interval, since these are estimated rather than measured until the identification of Section 4.3 closes.
- **Inertia and mass distribution:** I, m and r_{cm} perturbed by a few percent, covering print-density variation, fastener placement and harness routing.
- **Actuation:** K_m within motor tolerance, plus a current saturation limit and a commanded-torque delay, so the policy cannot assume an ideal actuator.
- **Contact:** pivot friction and restitution, and a small random ground slope — the term standing in for the surface variation that motivates this module.
- **Sensing:** gyro bias walk, accelerometer noise and an injected vibration component, matching the estimator degradation identified in Section 1.1 as the practical limit on balancing.

The sensing term requires particular care. The accelerometer corruption on this platform is generated by the controller's own wheel activity, so it is correlated with the policy's actions rather than independent of them. A randomization scheme that injects only white sensor noise will understate the difficulty, and a policy that transfers under it should not be assumed to transfer on hardware. Whether the randomized distribution covers the real cube is an empirical question, settled by bench testing; the principal failure mode of this module is a distribution selected for convenience rather than representativeness.

10.2.5 Module 3: Jump-Up Energy Optimisation

The jump-up analysis of Section 5.1.1 treats the manoeuvre as a spin-up to $\omega_{\max}$ followed by an abrupt stop, with all deviation from the ideal absorbed into a single brake-amplification factor $\beta = \tau_b/(\tau_b - \tau_g)$. That model is adequate for sizing and is deliberately conservative, but it is a poor description of what the mechanism does: the servo-driven barrier has finite travel, the collision is not instantaneous, and the resulting torque profile is neither a step nor known in advance.

The proposal is to treat the torque command over the manoeuvre window as a profile to be optimised rather than a constant. Parameterizing the pre-brake and post-impact torque commands by a small vector ψ — for instance the timing of brake release relative to the wheel-speed trajectory, and the shape of the recovery torque applied by the remaining wheels immediately after impact — the objective is

$$\min_{\psi} \quad \mathbb{E}\Big[\, \|\theta(t_f) - \theta^\star\|^2 + \mu\, \|\omega_b(t_f)\|^2 \,\Big] \quad \text{subject to} \quad \omega_w \le \omega_{\max}, \quad |u_i| \le u_{\max}, \tag{64}$$

where t_f is the instant the controller hands over to the balancing law at $|\hat{\theta}_b| < \theta_{\text{th}}$. Minimising residual body rate at handover is the quantity of interest, because a jump that reaches the equilibrium with excess rate demands recovery authority the balancing controller may not have. Because each evaluation of (64) is a physical jump, the sample budget is small and the appropriate method is Bayesian optimisation rather than a policy-gradient method, with the initial design taken from the analytical h_w of Equation (10).

Two constraints on this module must be stated. First, it presupposes the measurement described in Section 5.1.1: until the spin-down test reports both the impulse-equivalent mean $\bar{\tau}_b = h_w/\Delta t$ and the peak torque, there is no validated model to optimise against and no way to distinguish an improved profile from a mis-measured one. Second, the structural caveat carries over unchanged — optimisation must not be permitted to search toward shorter stopping times, since τ_b is simultaneously a load on the barrier, bolt, bracket and spokes, and the sizing value was chosen for the dynamics and not for that load path. The search space must be bounded by the structural case, not by the dynamic one.

10.2.6 Relevance Beyond the Bench

Two application claims follow, and both are narrower than the general enthusiasm for learned control would suggest.

For **CubeSat attitude control**, the relevant fact is that a spacecraft's inertia tensor is known imprecisely at integration and changes in flight — propellant depletion, deployable appendages, thermal distortion — while reaction-wheel friction rises with bearing age. These are the orbital analogues of the parametric drift of Section **??**, and they are conventionally handled by conservative fixed gains with margin, at a cost in pointing performance. A tuner that adapts Q and R against measured performance addresses exactly this gap. The qualification barrier for flying learned control is high and unaddressed here; the architecture of (62) is chosen partly because a bounded outer loop over a classical inner loop is the form most likely to survive that scrutiny.

For **low-gravity surface mobility**, the case is stronger, because the plant is genuinely unknown before arrival. Regolith mechanical properties on a small body are not characterised to the precision a hopping controller would want, and the flight-proven hoppers cited in Section 1.2 [**yoshimitsu1999minerva**, **ho2017mascot**] move ballistically rather than under closed-loop control for this reason. The contact randomization of Module 2 and the profile optimisation of Module 3 map onto that problem directly: a hop whose landing attitude is controlled, on a surface whose properties were sampled from a distribution rather than assumed, is the capability that separates directional walking from a ballistic tumble [**spiridonov2024spacehopper**]. The cube tests this under 1 g, where the destabilising torque is larger and never relents — a harder case in that one respect, as argued in Section 1.2, though it does not reproduce the low-gravity contact dynamics that would decide the real manoeuvre.

A Final Specification Sheet

This appendix is the single place where the finished design is stated as a set of numbers: what the cube is, what it weighs, what the wheel is, and what operating point it was designed to. Every figure below is *quoted*, never derived — the last column of each table names where it comes from, and the derivation stays in the section that owns it. Nothing here is new work, and nothing here may disagree with its source: all recurring values are drawn from the single definition file `sections/00_values.tex`, which in turn mirrors the run of `analysis/SIZING.py`, so a value cannot be updated in one table and left stale in another.

Two distinct masses appear throughout and are not interchangeable. The *sizing basis*, 1.800 kg, is the mass the reaction wheels were designed against and the mass at which PER-06 is derived; it is the heaviest case the design admits. The *as-built* mass, 1.626 kg, is what the integrated article weighs. The 174 g between them is realised margin, quantified in Appendix C.6, and is deliberately not folded back into the sizing.

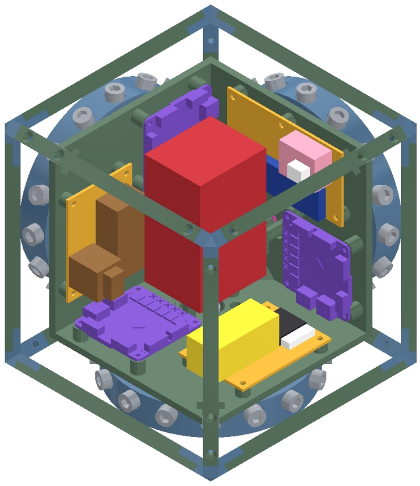

Figure 18: Configuration the tables below describe: an outer frame of edge rails and corner contact features enclosing the inner structure, the three orthogonal reaction-wheel modules and the electronics. The dimensioned general-assembly drawing is Figure 20.

A.1 Configuration and Principal Dimensions

A.2 Mass Decomposition

Table 58 is the itemised roll-up behind the six-line summary of Table 15. It is grouped by the same categories, so the two reconcile line for line. The *basis* column is the important one: it separates what the design knows (weighed parts and datasheet figures) from what it still projects (the structural allowance), which is 24.0 % of the projected total and the dominant uncertainty in the budget.

Table 58: Itemised mass decomposition at the converged design point, rolled up by `analysis/SIZING.py` (Stage 6). Quantities are per assembled cube.

Item	Unit (g)	Qty	Total (g)	Basis
Wheels — 702.4 g, 39.0 %				

Continued on next page

Table 58 (continued)

Item	Unit (g)	Qty	Total (g)	Basis
Reaction wheel (PET-CF, 120 mm, 15× M6 radial)	234.14	3	702.42	Sized, Sec. 5.3
Motors and brake servos — 245.4 g, 13.6 %				
T-Motor MN4006 KV380	68.00	3	204.00	Datasheet
TowerPro MG92B brake servo	13.80	3	41.40	Datasheet
Power — 279.0 g, 15.5 %				
Tattu R-Line V5.0 6S 1550 mA h LiPo	254.00	1	254.00	As weighed
XT60 connector pair	15.00	1	15.00	As weighed
LM2596 step-down regulator	10.00	1	10.00	As weighed
Electronics — 95.2 g, 5.3 %				
mjbots moteus-n1 driver	14.60	3	43.80	Datasheet
mjbots MA600 breakout + magnet	6.00	3	18.00	Datasheet
Teensy 4.1	11.00	1	11.00	Datasheet
CAN-FD adapter for Teensy 4.1	6.00	1	6.00	Datasheet
mjcanfd-usb-1x	3.40	1	3.40	Datasheet
Seeed XIAO ESP32-C6 + antenna	8.00	1	8.00	Datasheet
SparkFun BMI270 IMU (Qwiic)	5.00	1	5.00	Datasheet
Wiring, perfboard and passives — 45.5 g, 2.5 %				
Protoboard 15 cm × 9 cm double-sided	25.00	1	25.00	As weighed
JST PH3 cable	3.00	3	9.00	Estimated
100 µF 16 V electrolytic cap	1.00	3	3.00	Estimated
100 nF 50 V cap	1.00	3	3.00	Estimated
1N5819 Schottky diode	3.00	1	3.00	Estimated
4.7 nF cap (CAN)	1.00	2	2.00	Estimated
60.4 Ω 1/2 W resistor (CAN)	0.25	2	0.51	Estimated
Structure — 432.5 g, 24.0 %				
Outer frame / contact features	212.80	1	212.80	**Allowance**
Inner structure / motor subframe	176.90	1	176.90	**Allowance**
Fasteners / wiring harness	42.80	1	42.80	**Allowance**

A.3 Reaction Wheel

A.4 Operating Point

What this sheet does not yet carry. The plant parameters of Table 9 — the body inertia I_b about the pivot, the centre-of-mass distances l and l_b, the damping coefficients C_b and C_w — have determination methods assigned but no measured values, and are recorded here as outstanding rather than estimated. They close on the CAD mass properties and the 1-DoF identification run of Section 4.3, and belong in this sheet once measured. The same applies to the modelled-versus-measured wheel mass and inertia rows of Table 66, which are the check that the built wheel is the wheel specified above.

Table 57: Principal dimensions and configuration of the three-axis cube.

Symbol	Quantity	Value	Source
Body			
L	Outer edge length (frozen)	150 mm*	CON-01, gate M2
t_w	Frame wall thickness	5 mm	Tab. 63
M	Total mass, sizing basis	1.800 kg	Tab. 12
M	Total mass, as built	1.626 kg	Tab. 66
	Architecture	Two-part: outer frame + inner structure	Sec. 6.1
	Corner contact features	8	Tab. 64
Reaction wheel (×3, mutually orthogonal)			
D_w	Ring outer diameter	120 mm	Tab. 16
D_i	Ring inner diameter	100 mm	Tab. 16
w_r	Ring radial width (= hole depth)	10 mm	Tab. 16
t	Wheel axial thickness	20 mm	Tab. 16
n_s, w_s	Spoke count and width	3, 10 mm	Tab. 16
N	Ballast stations populated (of 37)	15	Tab. 16
d_h	Ballast hole diameter (M6 clearance)	6.4 mm	Tab. 16
R_{mean}	Ballast centroid radius	55.0 mm	Tab. 16
Actuation and power			
	Motor (×3)	T-Motor MN4006 KV380	Tab. 62
	Driver (×3)	mjbots moteus-n1	Tab. 62
	Brake servo (×3)	TowerPro MG92B	Tab. 62
	Battery	6S LiPo, 22.2 V, 1550 mA h	Tab. 62
	Flight computer	Teensy 4.1	Tab. 62

* The sizing analysis of Section 5 is run at 149 mm, one millimetre under the frozen value. The discrepancy is material at the present design point: since $h_w \propto ML^{3/2}$, closing at 150 mm raises $I_{w,\text{target}}$ by 1.22 % and leaves the selected wheel at 99.3 % of its own requirement rather than 100.6 %. The reconciliation is an open decision, recorded in Section 2, and is to be taken before the wheel is committed to manufacture.

Table 59: Mass reconciliation. The budget closes on the sizing basis by construction; the weighed article is lighter, and the difference is realised margin rather than an unclosed loop.

Line	Mass	Status
Specified hardware (wheels, motors, power, electronics, wiring)	1367.5 g	Measured or datasheet
Structure allowance	432.5 g	Pending CAD closure
Budget total = sizing basis	**1.800** kg	Converged, $\delta = 0.0$ g
As built, weighed	**1.626** kg	App. C.6
implied structure mass	258.5 g	Outcome, to be confirmed by CAD
realised margin	−174 g	Not spent
CON-02 cap	1.8 kg	Met with 174 g to spare

Table 60: Reaction-wheel specification, per wheel. The wheel is a printed ring on three spokes, tuned to its inertia target by M6 bolt-and-nut sets in radial ID-to-OD through-holes.

Quantity	Value	Source
Ring material	PET-CF, 1290 kg/m^3	Tab. 16
Mass, unballasted ring + spokes	127.86 g	`SIZING.py`
+ M6 steel, 15 stations	112.50 g	`SIZING.py`
− plastic displaced, net PET-CF	121.63 g	`SIZING.py`
Mass as ballasted	**234.14** g	Tab. 16
Inertia, unballasted	3.0451×10^{-4} kg m^2	`SIZING.py`
Net inertia per station	2.151×10^{-5} kg m^2	Tab. 16
Inertia as ballasted I_w	$\mathbf{6.2715 \times 10^{-4}}$ kg m^2	Tab. 16
Requirement $I_{w,\text{target}}$ at the sizing basis	6.2352×10^{-4} kg m^2	PER-06
margin delivered	100.6 %	Tab. 14
Requirement at the as-built mass	5.4445×10^{-4} kg m^2	App. C.6
margin delivered	115.2 %	App. C.6
Speed at which the built wheel still closes	3473 rpm	App. C.6
Centrifugal load per station at 4000 rpm	72 N	`SIZING.py`
against M6 class 8.8 proof load	0.57 %	`SIZING.py`
Edge-to-edge gap between stations	14.39 mm	`SIZING.py`
Axial wall, each face	6.80 mm	`SIZING.py`

Table 61: Design operating point and the derived performance figures that follow from it. Values at the sizing basis unless noted.

Quantity	Value	Source
Design wheel speed $\omega_{\max}$	4000 rpm	Tab. 12
as a fraction of the derived ceiling	57.8 %	PER-08
Stored momentum per wheel h_w	0.2612 kg m^2/s	Tab. 12
Stored energy per wheel	55 J	Sec. 5.1.1
Brake torque τ_b (sizing value)	5 N m	Tab. 12
Tip-over torque $\tau_g = MgL/2$	1.3155 N m	Tab. 12
at the as-built mass	1.1884 N m	App. C.6
Brake amplification β	1.3570	Tab. 12
Spin-up to design speed at 9 A	1.2 s	PER-12
floor set by τ_g	0.20 s	Sec. 5.1.1
Brake arrest time (requirement)	$\leq$ 100 ms	PER-10
Control and estimation loop rate	1 kHz	PER-05
Driver continuous current	9 A	PER-09
Motor peak current (short duration)	16 A	Tab. 62
Motor torque constant $k_t = 60/(2\pi K_v)$	0.025,13 N m/A	K_V, Tab. 62
Drive torque at 9 A continuous	0.226 N m	Sec. 5.1.1
Pack energy	$\approx$ 35 W h	Tab. 62

B Bill of Materials

The components supplied for this project constitute an *electronics* bill of materials: the actuation chain, wheel and attitude sensing, on-board compute, communications and power distribution are specified, but no structural or mechanical hardware is. The reaction wheels, the frame, the motor mounts, the fasteners and every printed bracket are the team's design responsibility, and are treated as such in Section 6. Table 62 lists the supplied items grouped by subsystem. The actuation figures derived from these parts — the torque constant, the achievable torque and wheel speed, and the inertia that sizes the wheel — are developed where they are used, in Sections 5.1.1 and 5.3.

The architecture the BOM implies is a two-level control stack: each mjbots moteus-n1 closes a high-rate field-oriented current loop on its motor, using the MA600 encoder for rotor angle, while the Teensy 4.1 runs the outer attitude-estimation and state-feedback loop over CAN-FD — the same inner-tachometer / outer-attitude layering used in spacecraft reaction-wheel assemblies. A mechanical wheel brake, following the original ETH Cubli [**gajamohan2013cubli**], provides the near-impulsive momentum transfer required for the jump-up.

Procurement is the single largest schedule risk. The long-lead items are not electronic but mechanical: the frame stock and the printed structure gate the entire build, and with three actuation sets and no spare, the loss of a motor or driver in service would force the reduced-actuation one-wheel variant [**hofer2023onewheelcubli**]. These items are ordered first. Consumables and bench equipment not covered by the supplied set — wire, connectors, fasteners, filament, the current-limited bench supply and the battery-safety kit — are tracked separately in the build documentation.

Table 62: Supplied bill of materials, grouped by subsystem.

Ref	Item	Qty[a]	Key specifications	Function
Actuation				
1	T-Motor MN4006 KV380	3	18N24P outrunner; 380 rpm/V; 68 g; ⌀44.35 mm × 21 mm; 194 mΩ; peak 16 A, 380 W	Reaction-wheel torque source
2	mjbots moteus-n1	3	10–54 V; 100 A peak / 9 A cont.; 15–30 kHz FOC; 5 Mbit/s CAN-FD; 14.6 g	FOC current/torque driver (inner loop)
11	TowerPro MG92B servo	3	metal-gear micro servo; ≈0.29 N m (≈3 kg cm) at 6 V; 50 Hz PWM	Wheel brake actuator (jump-up)
Sensing				
3	mjbots MA600 breakout + ring magnet	3	TMR absolute angle; SPI 6 MHz, 3.3 V; 16-bit (65,536 counts/rev); air-gap ≤ 1.5 mm	Wheel angle: commutation + state feedback
10	SparkFun BMI270 6-DoF IMU	1	3-axis accel + 3-axis gyro (no magnetometer); I^2C 400 kHz (Qwiic); ODR up to ≈ 1 kHz	Body attitude reference (tilt + rates)
Compute and Communications				
4	Teensy 4.1	1	600 MHz Cortex-M7 + FPU; native CAN-FD (CAN3); 5 V supply	Flight computer: estimator + control law
5	CAN-FD adapter (Teensy 4.1)	1	CAN-FD transceiver, logic-level → differential; 5 Mbit/s	Bus transceiver for the MCU

Continued on next page

Table 62 (continued)

Ref	Item	Qty[a]	Key specifications	Function
6	mjbots mjcanfd-usb-1x	1	USB ↔ CAN-FD, 5 Mbit/s	Bench interface (calibration, tuning)
7	JST-PH3 CAN cables	≥4	2 A AC/DC; 3-conductor daisy-chain	CAN bus interconnect
8	Seeed XIAO ESP32-C6 + antenna	1	Wi-Fi; 3.3 V UART to MCU; TX burst ≈ 300 mA	Wi-Fi telemetry / command link
9	60.4 Ω 0.5 W resistor	4	Split termination, $2 \times 60.4\,\Omega \approx 120\,\Omega$ bus impedance	CAN bus termination
17	4.7 nF capacitor	2	common-mode filter, bus midpoint to ground	CAN split-termination filter
Power				
12	Tattu R-Line V5.0 6S LiPo (XT60)	1	22.2 V; 1550 mA h; 150C; ≈ 35 W h; ≈ 254 g	Primary energy store
13	LM2596 buck module	1	step-down to 5.0 V; rated 2–3 A (derates without heatsink)	5 V logic rail
14	XT60 connector pair	1–2	rated ≈ 60 A	Battery power connector
15	100 µF 16 V electrolytic	2–4	16 V part — 5 V rail only	5 V-rail bulk reservoir
19	1N5819 Schottky diode	1–2	1 A, 40 V	5 V-rail back-feed protection
Integration				
16	100 nF 50 V capacitor	6–10	local decoupling at each IC	Per-IC supply decoupling
18	Double-sided perfboard	1	15 cm × 9 cm, double-sided	Power / signal distribution board

[a] Quantity per assembled cube; motors, drivers, encoders and brake servos are 3 each — one per wheel axis.

C CAD and Construction Detail

This appendix holds the mechanical detail deliberately kept out of Section 6.1: the model tree, the drawing set, the fabrication settings and the assembly sequence. The rationale for the architecture — why the frame is split, what fixes the envelope, how the wheel is tuned — stays in Section 6.1 and is not repeated here. Anything in this appendix may change without the design intent changing; anything in Section 6.1 may not.

C.1 Design Parameter Table

Table 63 is the single source of the driving dimensions referenced by every part in the model, per convention 1 of Section 6.1.1. It is the CAD-side mirror of the envelope closure of Section 5.2 and is updated on each iteration. It lists only what the CAD skeleton drives; the complete set of final values is Appendix A, and this table is subordinate to it.

Table 63: Driving CAD parameters. Every dependent feature references these; no part carries a duplicated hard-coded value. Values are drawn from the specification sheet of Appendix A.

Symbol	Parameter	Value
L	Cube edge length (outer)	150 mm
t_w	Frame wall thickness	5 mm
D_w	Wheel outer diameter (Sec. 5.3)	120 mm
t_r	Wheel axial thickness (Sec. 5.3)	20 mm
d_m	Motor axis offset from cube centre	TBD
h_m	Motor mounting face height on its axis	TBD
c_w	Wheel-to-subframe axial clearance	TBD
c_o	Clearance between orthogonal wheel modules	TBD
$V_{\min}$	Minimum internal clear volume	TBD
r_c	Corner contact feature radius	TBD

C.2 Model Tree and Part List

Table 64 lists the modelled parts by assembly, with their fabrication route and current status. It is the mechanical counterpart to the electronics bill of materials of Appendix B: the parts here are the team's design responsibility and are not supplied.

C.3 1-DoF Prototype Construction

The general-assembly drawing of the 1-DoF rig is given in Figure 19. This is the final released design of the single-axis prototype: the sheet carries the orthographic set — front and side elevations of the rig on its base, and a plan and edge view of the reaction-wheel assembly — from which the plate geometry, the pivot bearing seat, the motor bolt circle and the hard-stop placement are taken. The rig as built from this drawing is shown in Figure 9.

Table 64: CAD model tree and part list. Quantities are per assembled cube unless noted; the 1-DoF rig parts are listed separately.

Ref	Part	Qty	Fabrication route	Status
Inner structure				
M1	Motor subframe	TBD	TBD	TBD
M2	Motor mount (×axis)	3	TBD	TBD
M3	Battery cradle / retention	1	TBD	TBD
M4	Electronics tray	1	TBD	TBD
M5	Brake servo bracket	3	TBD	TBD
Reaction wheel assembly				
W1	Ballasted ring (Sec. 5.3)	3	TBD	TBD
W2	Wheel hub / rotor adapter	3	TBD	TBD
W3	Ballast bolt and nut set (M6, 15 per wheel)	45	Standard hardware	TBD
W4	Brake strike feature	3	TBD	TBD
Outer frame				
F1	Frame member / edge rail	TBD	TBD	TBD
F2	Corner contact feature	8	TBD	TBD
F3	Face panel / wheel guard	TBD	TBD	TBD
1-DoF rig (Sec. 6.1.2)				
R1	Face plate	1	TBD	TBD
R2	Pivot bearing mount	1	TBD	TBD
R3	Rig motor mount	1	TBD	TBD
R4	Hard stops / containment	TBD	TBD	TBD

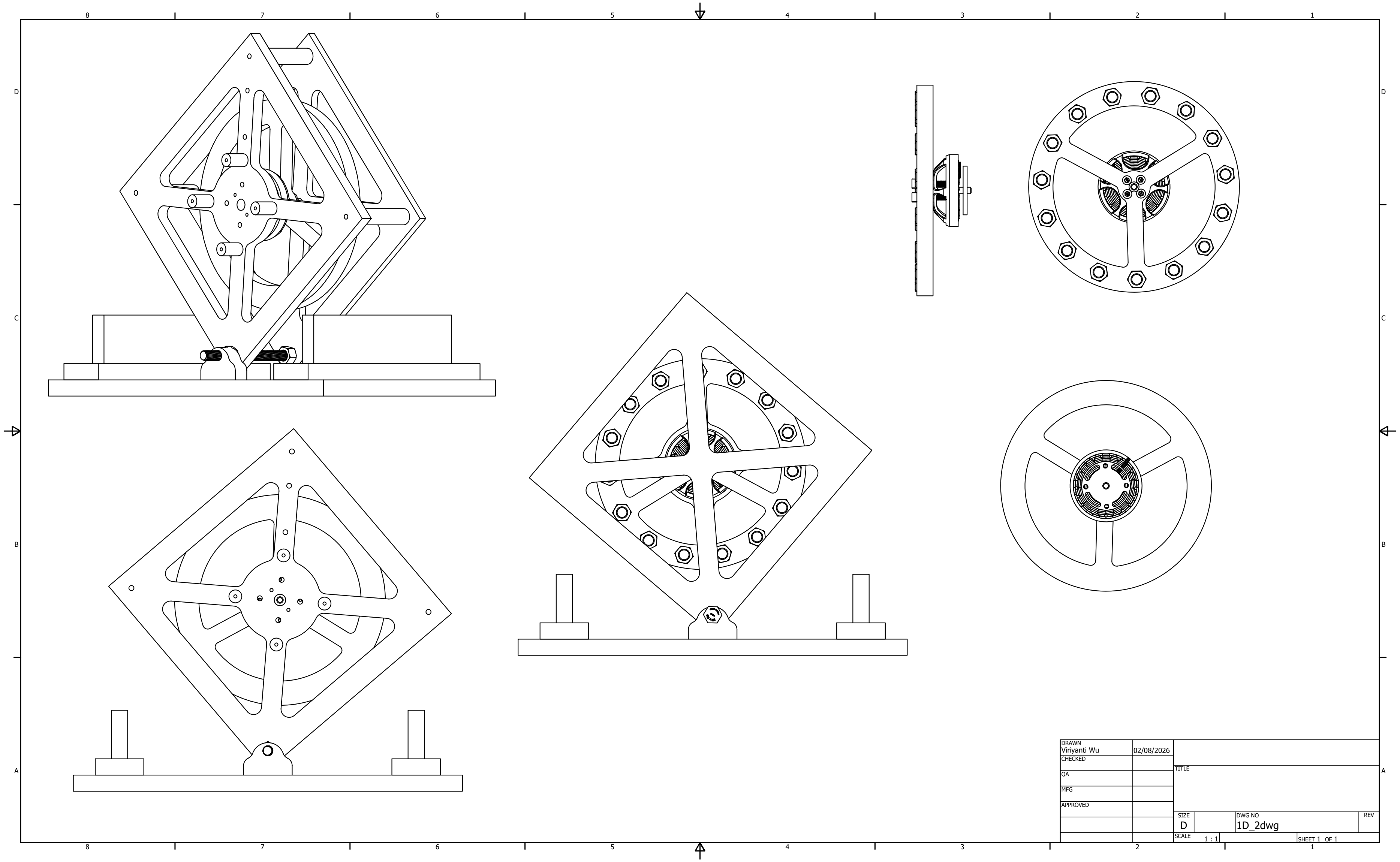

Figure 19: General-assembly drawing of the 1-DoF prototype rig — the final released design of the single-axis article — sheet 1 of 1 at 1:1. Upper left and lower left: front elevation of the rig mounted on its base, showing the diagonal plate orientation, the corner pivot and the two hard stops. Centre: side elevation through the wheel axis. Right: plan and edge views of the reaction-wheel and motor assembly, giving the spoke pattern and the ballast bolt circle used for the balancing adjustment of Section 5.3.

C.4 3D Cube Construction

The general-assembly drawing of the three-axis cube is given in Figure 20. The frame realises the two-part architecture of Section 6.1: an outer frame of edge rails and corner contact features encloses an inner structure carrying the three orthogonal reaction-wheel modules and the electronics tray. The sheet carries an isometric cutaway of the inner structure with the individual boards called out (upper left), an isometric view of the closed assembly with the three wheel modules visible through the open frame faces and the ballast bolt set annotated (upper centre), a front elevation of a single motor/wheel module in section, with the IMU location called out (right), and two front orthographic views of individual face modules giving the motor bolt circle and corner fastener pattern (lower left and centre). This drawing supersedes the concept prototype of Figure 10, which validated the two-part architecture and internal packaging but did not carry final dimensions; the edge length L and wheel geometry realised here follow the closure of Section 5.3.

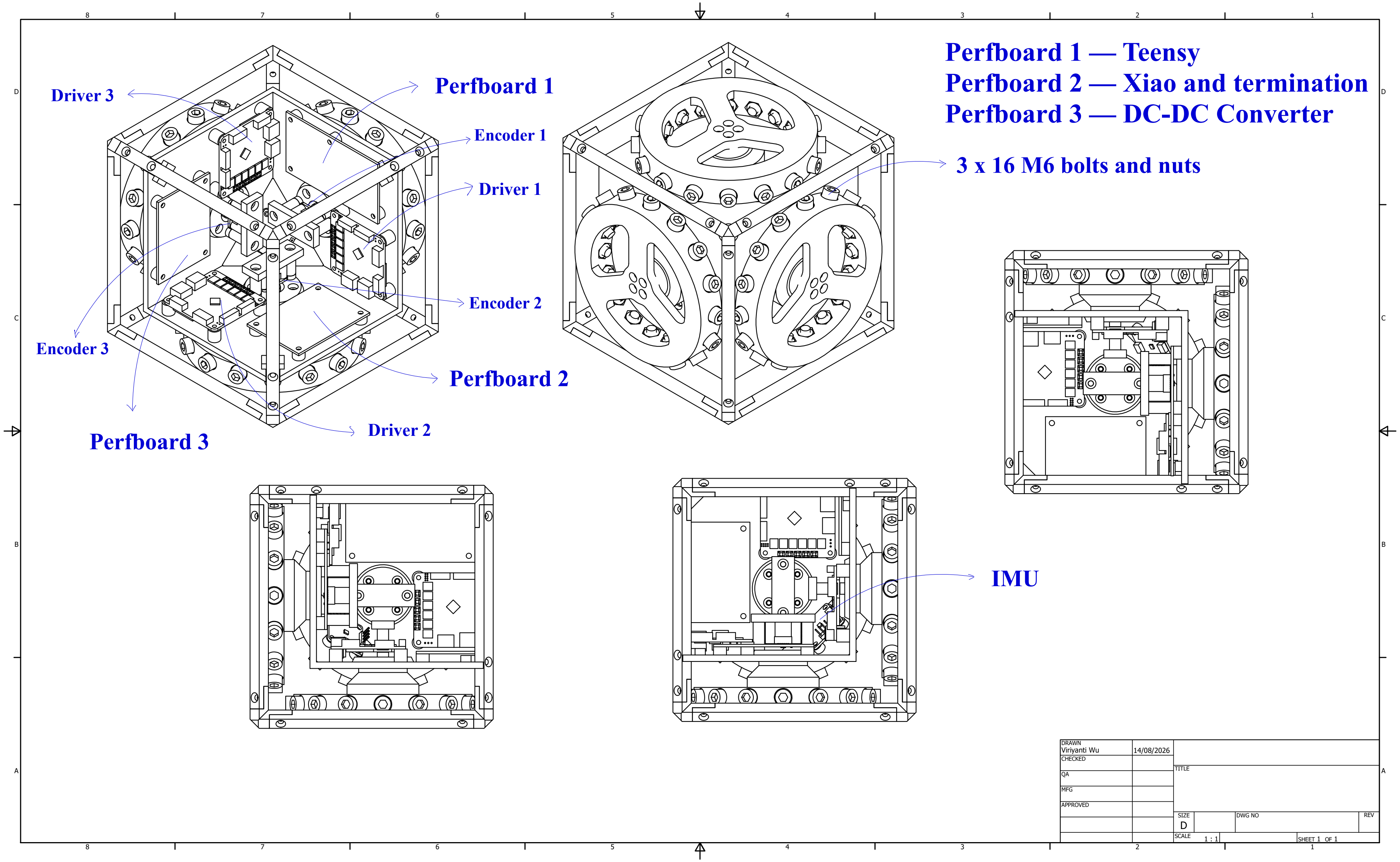

Figure 20: 3D visualization of the final three-axis cube flight article. The render highlights the internal structure containing three motor drivers, three encoder boards, and three core perfboards (Teensy FC, Xiao, DC-DC converter), integrated alongside three orthogonal reaction-wheel assemblies, the inner IMU, and frame ballast hardware.

The IMU callout above marks its physical location only: its sensing axes are not aligned with the body frame $\{B\}$ of Figure 7, and the fixed correction applied in software is given in Section ??.

C.5 Fabrication and Assembly

C.5.1 Assembly Sequence

The sequence of Table 65 is ordered by access: each step is one whose verification becomes impossible or destructive once a later step is complete. The encoder air gap in particular is re-checked after mechanical assembly, since it can shift when the structure is bolted up (Table 19).

Table 65: Assembly sequence. Ordering is set by access: each check must be completed before the step that encloses it.

Step	Operation	Check before proceeding
1	Wheel assembly: ring, hub, ballast bolts to the selected count	Balance check; bolt retention per Sec. 5.3
2	Motor to mount, encoder magnet to rotor	Encoder air gap and centring to the assembly spec
3	Motor modules to inner structure, three axes	Axis orthogonality; wheel clearance c_w, c_o
4	Battery and electronics tray into inner structure	Pack centring; board keep-out clearances
5	Harness install and dressing	Continuity and isolation (Sec. 7.7)
6	Inner structure into outer frame	Encoder air gaps re-checked after bolt-up
7	Contact features and guards	Contact geometry; wheel guard clearance
8	Mass properties measurement	Measured mass against Table 15

C.6 Mass Properties Verification

The CAD roll-up is an estimate until the assembly is weighed. Step 8 above closes the loop of Section 5.2: the measured mass is compared against the modelled value, and any discrepancy is carried back into $I_{w,\text{target}}$ and absorbed, where possible, by the ballast trim of Section 5.3 rather than by a re-design.

Table 66: Modelled versus measured mass properties. A discrepancy here is absorbed by ballast trim where the margin allows.

Quantity	CAD value	Measured	Discrepancy
Total cube mass M	1.800 kg*	1.626 kg	−174 g
Wheel mass m_w (each)	TBD	TBD	TBD
Wheel inertia I_w (each)	TBD	TBD	TBD
CoM offset from cube centre	TBD	TBD	TBD

* Not a closed CAD roll-up: the 1.800 kg is the budget of Table 15, whose structural line is still an allowance. It is quoted here as the value the measurement is compared against.

As-built mass, and what the underrun is worth. The integrated cube weighs 1.626 kg — 174 g below the 1.800 kg sizing basis of Table 12, and inside the 1.8 kg cap of CON-02. The difference is entirely structural: the specified hardware of Table 15 totals 1367.5 g and is measured or taken from datasheets, so the realised structure comes to 258.5 g against the 432.5 g allowance the budget carries.

The sizing basis is deliberately *not* revised to match. Section 5 sizes against the heaviest mass the design admits rather than the lightest it achieves, so 1.800 kg remains the figure PER-06 is derived at and the wheel is built to. What the underrun changes is margin, not requirement. Re-evaluating Equations (10)–(11) at 1.626 kg gives $\tau_g = 1.1884\,\mathrm{N\,m}$, $\beta = 1.3118$ and $h_w = 0.2281\,\mathrm{kg\,m^2/s}$, hence $I_{w,\mathrm{target}} = 5.4445 \times 10^{-4}\,\mathrm{kg\,m^2}$: the fifteen-station wheel delivers 115.2 % of what the built cube actually demands, against the 100.6 % it delivers against the basis it was sized on. Equivalently, the same wheel closes the corner jump-up at any speed down to about 3470 rpm, so the 4000 rpm design point carries some 530 rpm of speed margin on the article as built. That margin is realised, and is not to be spent by relaxing the sizing basis or by allowing the structure to grow back into its allowance.

D Electrical System Detail

This appendix holds the electrical detail kept out of Section 7: the full schematic capture, the connector and pin assignments, the harness drawings, the board layout and the bring-up checklists. The architectural decisions — two rails behind one E-stop, linear bus topology, perfboard rather than fabricated board — remain in Section 7. The part specifications are in Appendix B and are not repeated.

D.1 Schematic Capture

The system is captured on one flat KiCad sheet, reproduced at page size in Figure 21 and listed by functional block in Table 19. That capture is the working drawing the build is wired from. The blocks it carries are: power entry and protection (pack connector, anti-spark inrush path, combined E-stop and fuse, split to the motor bus and the logic rail); the 5 V logic rail off an LM2596 converter with back-feed protection and per-device decoupling; the CAN-FD bus as a linear chain across the three moteus drivers and the flight computer, terminated at the two physical ends only; the per-axis encoder SPI interfaces; the Teensy 4.1 flight computer with the BMI270 IMU; the ESP32 telemetry link; and the three brake servo drives.

Note: the MA600 breakout is paired with a diametric ring magnet (4 mm thick; Appendix B). The datasheet-recommended air gap for the 20 mT–80 mT nominal magnetic field window is 2 mm; the gap set on the build, and used throughout this document, is 1.5 mm.

Each block of the capture is checked against its own layout, mounting and connection constraint before bring-up. That constraint table (Table 19) is given in Section 7.2, ahead of the schematic itself, and is not repeated here. Component current ratings are given in Appendix D.2 (Table 67).

D.2 Component Current Ratings

Table 67 states the current each component on the 5 V logic rail and the pack rail is rated to draw. It replaces a measured-consumption table: at this stage no bench data exists, and a table of datasheet ratings is a stronger design input than an empty one, since the 5 V rail is sized against the coincident worst case — telemetry transmit burst, servo stall and full sensor load at once — and not the sum of typical values. Two entries are marked as unavailable rather than estimated: neither manufacturer publishes a stall or supply-current figure for that part, and a fabricated number would be worse than an honest gap. Both are closed by the bench measurement of Table 69 during electrical bring-up (Section 7.7).

Table 67: Component current ratings, by rail. Datasheet figures where published; measured values replace these per Table 69.

Component	Rail	Rated current	Basis
Teensy 4.1	5 V	≈ 100 mA	Board only, 600 MHz, no shields (PJRC community data)
XIAO ESP32-C6 telemetry	5 V	≈ 300 mA	TX burst (Table 62)
BMI270 IMU	5 V	≈ 0.7 mA	Typical, full 6.4 kHz ODR (Bosch datasheet)
MG92B brake servo (each, 3×)	5 V	Not published	TowerPro does not publish a stall-current figure; measure during bring-up
MA600 encoder (each, 3×)	5 V	Not published	MPS datasheet does not state supply current in the accessible electrical table; measure during bring-up
moteus-n1 driver (each, 3×)	Pack	100 A peak / 9 A cont.	Datasheet (Table 62)
MN4006 motor (each, 3×, via driver)	Pack	16 A peak, thermally limited	Manufacturer bench data (Sec. 8.2)

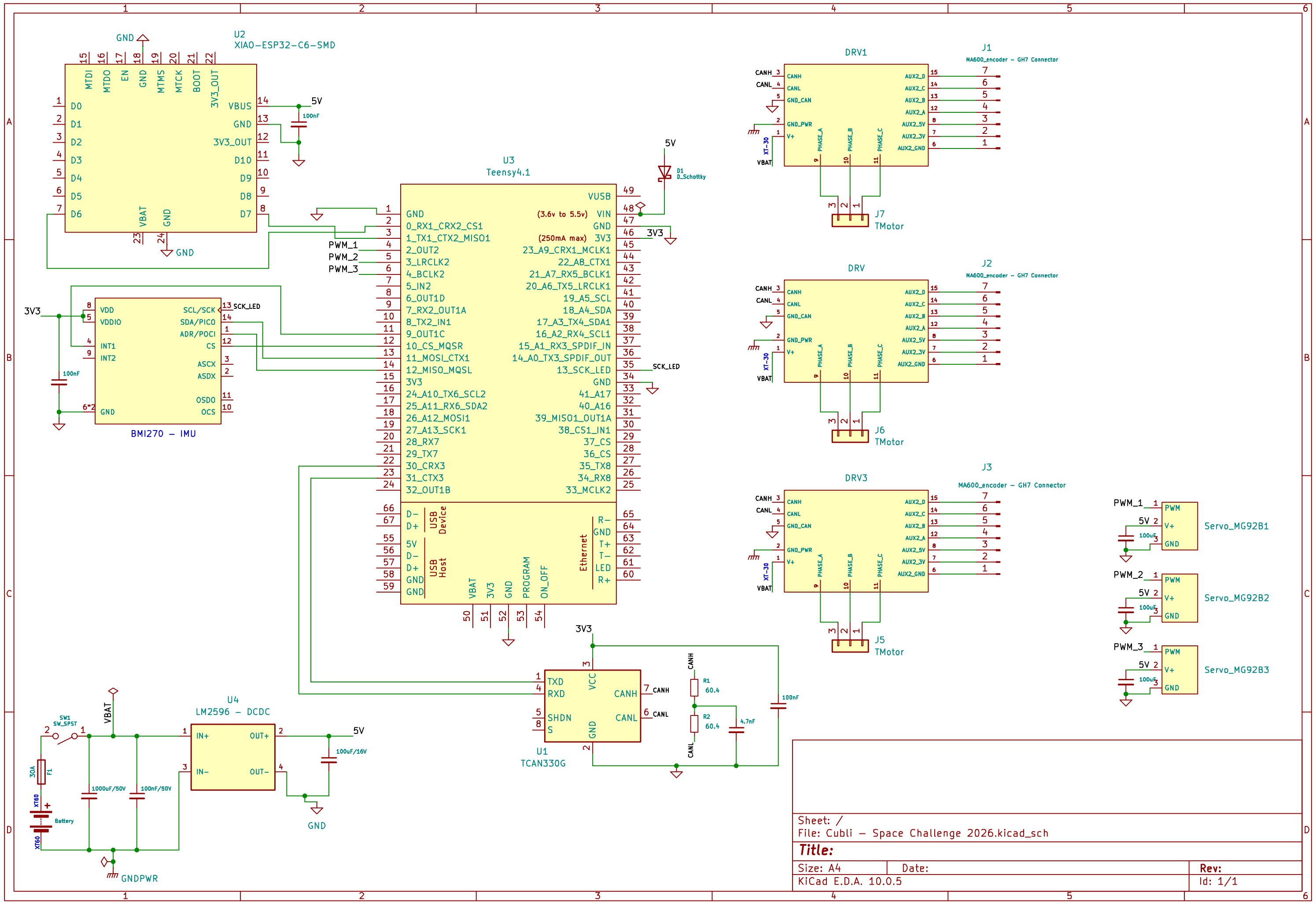

Figure 21: Full electrical schematic as captured, reproduced at page size. Covers the flight computer (Teensy 4.1) and its ESP32 telemetry module, the BMI270 IMU, the three moteus driver and motor connectors, the CAN-FD transceiver and termination, the LM2596 5 V converter from the pack, and the three brake servos.

D.3 Connector and Pin Assignments

Table 68 is the connector schedule. It exists so that a harness can be built and re-built from this document alone, and so that a mis-mate is a documented impossibility rather than a discovered one: no two connectors carrying different voltages share a housing type on this machine.

Table 68: Connector schedule. Keying and housing types are chosen so that two connectors carrying different voltages cannot be mated. Note: values have not been finalized at this preliminary stage.

Ref	From	To	Signals	
J1	Battery pack	Protection / E-stop	TBD	Pack +, pack −
J2	Protection	Distribution board	TBD	Motor bus
J3	Distribution board	Driver, axis 1	TBD	Motor bus
J4	Distribution board	Driver, axis 2	TBD	Motor bus
J5	Distribution board	Driver, axis 3	TBD	Motor bus
J6	Board	CAN chain, end A	TBD	CAN-H, CAN-L, GND
J7	CAN chain, end B	Termination network	TBD	CAN-H, CAN-L, GND
J8	Driver, axis i	Encoder, axis i	TBD	SPI, 3.3 V, GND
J9	Board	IMU	TBD	Serial bus, supply, GND
J10	Board	Telemetry module	TBD	UART, supply, GND
J11	Board	Brake servo, axis i	TBD	PWM, 5 V, GND

D.4 Harness Drawings and Wire Schedule

The harness is organised into seven runs: pack to protection, protection to board, board to the three driver drops, the three motor phase leads, the CAN chain segments, the three encoder SPI runs, and the three servo drives. Wire gauge on the motor bus (pack-to-protection, protection-to-board, driver drops and motor phase leads) is sized on spin-up current rather than the balancing current, since spin-up is the sustained high-current phase; the CAN and encoder SPI runs are sized on length limits rather than current (Table 19); the servo drives are sized on stall current. Gauge and run length are recorded once measured on the assembled harness.

D.5 Board Layout

The floorplan is constrained before it is aesthetic: the driver positions follow from the encoder cable limit, the bus route follows from the driver positions, and the board is placed around what remains (Section 7.5). Rail separation is maintained physically on the board, not only schematically, so that motor-current return does not share copper with the sensing ground.

D.6 Bring-Up and Isolation Checklists

These checks are performed on every re-assembly, not only on first build, since the encoder gaps and connector seating are exactly what mechanical work disturbs.

Before any power is applied.

1. Isolation: motor bus to chassis, logic rail to motor bus, each rail to ground.
2. Capacitor voltage ratings confirmed by position — no low-voltage part on the pack bus.

3. Polarity of every fabricated connector verified against Table 68.
4. Protection element in circuit: E-stop functional, fuse fitted and rated, anti-spark path intact.
5. Encoder air gaps re-measured after mechanical assembly.

First power-on, current-limited bench supply.

1. Current limit verified against a known load before connecting the assembly.
2. Logic rail voltage and ripple under load; converter thermal check.
3. All drivers enumerate on the bus; encoders read back with no error flags; IMU responds.
4. Bus integrity at the full data rate: sustained soak with all nodes addressed, error-frame count logged.
5. Open-loop wheel motion, wheels contained, one axis at a time.

First battery power-on. Performed only after the bench sequence passes in full, with the pack protection verified, cell monitoring active and the assembly in a fire-safe area. Inrush is measured against the bulk capacitance on connection.

D.7 As-Built Measurements

Table 69 records the measured electrical figures that replace the estimates used during design. These are the numbers reported back to the modelling and control work, and they are the reason the parameter identification of Section 4.3 runs on measured rather than assumed values.

Table 69: As-built electrical measurements. Each replaces an estimate used earlier in the design. Note: values have not been finalized at this preliminary stage.

Quantity	Measured	Replaces / feeds
Motor torque constant K_m	TBD	Replaces 0.025,13 N m/A from datasheet K_V; Table 9
Wheel inertia I_w (spin-down)	TBD	Table 9, CAD value
Viscous and Coulomb friction	TBD	C_b, C_w estimates
Command-to-encoder latency	TBD	Loop timing, Sec. ??
Achievable wheel speed on bus	TBD	$\omega_{\max}$, Sec. 5.1.1
Logic rail load, worst case	TBD	Table 67
MG92B servo stall current	TBD	Table 67, not published
MA600 encoder supply current	TBD	Table 67, not published
Runtime per charge under balance load	TBD	Demonstration planning

E Control Algorithm Detail

F Requirements Verification Matrix

This appendix is the working record behind Table 4: for each requirement, the procedure that verifies it, the criterion that closes it, the evidence to date and the current status. Section 2 states *what* must be true; this appendix states *how it is shown*, and is updated as testing proceeds rather than written once.

Status values are: **Verified**, evidence exists and meets the criterion; **Partial**, evidence exists on a representative article but not on the delivered one; **Open**, no evidence yet. A requirement verified on the 1-DoF rig is *Partial* with respect to the cube, because the rig is a different article — the distinction is kept deliberately, since collapsing it would overstate what the integrated machine has been shown to do.

Table 70: Requirements verification matrix. Methods: A analysis, I inspection, T test, D demonstration.

ID	V	Procedure	Acceptance criterion	Status
Functional				
FUN-01	T	Release the cube at the edge equilibrium; run the balancing controller with no operator contact.	Balance sustained; no external support at any point.	Open
FUN-02	T	As FUN-01, at the corner equilibrium, with the three-axis gain set.	Balance sustained on a point contact.	Open
FUN-03	T	Command a slew between two defined attitudes from a stable balance.	Target attitude reached and balance re-captured without a fall.	Open
FUN-04	T	Apply a repeatable impulse to the body during balance.	Return to the equilibrium without a fall.	Partial (rig)
FUN-05	T	Compare the estimator output against a known reference attitude, statically and while moving.	Estimate tracks the reference within the band recorded in Sec. 9.1.1; no external reference used in flight.	Open
FUN-06	T	Balance until a wheel reaches the desaturation threshold; log the commanded bias torque.	Desaturation torque commanded; wheel speed bounded without loss of balance.	Open
FUN-07	D	Exercise every transition of Table ??, including at each threshold boundary.	All transitions taken as specified; no chattering at any boundary.	Open
FUN-08	D	Issue a disarm command from each operating mode in turn.	Drivers de-energised from every mode without exception.	Open
FUN-09	D	Induce a fall from balance and observe the response.	Safe state entered with wheels commanded down; no continued actuation.	Open
FUN-10	T	Spin up to the design speed from a face rest; actuate the brake; attempt capture.	Cube reaches an edge or corner and captures into balance.	Open
FUN-11	T	Execute consecutive jump-ups with re-stabilisation between each.	Two or more consecutive hops with balance re-established after each.	Open
FUN-12	D	Run the control loop with telemetry enabled and disabled; compare cycle timing.	No change in worst-case cycle time attributable to telemetry.	Open
Performance				

Continued on next page

Table 70 (continued)

ID	V	Procedure	Acceptance criterion	Status
PER-01	T	Initialise at rest at the upright equilibrium; release; time the balance. Run on the 1-DoF rig, then on the integrated cube at the edge equilibrium.	Duration $\geq 60\,\mathrm{s}$; no evident wheel-speed drift, on each article.	Partial (rig verified; cube open)
PER-02	T	As PER-01, on the integrated cube at the corner equilibrium.	Duration $\geq 60\,\mathrm{s}$; no evident drift.	Open
PER-03	T	Displace the rig by hand to a measured angle from the support; release.	Recovery from $\geq 12°$, then PER-01 satisfied.	**Verified** (to 24°)
PER-04	T	As PER-03, on the cube, in each of the principal tilt directions.	Recovery from $\geq 5°$, then PER-01 or PER-02 satisfied.	Open
PER-05	T	Instrument the control task; log per-cycle execution time over a full balancing run.	Loop rate 1 kHz; worst-case cycle $\leq 1\,\mathrm{ms}$, not merely the mean.	Open
PER-06	A, I	Evaluate wheel inertia from the CAD mass properties and from the sizing calculator; confirm the ballast station count on the built wheel.	$I_w \geq 6.2352 \times 10^{-4}\,\mathrm{kg\,m^2}$; CAD and calculator agree.	**Verified** (analysis, 100.6 %)
PER-07	A, T	Spin-up test on the built wheel at end-of-discharge pack voltage (WBS E10).	Design speed reached and sustained on a depleted pack, not only a charged one.	Open
PER-08	A	Derive the achievable ceiling from K_v, winding resistance, modulation derate and end-of-discharge bus voltage; compare against the design speed.	Design speed $\leq 80\,\%$ of the derived ceiling.	**Verified** (analysis, 57.8 %)
PER-09	T	Log driver current throughout balancing and disturbance recovery.	Peak within the 9 A continuous rating.	Partial (rig)
PER-10	T	Spin-down test: brake from the design speed with the event logged at a rate resolving a few-millisecond transient.	Arrest within 100 ms.	Open
PER-11	T	Record a telemetry session; count logged samples per second of each field.	$\geq 50\,\mathrm{Hz}$ for state, command and wheel speed.	Open
PER-12	A, T	Compute $t_{\mathrm{spin}} = I_w \omega_{\max} / k_t i$ from the K_V-derived torque constant; confirm on the built wheel by logging speed against time through a spin-up at the continuous current limit (WBS E10).	Design speed reached within 2 s without exceeding 9 A.	Open (analysis: 1.2 s)
PER-13	A, T	Compare the commanded spin-up torque against $\tau_g = MgL/2$ at the as-built mass; observe the body on a face rest through a full spin-up.	Reaction torque $\leq 0.5\,\tau_g$; no body motion before the brake event.	Open (analysis: 17.2 %)
Design constraint				
CON-01	I	Measure the built frame against the CAD driving parameter.	Outer edge 150 mm within the stated tolerance.	Open
CON-02	I	Weigh the integrated cube, ready to run.	$\leq 1.8\,\mathrm{kg}$.	**Verified** (1.626 kg, App. C.6)
CON-03	I	Inspect the assembled machine against the CAD model tree.	Three wheels on mutually orthogonal axes through a common centre.	Open

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Table 70 (continued)

ID	V	Procedure	Acceptance criterion	Status
CON-04	I, D	Inspect the power architecture; run a full demonstration on pack power alone.	No tether or bench supply connected during the demonstration.	Open
CON-05	I	Inspect the bus route and termination against the layout drawings.	Linear chain, short stubs, termination at exactly the two ends.	Open
CON-06	A, T	Sum the coincident worst-case rail load (telemetry burst, servo stall, full sensor load); measure at bring-up.	Within the regulator's derated rating as installed.	Open
CON-07	I	Check the assembled electronics against the frozen mount interface.	Fits the outline, hole pattern and keep-out heights without modification.	Open

Evidence pointers. Verified and partial entries are supported by Section 9.1.1 (1-DoF balance duration, recovery angle and driver current), Section 5.2 and Table 16 (wheel inertia against target), and the wheel-speed derivation behind Table 12. Section 9.1.2 carries the result templates for the cube; every requirement listed Open above resolves through one of them, and the templates are laid out to mirror the 1-DoF quantities so the two articles remain directly comparable.

G Risk Register and Contingency Plan

Table 71 lists the risks judged material to the project, with the trigger that realises each one, its impact, severity (S) and likelihood (L) on an H/M/L scale, the accountable owner, and the mitigation. Table 72 fixes, in advance, the order scope is cut if the schedule runs short, and the note that follows covers how the team backs up whoever is stuck on a critical-path task.

Table 71: Risk register: trigger, impact, severity (S), likelihood (L), owner and mitigation.

ID	Risk	Trigger	Impact	S	L	Owner	Mitigation
Electrical and Power							
RE1	16 V capacitor on the 25 V bus	Build error: the BOM 100 µF parts are 16 V and fit the bus rail	Capacitor vents; board destroyed and bus shorted; ≥ 3 day recovery through the 9 Aug gate	H	M	Pablo	16 V parts segregated at build, 5 V rail only; bulk cap is a separate 470–1000 µF, ≥ 50 V item; checked before any power
RE2	No bulk capacitance at regenerative braking	Hard wheel deceleration returns ≈ 72.8 J per wheel (218 J across three) to the bus	Bus overvoltage damages a moteus-n1; loses one axis and the bench-rig spare	H	M	Pablo	Fit and verify the 50 V bulk capacitor before any commanded deceleration; driver bus-overvoltage fault limit
RE3	MA600 air gap out of specification	Gap > 0.5 mm or chip off-centre, at build or disturbed later at assembly	Angle dropout and degraded commutation; presents as a control fault and misdirects debugging	H	H	Pablo	Shim value fixed on the bench and re-verified per axis at every assembly step; encoder error flags logged in every balance run
RE4	SPI run exceeds 20 cm	Repackaging moves a driver away from its encoder	Encoder corruption at 6 MHz; same signature as RE3	H	M	Pablo	Driver positions derived from encoder positions and frozen in the board outline; later moves are electrical change requests
RE5	CAN-FD unreliable at 5 Mbit/s	Stub > 10 cm, mis-placed termination, or star topology at integration	Error frames and dropped commands under vibration; intermittent and hard to diagnose	H	M	Pablo	Linear topology; split termination at the two physical bus ends only; scope and soak test before integration
RE6	Board outline late to CAD	Layout slips past 1 Aug	Mass freeze and frame print slip, cascading to the 9 Aug integration gate	H	M	Pablo	Two days of float; if bring-up slips, issue the outline with conservative keep-outs rather than delay it
RE7	Rig parameters late to control	Wheel balance or rig assembly slips past 2 Aug	Blocks the sequential control chain; propagates to 20 Aug with no recovery	H	M	Pablo	Escalate if off track; fallback delivers K_t and loop latency from bench measurement without the wheel

ID	Risk	Trigger	Impact	S	L	Owner	Mitigation
RE8	5 V rail undersized	Teensy, XIAO, IMU and servo stall draw together	Rail sag and MCU brown-out mid-balance; misread as a control failure	M	M	Pablo	Sum the load budget before build; heatsink the regulator; bulk cap plus back-feed diode; measure sag under worst-case load
RE9	Battery bring-up incident	Battery connected before the bench sequence completes, or without anti-spark switch and fuse	Arc or LiPo fire; injury and total project loss	H	L	Pablo	All bring-up on a current-limited supply; battery introduced only at the final power-on, with anti-spark switch, fuse, cell alarm and a second person present
RE10	Power harness fails under load	Cold joint or under-gauge wire on a high-current bus	Voltage drop, connector melt, or open circuit mid-demonstration	M	M	Pablo	14 AWG minimum, pull-tested and thermally imaged; spare harness in the demonstration kit
RE11	No spare actuation set	Any motor, driver or encoder fails after cube integration	Loses an axis with no resupply lead time before 20 Aug; demonstration cut short	H	M	Pablo	All three sets treated as flight hardware from integration on; no cannibalising for bench tests afterward; reseat and inspect at every touch
RE12	Brake work displaces edge balance	Brake build runs concurrently with the edge-balance gate	Missing the gate cuts the jump-up stretch goal, which the brake exists to serve	M	M	Pablo	Brake support is the designated sacrificial task; conflicts resolve to edge balance by standing decision
RE13	Vibration loosens connectors	Sustained running at high wheel speed	Intermittent axis dropout; cube falls and damages structure	M	H	Pablo	Strain relief, labelling and connector retention; reseat check as the first action of every support call
RE14	Perfboard rework consumes its float	Wiring error found at bus test; the build step has no slack	Board late to integration, cascading to the 9 Aug gate	M	M	Pablo	Wiring checked against the floorplan before power; rails brought up incrementally so faults localise
Programme and Technical							
PR1	Final week has almost no slack	Any slip after 8 Aug	No days left to recover before the 20 Aug presentation	H	M	Pablo	Go/no-go on the jump-up called at end of day 11 Aug; nothing new started after that date, only debugging what already runs
PR2	Structure heavier than the mass allowance	Printed frame and fasteners weigh in over the 374 g allowance	Wheel undersized against the jump-up sizing; momentum budget missed	M	M	Nasia	181 g of headroom before re-ballasting is needed; weigh at frame print and re-run the sizing script if over
PR3	Brake torque never measured on the bench	Brake work stays sacrificial through the final week	The inertia-requirement safety margin ships unverified	M	H	Niccolo	Run a standalone spin-down test in any open window; otherwise state explicitly that the figure is unmeasured

ID	Risk	Trigger	Impact	S	L	Owner	Mitigation
PR4	Programme lead is a single point of failure	Pablo unavailable during a critical-path week	Schedule, most electrical tasks and several software tasks stall together	H	L	All	Deyan holds the document and schedule state; Niccolo and Andrea can act on the electrical notes directly; no single day depends on one person

Table 72: Contingency: order in which scope is cut if the schedule runs short.

Order	Scope cut
1	Jump-up demonstration (S3) — cut first; standing decision, see §3.3
2	Brake mechanism build and its bench test — cut with the jump-up, since it exists only to serve it
3	Corner balance and commanded slew (S2b/c) — cut only if edge balance is not solid by 17 Aug
4	Edge balance (S2a) and the 1-DoF rig result — never cut; the protected demonstration and the document's primary evidence

Parallel work and backup coverage. The five workstreams (documentation, control/software, hardware build, 3D design, electrical) run in parallel by design, so a stall in one does not by default stall the others — but only if someone is deliberately free to help. The practice already applies twice in this plan: Nasia's fallback on any day the sizing numbers are late is CAD review and tolerance work rather than idle waiting, and Andrea's fallback if the rig is not ready on time is bench measurement of K_t and loop latency, so she is never simply blocked (RE7). The same rule holds for the rest of the schedule: whoever is not the bottleneck on a given day supports whoever is, rather than working ahead alone on their own next task.